MCK Region Transitions: Restricted Realization Mechanisms and Their Limits

Davide Bragetti*
DIAG, Sapienza University of Rome
ORCID: 0009-0008-7645-8241
Carmelo Asaro
DIAG, Sapienza University of Rome
Edoardo Palumbo
Don Carlo Gnocchi Foundation
ORCID: 0009-0004-4527-5830
Giuseppe Francesco Italiano
LUISS Guido Carli, Rome
ORCID: 0000-0002-9492-9894

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Abstract

The MCK framework partitions artifacts into seven regions by three predicates — meaning, opacity, function — and its region transitions form a group acting on the region set. The result is algebraic: it exhibits the transition between any two regions and supplies no procedure that moves an artifact along one. We study four restricted realization families and fix, for each, what it covers and where it stops.

Assimilation reclassifies pure opacity and nothing else; we prove that the residue — the opacity co-present with meaning or with function — is a floor no comprehension strength lowers, so an agent that possesses symbols cannot assimilate its way out of possessing them. Operator admission seats a candidate in the fully intersecting region without extending structural reach, so any extension of reach under a fixed realization witnesses a changed basis rather than a new name. Denotation admits as sources exactly the symbol domain and is not recoverable from the predicates: perfect recovery of a region label leaves denotation undetermined. The reconstruction certificate is strictly weaker than the discovery predicate, the gap being a key-membership condition we show unclosable in general.

Each mechanism is studied for a declared observer; no universal artifact-level realizer follows. The realization is ideal and exact; whether a trained representation supports an approximate counterpart is not addressed here, and no infinite-reachability claim rests on finite evidence. Five temporal conjectures over locally declared runs mark open trace-level questions; no monitor is constructed.

1 Introduction

The MCK framework classifies every artifact by a triple of binary predicates — semantic content M, opacity C, and functionality K — and partitions the admissible triples into seven regions $R_1, \ldots, R_7$ [3]. That framework is stated over an abstract artifact domain and says nothing about *where* artifacts sit in a geometric representation. Modern learned systems, however, carry exactly such a representation: an embedding space in which every string and substring has a position, and in which regularities (difference vectors, lattice-like relations) propose where *new* meanings could live.

*Corresponding author: davide.bragetti@uniroma1.it

This paper asks a single question: what happens when the MCK predicates are overlaid on an embedding space as *envelopes* of that geometry? Reading M, C, K geometrically makes several things newly expressible. The paper's contributions are the following seven.

1. **A restricted realization architecture.** We separate the imported algebraic transition object from artifact-level mechanisms and study four of the latter — assimilation, operator admission, denotation, and reconstruction certification — each with its own coverage and its own stopping point. The envelopes of Sec. 3 and the three-case verifier dynamic of Sec. 4 are the setting in which they are stated; no new region primitive is introduced.

2. **Assimilation stops at a floor** (Thm. 5.6, Prop. 6.5). An assimilation step reclassifies pure opacity and nothing else, and the residue $\Theta \cup R_6$ is a lower bound no comprehension strength lowers. An agent that possesses symbols cannot assimilate its way out of possessing them.

3. **Admission preserves reach** (Prop. 4.4). Seating an admitted candidate operator leaves structural reach unchanged, so an observed extension of reach, under a fixed realization, witnesses a changed basis rather than a new name.

4. **Denotation is constrained at the source and unrecoverable from the predicates** (Prop. 9.3, Cor. 9.11). Θ is exactly the set of admissible sources — instantiation is a further question, and degree zero stays permitted; perfect recovery of an artifact's region label still leaves its denotation undetermined, so label recovery and semantic recovery are two problems and the first does not reduce the second.

5. **The reconstruction certificate is strictly weaker than discovery** (Prop. 5.9). The two differ by a key-membership condition which the construction supplies no ground for and which is unclosable in general.

6. **The regularity notion has an exact boundary** (Rem. 6.1, Rem. 6.2). Separation forces the offset condition to be exact rather than tolerant, and cosine similarity admits no separating radius at all. The empirical literature reports a related finding by a different route — that offset-based recovery tracks neighbourhood similarity — and Sec. 10.3 states the relation between the two carefully.

7. **Composite status is not determined by component status** (Prop. 7.3, Sec. 7). Classifying the artifacts of a computation separately does not classify the computation: the justification relations between them carry a region of their own, and a procedure every one of whose components is resolved may still have an unresolved relation among them. Sec. 7 makes this the basis of a diagnostic reading of the regions and works it through on Hensel–Newton refinement.

Formal and systematisation contributions, and what is excluded. Items 2–5 and 7 above rest on the proved statements Thm. 5.6, Prop. 6.5, Prop. 4.4, Prop. 9.3, Cor. 9.11, Prop. 5.9 and Prop. 7.3, supported among others by Prop. 5.11 and Prop. 6.6; Prop. 3.6 records the envelope construction. Most of them are limitative: assimilation leaves Θ invariant, so a symbol-bearing agent stays symbol-bearing (Thm. 5.6, Cor. 5.7); the opacity in $\Theta \cup R_6$ survives every assimilation step (Prop. 6.5(ii)); the reconstruction certificate leaves the key-membership condition open (Prop. 5.9); the XOR certificate is inadmissible (Prop. 5.11); admitting an already reachable point leaves reach unchanged under a fixed basis and realisation (Prop. 4.4); region labels leave denotation undetermined (Cor. 9.11); and component statuses leave composite status undetermined (Prop. 7.3).

Beyond these formal contributions, we use the envelope + region + admissibility apparatus to locate and distinguish (i) the vocabulary-axis-extension family of engineering practice — vocabulary expansion, embedding initialisation, and continued pretraining (Sec. 10) — as the vocabulary-extension reading of the mint of Def. 4.2 (Rem. 4.1), and (ii) the fixed-vocabulary parametric editing family — ROME [28], MEMIT [29], EMMET [16], MEND [33], Task Arithmetic [19], and KnowledgeEditor [8], with the ripple-effect evaluation of Cohen et al. [7] — as *adjacent* operators distinct from the mint (they preserve the input-token axis; the mint, in its language-model reading, extends it). Context-only learning [32] and external-memory persistence [23, 43] are explicitly excluded from the present formal substrate.

Section 6 then turns the structural and the operational side of contributions 2 and 6 into two *measurable* axes — the offset regularity of the embedding and the strength of an agent's comprehension function — and asks what each one bounds. Two of the paper's results live there: that strengthening the comprehension function lowers residual opacity while assimilation nonetheless stops at a floor no strength can pass, and that the two axes bound different things, neither constraining the other.

Because the geometric side is finite (a finite domain, a finite vocabulary) while the concept space it proposes is not, the paper straddles a finite substrate and an infinite target. Section 8 states the discipline that governs every finite-to-infinite crossing and explicitly refuses to inflate finite evidence into an infinite-reachability claim. The unifying thread across these outputs and the companion papers is kept implicit; the present manuscript is self-contained and rests only on the foundations paper [3].

What is imported, and what this paper adds to it. The foundations paper proves that the region transitions form a group acting simply transitively, so that for any source and target cell a unique transition operator exists, and it is explicit that this operator is there *a purely algebraic object*: it fixes which transitions are admissible and between which cells, and records the quantitative question — how far a realisation carries, at what cost, and along what trajectory — as open, noting that the present companion studies restricted realisation mechanisms of its own rather than movement along the transition operator [3]. That import is a statement about *structure*, not about artifacts: it exhibits, for each pair of cells, the transition that would take one to the other, and it supplies no procedure that moves any given artifact. Nothing below closes that gap in general. What this paper does instead is study four *restricted* realisation families and establish what each covers and where each stops.

Structural target	Realisation family	Proven coverage	Formal limitation
$R_2 \to R_1$	assimilation	removes only C_0; the residue $\Theta \cup R_6$ is a floor no strength lowers (Prop. 6.5)	the selected set may be empty (Prop. 5.9(iii))
into R_7	operator admission	seats an admitted candidate (Def. 4.2)	reach is unchanged (Prop. 4.4)
Θ targets	denotation	Θ is exactly the admissible source set (Prop. 9.3)	not recoverable from the predicates (Cor. 9.11)
$C_0 \to M_0$	reconstruction certificate	the construction discharges two of three conditions	key-membership is unclosable in general (Prop. 5.9)

Each row is an existence-and-limit pair, and no row is universal: every one is stated for a declared observer and a declared mechanism. This is why the paper is about the *limits* of ideal realization as much as about its mechanisms. The universal artifact-level realiser — a procedure that, given

any admissible structural transition, moves an arbitrary artifact along it — is not established here and remains open.

Every numbered statement carries exactly one epistemic tag — PROVEN, CONJECTURED, FOLKLORE, or DEFINITIONAL for definitions. Remarks follow the same rule with one refinement: a remark that *asserts* something of its own is tagged, while a remark that only explicates an already-tagged result inherits that result's status and is left untagged.

2 Preliminaries

This section recalls the MCK structures this paper builds on. Nothing here is new: all definitions are those of the foundations paper [3], recalled verbatim in substance and cited, not redefined. New primitives appear only from Section 3 onward.

2.1 Predicates and the artifact domain

The foundations paper fixes an *artifact domain* $\mathcal{A}$ — the totality of artifacts under discussion — and equips each $x \in \mathcal{A}$ with three binary predicates [3]:

- $M(x) \in \{0,1\}$: whether x carries structured *semantic content*;
- $C(x) \in \{0,1\}$: whether the internal structure of x is opaque (*opacity*) — the high-entropy end of the semantic/entropy axis of the foundations treatment [3];
- $K(x) \in \{0,1\}$: whether x has operator-like or transformative *functionality*.

We write M, C, $\mathsf{K} \subseteq \mathcal{A}$ for the corresponding extensions $\{x : M(x) = 1\}$, $\{x : C(x) = 1\}$, $\{x : K(x) = 1\}$. Following the foundations paper, the three predicates are logically independent: all eight predicate triples are admissible a priori, so in particular the purely functional cell $R_3 = \mathsf{K} \setminus (\mathsf{M} \cup \mathsf{C})$ (functionality without meaning or accessible structure) is a cell of the seven-region partition [3]; whether it is instantiated is a separate, conventional matter.

2.2 Regions R_1–R_7

The admissible predicate triples partition into the seven MCK regions [3]. We use them exactly as defined there and introduce no new region primitive. All seven are listed, since results below refer to each:

- $R_1 = \mathsf{M} \setminus (\mathsf{C} \cup \mathsf{K})$: *pure understanding*;
- $R_2 = \mathsf{C} \setminus (\mathsf{M} \cup \mathsf{K})$: *pure opacity*;
- $R_3 = \mathsf{K} \setminus (\mathsf{M} \cup \mathsf{C})$: the *pure functional / generative* region — functionality with neither exposed meaning nor accessible structure;
- $R_4 = (\mathsf{M} \cap \mathsf{C}) \setminus \mathsf{K}$: *partial integration*;
- $R_5 = (\mathsf{M} \cap \mathsf{K}) \setminus \mathsf{C}$: the *transparent operator* region — functionality with exposed meaning and no residual opacity;
- $R_6 = (\mathsf{C} \cap \mathsf{K}) \setminus \mathsf{M}$: the *opaque operator* region — functionality whose structure is present but whose meaning is not exposed;

- $R_7 = \mathsf{M} \cap \mathsf{C} \cap \mathsf{K}$: the *Full Triple* region where a minted operator carries meaning, structure, and function together.

We also write $M_0 = \mathsf{M} \setminus (\mathsf{C} \cup \mathsf{K})$ and $C_0 = \mathsf{C} \setminus (\mathsf{M} \cup \mathsf{K})$ (pure-message and pure-cipher); this notation is introduced here, and comparing the displays, these are the regions R_1 and R_2 of [3] under another name,

$$M_0 = R_1, \qquad C_0 = R_2, \tag{1}$$

and we use the two notations interchangeably.

2.3 The structured ambiguity domain Θ

The foundations paper singles out one further set, which is not a region but a union of two [3]: the *structured ambiguity domain*

$$\Theta \;=\; \mathsf{M} \cap \mathsf{C} \;=\; R_4 \cup R_7. \tag{2}$$

Its role there is semantic rather than merely combinatorial. Θ is exactly the domain of the denotation map: an artifact denotes — is a *symbol* — precisely when it lies in Θ, bearing meaning (M) while retaining unresolved reference (C). A symbol therefore cannot inhabit pure M. All of this is recalled, not restated as new: Θ, the denotation map, and the domain-restriction clause that confines denotation to Θ are defined in [3] and are used here exactly as defined there.

Three clauses of that denotation map are recalled explicitly, because Section 9 turns on them. The map denotes: $\Theta \to \mathrm{Ref}$ sending a symbol to a referent is stipulated there to be *static* (denotation does not change under the region dynamics), *single-valued* (one referent per application), and *domain-restricted* (only Θ-artifacts denote), with every denotation target lying in the classified domain $\mathcal{A}^+$ of Section 2.6 [3]. The multiplicity a symbol may carry is therefore not located in the map but in its *target set*, likewise recalled from [3],

$$T(s) \;=\; T_{\mathrm{direct}}(s) \,\cup\, T_{\mathrm{indirect}}(s), \tag{3}$$

the referents reachable from s by a single denotation edge together with those reachable along directed denotation paths all of whose internal nodes are symbolic. T is used below exactly as defined there, and the staticity clause in particular is a *stipulation* of that paper and not a theorem of it, a point Section 9.7 takes up. The graph on which T is computed, however, is not simply the graph of denotes: the axioms Section 9 supplies *extend* that map rather than transcribe it, and Section 9.1 states in what three respects and why the deferral leaves that choice open. The foundations paper defers the axioms governing the graph on which T is computed, and the classification of symbols by the shape of $T(s)$, to the present one; Section 9 supplies them.

2.4 A scoping convention that is *not* adopted here

The foundations paper carries a scoping convention under which no artifact is instantiated in R_3, stated there as an axiom and explicitly qualified as a choice of that exploration rather than a claim that R_3 is empty in general [3]. Because the convention is a scoping choice, a downstream paper must say whether it inherits it, and the answer changes which sets are equal to which.

This paper does not adopt it. The material below treats R_3 as an available region rather than a suspended one: it is the pure part of the generative potential of K (Rem. 3.4), and a paper that suspended it could not make that statement. Declining the convention admits R_3 as a region that *may* be inhabited; it does not assert that it is, and nothing below requires it to be. Two consequences are recorded now because later sections turn on them:

1. $\mathsf{K} \subseteq \mathsf{M} \cup \mathsf{C}$ is *not available* here. That disjunctive inclusion is not asserted by the independence axioms of [3], which deny only the four pointwise implications between K and M, C; it is imposed by the R_3 suspension over that convention's own scope, and it lapses with it.

2. Consequently $(\mathsf{M} \cap \mathsf{K}) \cup (\mathsf{C} \cap \mathsf{K}) = (\mathsf{M} \cup \mathsf{C}) \cap \mathsf{K} = \mathsf{K} \setminus R_3 = R_5 \cup R_6 \cup R_7$, and this set *may* be a proper subset of K here — it is one exactly when R_3 is inhabited, which declining the convention permits without asserting. The identity itself is unconditional and does not require $R_3 \neq \emptyset$; what the suspended reading adds is the collapse of the set to K outright. Prop. 5.9 turns on this set: its content is that the key of a functional certificate must belong to it, which under the suspended reading is the requirement of being K-bearing and here is the requirement of being K-bearing and not purely so. Neither reading makes it automatic, and the countermodel of that proposition places no artifact in R_3, so the separation it establishes holds under both.

2.5 Denotation, agents, and the semantic horizon

The foundations paper defines a *denotation* map that assigns a referent to a symbol, an *agent* as a triple $a = (M_a, C_a, K_a)$ of agent-relativized predicates, the induced agent-relativized regions, and the *semantic horizon*, the boundary of the known domain, relativized per agent [3]. We reuse all of these: a concept is *known to a* when it lies within a's semantic horizon, and we say a point p is *denoted* when p lies in the image of the denotation map.

A distinction the wording must not blur. "Denoted" is a set-theoretic condition — membership in the image of a map — and requires no metric. *Proximity* to p is a different condition: it would require a metric and, beyond that, a threshold that nothing here supplies. Only the first is meant and only the first is used, so only the first is named, and no term of this paper should be read as asserting the second. Wherever a statement below turns on whether some token denotes p, it turns on image membership and on nothing geometric; a genuinely metric notion of nearness would be a separate predicate, is not defined in this paper, and is not assumed by any result in it.

2.6 The classified domain $\mathcal{A}^+$ and its embedding

The foundations paper defines the *classified domain*

$$\mathcal{A}^+ = \{\, x \in \mathcal{A} : M(x) \vee C(x) \vee K(x) \,\} = \mathsf{M} \cup \mathsf{C} \cup \mathsf{K} \subseteq \mathcal{A}$$

as the set of artifacts on which at least one predicate fires; it is exactly the support of the seven-region partition [3]. We adopt $\mathcal{A}^+$ with this meaning throughout — it is *not* redefined here — and read it geometrically: for the embedding-space reading we work with the point set that realizes $\mathcal{A}^+$ in the embedding, in which every classified artifact (and, under Conj. 3.8, every tokenizable string) has a position. No new predicate is attached: the predicates M, C, K on the embedded points are exactly those of [3], and the envelopes of Section 3 are the geometric shapes they trace out over $\mathcal{A}^+$.

3 Embedding-Space Envelope Theory

We now read the MCK predicates as geometric *envelopes* over the embedded domain $\mathcal{A}^+$ (Sec. 2.6). The three envelope definitions are the paper's first new primitives; everything else in this section is either a definitional observation or a staged emb-* claim carrying the tag already assigned to it upstream.

Before the envelopes are stated, the object they live over is fixed. What Sec. 2.6 supplies is a point set and no more; the definitions below say what a realization is, which realization properties this paper assumes, and which it deliberately does not. Separating them matters because the results of Sections 5 and 9 are proved abstractly and transported here, and a transport is only as strong as the compatibility it is granted.

3.1 The ideal realization

Definition 3.1 (Ideal MCK realization)**.** *An* ideal MCK realization *is a triple* $\mathfrak{R} = (\mathfrak{M}_a, E, e)$ *in which* $\mathfrak{M}_a$ *is the observer-relative MCK structure of [3] — the domain* $\mathcal{A}$ *with the agent-relativized predicates* M_a, C_a, K_a*, the induced regions,* Θ_a*, and the denotation data — E is a declared representation carrier, and* $e\colon \mathcal{A}^+ \to E$ *is a* realization map *assigning each classified artifact a position. Write* $\lambda_a(x) = (M_a(x), C_a(x), K_a(x))$ *for the label triple.*

The map is not *observer-indexed: one representation is shared and the observer-relativity is carried by* λ_a *alone. An index* e_a *would be required only if two observers assigned* different positions *to the same artifact, and nothing in the foundations this paper builds on [3] licenses that; the predicates are relativized there, the artifacts are not. The weaker choice is therefore made deliberately and is recorded here so that a later paper needing* e_a *must argue for it rather than inherit it.*

Notation. *Section 6 writes* emb *for this same map, the notation being the natural one there; e and* emb *denote one object and not two, and no second realization is in play.*

Definition 3.2 (Realization levels)**.** *Four properties a realization may have, in strictly increasing strength:*

Labelled. *The extensions are represented by their images,* $e(\mathsf{M}_a)$*,* $e(\mathsf{C}_a)$*,* $e(\mathsf{K}_a) \subseteq E$*. No further condition. This is the level at which the envelopes below are defined.*

A note on where the envelopes live. *Sec. 3.2 writes the envelopes* $E_{\mathsf{M}}, E_{\mathsf{C}}, E_{\mathsf{K}}$ *as subsets of* $\mathcal{A}^+$ *rather than of E, and that is deliberate: the predicates are properties of artifacts, so their extensions are sets of artifacts, and every result in this paper is proved about those sets. The images above are what those envelopes realize, and the two are related by e and not identified. Where a statement below is about the carrier it is written with e explicitly (Cor. 5.8, Cor. 9.11); everywhere else the domain reading is the one in force. Sec. 2.6 reads* $\mathcal{A}^+$ *geometrically by transporting it along e, and this is the map that transport uses.*

Label-faithful. *No point identifies artifacts carrying incompatible labels:*

$$e(x) = e(y) \implies \lambda_a(x) = \lambda_a(y).$$

If e is injective this holds vacuously; injectivity is not *assumed here, and label-faithfulness is the weaker hypothesis actually used.*

Dynamically compatible. *For an abstract partial transition* $\tau_t\colon \mathcal{A}_t^+ \rightharpoonup \mathcal{A}_{t+1}^+$ *and a realized partial transition* $\hat{\tau}_t\colon E_t \rightharpoonup E_{t+1}$*,*

$$e_{t+1} \circ \tau_t \;=\; \hat{\tau}_t \circ e_t \qquad \textit{wherever both sides are defined.}$$

Recoverable. *There is a map* $\Psi_a\colon E \times \Gamma \to \{0,1\}^3$ *with* $\Psi_a(e(x), \gamma) = \lambda_a(x)$ *for a declared context parameter* $\gamma \in \Gamma$*.*

Remark 3.1 (What this paper assumes, and what it does not). This paper works at the *labelled* level throughout, and invokes label-faithfulness and dynamic compatibility only where a transported statement names them (Cor. 5.8, Cor. 9.11). *Recoverability is not assumed anywhere.* It is stated in Def. 3.2 so that it can be named, not used: whether the MCK labels can be reconstructed from a position together with contextual data is an empirical question about learned representations, and it is not settled — or approached — here. Cor. 9.11 is a limitative statement about what recoverability would *still* not deliver, and is therefore available without assuming it. A fifth level, *empirical* realization — a learned representation approximately satisfying declared realization properties — is delegated in full to a companion treatment and is not defined here.

Remark 3.2 (Structure commitments). The carrier E is not called geometric or topological without saying which structure is meant. Three declarations suffice for this paper, and the weakest sufficient one is used in each place:

- E_{set}, a bare carrier set. Sufficient for the envelopes (Sec. 3.2), for Prop. 3.6, and for both transported corollaries, all of which are set-theoretic.
- E_{vec}, a vector space with a declared norm. Required by Sec. 6 alone, where difference vectors, a tolerance, and a separating radius are all used (Def. 6.1).
- E_{top}, a topology. *Not required by any result below*, and correspondingly not assumed. No claim in this paper is topological, and the absence is deliberate: separation, connectedness, and margin statements about the envelopes belong to the empirical programme, not to this one.

Section 4 needs E_{vec}, but for a reason worth separating from the one usually given. Its *decision* does not: whether some token denotes a given point is membership in the image of the denotation map (Sec. 2.5), which is set-theoretic and needs no metric. Its *hypothesis* does, since the point it ranges over is one reached by difference-vector closure, and that closure is defined only over E_{vec} (Def. 6.1). So the vector structure enters Sec. 4 through what it quantifies over and not through what it decides — and the two transported corollaries, which quantify over neither, need only E_{set}.

The word *geometric* is used below in the weak sense of "read as positions in E", and never as a claim that a metric or topological property has been established.

3.2 The three envelopes

Definition 3.3 (M-envelope)**.** *The* M-envelope $E_{\mathsf{M}} \subseteq \mathcal{A}^+$ *is the region of the embedding space swept out by the semantic-content predicate:* $E_{\mathsf{M}} = \{\, p \in \mathcal{A}^+ : M(p) = 1 \,\}$*, i.e. the geometric realization of the extension* M *(Sec. 2.1).*

Definition 3.4 (C-envelope)**.** *The* C-envelope $E_{\mathsf{C}} \subseteq \mathcal{A}^+$ *is* $E_{\mathsf{C}} = \{\, p \in \mathcal{A}^+ : C(p) = 1 \,\}$*, the geometric realization of the opacity extension* C*.*

Definition 3.5 (K-envelope)**.** *The* K-envelope $E_{\mathsf{K}} \subseteq \mathcal{A}^+$ *is* $E_{\mathsf{K}} = \{\, p \in \mathcal{A}^+ : K(p) = 1 \,\}$*, the geometric realization of the functionality extension* K*. Note that* $E_{\mathsf{K}} \subseteq E_{\mathsf{M}} \cup E_{\mathsf{C}}$ *is* not asserted *here: that inclusion is imposed by the* R_3 *suspension of [3], a scoping convention this paper declines (Sec. 2.4). The part of* E_{K} *lying outside* $E_{\mathsf{M}} \cup E_{\mathsf{C}}$ *is the geometric realization of* R_3*.*

Proposition 3.6 (Envelope intersections reproduce the regions)**.** This records the envelope construction rather than establishing an independent fact. *The Boolean cells of the three envelopes* $E_{\mathsf{M}}, E_{\mathsf{C}}, E_{\mathsf{K}}$ *over* $\mathcal{A}^+$ *are in bijection with the seven MCK regions* $R_1, \ldots, R_7$ *of [3]: for each region* R_i *the corresponding cell is the intersection of the envelopes (or their complements) prescribed*

by R_i's predicate triple. In particular $E_{\mathsf{K}} \setminus (E_{\mathsf{M}} \cup E_{\mathsf{C}})$ is the geometric realization of R_3, and $E_{\mathsf{M}} \cap E_{\mathsf{C}} \cap E_{\mathsf{K}}$ is that of R_7.

Justification. Immediate from the definitions: $E_{\mathsf{M}}, E_{\mathsf{C}}, E_{\mathsf{K}}$ are the point sets on which M, C, K equal 1, so their Boolean combinations are the point sets on which the predicate triple takes each admissible value. The bijection with $R_1, \ldots, R_7$ is exactly the region partition of [3] pulled back along the embedding. No new region primitive is introduced; this is a re-description of the existing partition in geometric terms. □

3.3 Staged embedding claims

The following statements record how meaning is organized over the envelopes. They are staged at the epistemic status already assigned to them upstream; no statement is promoted here.

Remark 3.3 (Difference-vector concept discovery)*.* New meanings are computed as combinations of stable difference vectors over the embedding geometry; a meaning reached this way is either common knowledge, known by some agent, or genuinely novel (landing in K). The use of relation-associated vector offsets is received understanding of the field, reported here as folklore rather than as a result of this paper. Classical distributional-embedding work motivates that use and evaluates its empirical utility [31, 30]. What is cited is the *form* of the regularity — relations read as offsets between vectors — and not any particular similarity measure, and not the condition fixed in Def. 6.1. That condition is an idealization adopted here, and it is *not* inherited as an empirical theorem from those works: they assume that pairs sharing a relation are related by one offset and proceed to evaluate what follows, rather than establishing the sharing itself. The assumption there and the definition here have the same logical status, so the citation motivates the definition and does not ground it; Rem. 6.2 states what the definition does and does not claim about real representations.

Conjecture 3.7 (Region-relative meaning-direction)**.** *In region M the meaning-direction of a concept is exposed (word2vec-like); in C it is present but not emergent; in K it is undefinable.*

Conjecture 3.8 (Tokenization envelope)**.** *The $\mathsf{M}/\mathsf{C}/\mathsf{K}$ envelopes range over every possible tokenization, so every string and substring carries an embedding, yielding a hyper-rich embedding space spanning all of $\mathsf{M} \cup \mathsf{C} \cup \mathsf{K}$.*

Remark 3.4 (K as unemptiable generative potential)*.* K is an unemptiable generative potential holding answers to not-yet-conceived questions, and its pure part is the region $R_3 = \mathsf{K} \setminus (\mathsf{M} \cup \mathsf{C})$; C is mineable, K is not. Behaviourally this is GF new-concept-reachable, a property that is at once liveness *and* coliveness, so no finite prefix determines it [2]; we conjecture it rather than assert it as a result. *Scope of the claim:* it is unwitnessed precisely on the portion of the infinite concept space that no finite witness ever reaches — a finite observation of "a new concept was reachable" never establishes the GF over an infinite concept space, so the reachability-forever reading is not promoted.

Remark 3.5 (Knowledge snapshot)*.* A word2vec-plus-sentence2vec coherent total order is an immutable snapshot of human knowledge at a moment, spatially ordered by meaning over the $\mathsf{M}/\mathsf{C}/\mathsf{K}$ regions. One part of this is received: that a trained embedding fixes a picture of what is known at the moment it is trained is common understanding of the field, and is reported as such. The rest is not, and is claimed here as this paper's own — that a *single* coherent order can be carried across the word and the sentence level at once, and that the order it induces is laid out over the $\mathsf{M}/\mathsf{C}/\mathsf{K}$ regions rather than over an unstructured space. The conjunction is testable and is not established

here: it fails if word-level and sentence-level coherence cannot be had under one order, and it fails if the resulting order cuts across the regions instead of respecting them.

4 Verifier Transition Model and Controlled Dimensional Augmentation

The envelope theory of Section 3 is static. This section makes it dynamic: a *verifier* confronts a concept reached geometrically and either leaves the space unchanged or performs a *controlled dimensional augmentation*. It is the first of the four mechanism families.

4.1 The three cases

Definition 4.1 (Verifier transition)**.** *Let $p \in \mathcal{A}^+$ be a concept reached by difference-vector closure (Rem. 3.3). A* verifier transition *is the action of a verifier V on p, resolved into three cases by two successive decisions — whether a token already denotes p and, when none does, whether V* accepts *p:*

Case A (no augmentation). *Some token denotes p (Sec. 2.5). Then V certifies p within the current embedding space; the envelopes $E_{\mathsf{M}}, E_{\mathsf{C}}, E_{\mathsf{K}}$ and the region partition (Prop. 3.6) are unchanged.*

Case B (controlled dimensional augmentation). *No token denotes p* and *V* accepts *p. Then the acceptance makes a dimensional augmentation admissible (Def. 4.2): a candidate operator for p is admitted and seated in R_7, and the space is recomputed so that p becomes denoted. How that operator comes to be held is not modelled.*

Case C (rejection). *No token denotes p* and *V* declines it*. Then no augmentation is performed — nothing is minted and the space is not recomputed — and the envelopes and the region partition are again unchanged. The concept is neither certified nor named: it stays in the token-less region (Sec. 8), where it is counted in the uncovered remainder, and it remains available to a later transition.*

Fig. 1 draws the branch.

Two features of the second decision are deliberately weak, and saying so is part of the definition rather than a caveat on it. First, acceptance is a decision of V: the definition records that the branch exists and where each of its outcomes leaves p, and supplies no criterion by which V decides. Second, Case C is stated as a *non*-transition. A declined concept is left uncertified and unnamed, which is not the same as being moved into C; the stronger reading — that declining assigns the concept to the opacity extension — would require an operation that supplies the opacity predicate to an artifact that lacked it, and no such operation is modelled here: the assimilation step of Sec. 5.4 only removes opacity, and Def. 4.2 states an admissibility condition rather than supplying anything to p. What is consequently left open is the status of a declined concept that was not already opaque.

Definition 4.2 (Controlled dimensional augmentation)**.** *Let $p \in \mathcal{A}^+$ be token-less — no token of the agent's current vocabulary denotes p, which is the guard of Case B — and let κ_p be a functional operator* candidate *for p: an operator, not among the agent's admitted vocabulary, that* would *denote p if admitted. The two quantifications must be kept apart. If κ_p already denoted p in the current vocabulary then a token would denote p, and Case B could not have been reached;*

what is token-less is p relative to the admitted *vocabulary, while κ_p is a candidate from outside it, and admission is precisely the event in question. A* controlled dimensional augmentation *at p is the condition under which seating κ_p in region $R_7 = \mathsf{M} \cap \mathsf{C} \cap \mathsf{K}$ is admissible, together with what admitting it requires of the space: the region transition into R_7 is* witness-based*, not stable — admissible only when the denotation-absence witness that triggered Case B is discharged, so that κ_p carries meaning, structure, and function together the moment it is seated — and the envelopes are then recomputed so that p lies in the image of the denotation map. We call an admitted seating a* mint*. The augmentation is* controlled *in that exactly one operator per triggering point is admitted; it is* dimensional *in that the recomputed space may require an enlarged representation to seat κ_p. The definition fixes* when *the transition is admissible and not* how *κ_p comes to be held: no operation furnishing κ_p — by an agent's own generation or from outside — is defined here, and none is available from the foundations this paper builds on, whose step relation holds the agent's operator basis fixed and models no acquisition [3]. Where κ_p comes from is deferred to a treatment of acquisition that neither paper provides, and nothing below is proved from its availability.*

Remark 4.1 (A language-model reading of the mint)*.* Def. 4.2 fixes when seating κ_p is admissible, not how it is realised. In a language model the natural realisation is vocabulary-axis extension, in which the embedding matrix acquires an initialised row: a change of the realisation together with $\mathcal{A}^+$, i.e. the case Rem. 4.2 identifies as excluded by the hypothesis of Prop. 4.4. Two distinct kinds of parametric transformation are in play in the surrounding literature:

Vocabulary-axis extension. A new lexical item is added to the tokenizer; a new embedding row is initialised; the model's vocabulary axis is extended by one (or more). This is the reading of the mint used in Sec. 10.6. OFA [27], HYPEROFA [37], Mundra's CW2V [35], and Tejaswi et al. [42] are engineering realisations in the same architectural family; VRCP [36] is an adjacent case that replaces vocabulary items at fixed total vocabulary size rather than extending the axis (Sec. 10.6).

Fixed-vocabulary weight edit. The tokenizer and the embedding matrix stay dimensionally fixed; targeted weight updates act on internal parameters (MLP keys/values, task-vector directions, hypernetwork editors). ROME [28], MEMIT [29], EMMET [16], MEND [33], Task Arithmetic [19], and KnowledgeEditor [8] are instances; the ripple-effect analysis of Cohen et al. [7] evaluates such edits.

Both are parametric transformations; the first changes the input-token axis, the second preserves it. The formal results of this paper concern the mint of Def. 4.2, not either engineering family.

4.2 Minting and recomputation

Conjecture 4.3 (Minting forces recomputation)**.** *A geometrically-reached novel concept carries no token; minting a token for it forces a recomputation of the embedding space (a witness-based transition into R_7, Def. 4.2). Behaviourally this is $\mathsf{G}(\mathit{mint} \rightarrow \mathit{recompute})$ over runs of the transition model of Def. 4.1, whose positions are its accepted triggering points and whose alphabet is $\{\mathit{mint}, \mathit{recompute}\}$; over that alphabet the property is at once safety* and *coliveness [2] — with the scope of this and the paper's one other temporal formula fixed in Sec. 5.3: a single mint that fails to force a recomputation refutes it on a finite prefix, but no finite prefix ever confirms it. We conjecture it rather than assert it as a result.* What the claim quantifies over: *mints, not token-less concepts. By Case C of Def. 4.1 a token-less concept need not be minted at all, so the antecedent ranges over accepted triggering points only, and carrying no token is not by itself a sufficient condition for a*

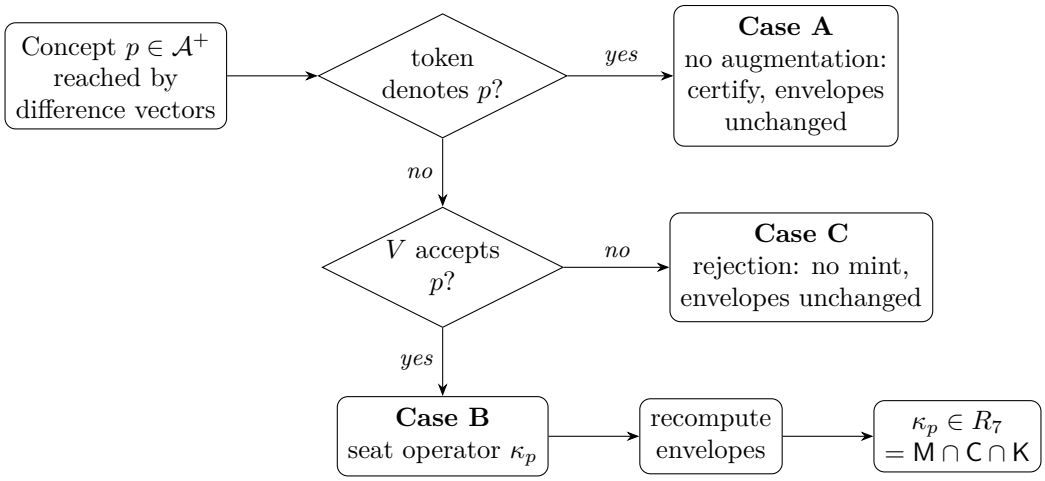


Figure 1: The verifier transition (Def. 4.1), a branch on two successive decisions. In Case A a token already denotes p and the envelopes are unchanged. If none does, the verifier decides. In Case B it accepts, which makes the augmentation admissible: an operator κ_p is seated in region R_7 by a witness-based transition and the envelopes are recomputed (controlled dimensional augmentation, Def. 4.2). In Case C it declines: nothing is minted, the envelopes are again unchanged, and p stays token-less. This figure is deliberately *not* an $M/C/K$ Venn diagram; the static region picture lives in [3].

recomputation. Scope of the claim: *the recomputation does* not *unlock all concepts — an* uncovered remainder *persists after each step, and with the rejection branch in place it has two sources: the still-token-less region of $\mathcal{A}^+$ that the recompute did not reach, and the points that were reached and declined, which no number of further steps removes. Consequently no finite sequence of mints is claimed to reach all of the concept space.*

Proposition 4.4 (Admission alone never extends structural reach)**.** *Let Φ be a regularity basis,* Den_a *the set of points at which some token of a denotes, and* $\mathrm{Reach}_\Phi(a) = \mathrm{cl}_\Phi(\mathrm{Den}_a)$ *(Def. 6.2). Let p be a point reached by Φ-closure and admitted, so that* $\mathrm{Den}'_a = \mathrm{Den}_a \cup \{p\}$ *while the basis, the realization* emb*, and the classified domain $\mathcal{A}^+$ are all unchanged. Then*

$$\mathrm{Reach}_\Phi(a') \;=\; \mathrm{Reach}_\Phi(a).$$

Consequently, if $\mathrm{Reach}_{\Phi'}(a') \supsetneq \mathrm{Reach}_\Phi(a)$ *with* emb *and $\mathcal{A}^+$ held fixed, then $\Phi' \neq \Phi$:* under a fixed realization, any extension of structural reach requires a change of basis, and no admission of an already reachable point can supply one.

Remark 4.2 (What the fixed-realization hypothesis excludes)*.* The hypothesis is not idle, and naming what it excludes sharpens the result rather than weakening it. $\mathrm{Reach}_\Phi(a) = \mathrm{cl}_\Phi(\mathrm{Den}_a)$ is computed by steps that displace *embedded* points and keep only those landing in $\mathcal{A}^+$ (Def. 6.1), so it depends on emb and on $\mathcal{A}^+$ as well as on (Φ, Den_a) — a dependency Sec. 6 states outright and calls ineliminable. And Def. 4.2 contemplates exactly the excluded case: seating κ_p *may require an enlarged representation*, and κ_p is itself classified, so both emb and $\mathcal{A}^+$ may differ across the step. The proposition therefore does not say that admission never extends reach. It says that admission does not extend reach *by the act of naming*: any extension must come from the recomputation, which is a change of the realization and not of the vocabulary.

Proof. By hypothesis p was reached by Φ-closure from Den_a, so $p \in \mathrm{cl}_\Phi(\mathrm{Den}_a)$. Since cl_Φ is a closure operator it is idempotent and monotone, whence $\mathrm{cl}_\Phi(\mathrm{Den}_a \cup \{p\}) = \mathrm{cl}_\Phi(\mathrm{Den}_a)$. The second claim follows from the first by contraposition, the hypothesis having fixed every argument of Reach other than the basis. □

Conjecture 4.5 (Minting-unlocks, conditional form)**.** *Whether minting can extend reach at all is the converse question, and it is open. Writing Φ' for the basis after the step, the claim entertained is the* conditional

$$\mathsf{Minted}(s) \wedge \mathsf{Acquired}_a(s) \wedge \mathsf{Integrated}_a(s) \wedge \mathsf{ReachExtension}(s) \implies \mathrm{Reach}_{\Phi'}(a) \supsetneq \mathrm{Reach}_{\Phi}(a),$$

and every conjunct is load-bearing. Prop. 4.4 settles the negative half and locates what the antecedent must supply: a change of basis, which is what ReachExtension *asserts. Two of the remaining conjuncts are not modelled anywhere in this paper —* $\mathsf{Acquired}_a$ *and* $\mathsf{Integrated}_a$ *concern how the operator comes to be held and used, and Def. 4.2 declines that question explicitly.* Scope of the claim: *what remains unwitnessed is the set of concepts alleged to become reachable after a mint but never exhibited. Folklore is received understanding of a field, and no identifiable claim of this form is attributable to one; this is the paper's own conditional claim, and is tagged accordingly.*

5 The Functional-Understanding Hierarchy

This section stratifies *functional understanding* and connects it to the region dynamics of Section 4. Understanding is modelled as semantic reclassification: an artifact is understood once it can be represented as the output of a known function on known arguments.

Remark 5.1 (Two levels of understanding)*.* We conjecture that understanding blends two levels that must never be conflated: a *structural* level — stable embedding relations, the closure of the latent manifold under a basis of relational maps in M — and an *operational* level — validated executions and constructive witnesses. The three levels U_0, U_1, U_2 below are the stratification this split already presupposes: $U_0(f)$ is structural comprehension (the structural level), while $U_1(f)$ and $U_2(f)$ — self-oracular and external-oracular execution — are operational discovery and completeness (the operational level). Reading a structural regularity as if it were an operational witness, or conversely, conflates the two levels and inflates the epistemic status of the artifact; the region promotions of Section 4 are exactly the disciplined crossing from the structural to the operational level.

5.1 The three levels U_0, U_1, U_2

Definition 5.1 (Functional-understanding hierarchy)**.** *For a function f, define three levels of functional understanding:*

$U_0(f)$ — **structural understanding.** *The embedding space exhibits stable relational regularity for f (geometric / representational comprehension, in the sense of* M*).*

$U_1(f)$ — **self-oracular execution.** *The system can internally simulate or execute f.*

$U_2(f)$ — **external-oracular delegation.** *The system can invoke a trusted external implementation or oracle for f.*

The indices order the levels by kind*, not by strength, and no implication among them is asserted here. In particular $U_2(f)$ is not asserted to imply $U_1(f)$, and should not be: delegating f to a trusted external implementation is what a system does when it cannot execute f itself, so the two*

operational levels are to be read as incomparable rather than as a chain. Nor is either operational level claimed to imply the structural one: by Rem. 5.1 the structural and operational levels are not to be read off one another, and Prop. 6.6 exhibits a configuration separating the formal reading of U_0 fixed in Sec. 6 from the operational side. Only U_0 is used in the results below, and only under the reading of Def. 6.2.

5.2 The discovery predicate and functional discovery

Definition 5.2 (Discovery predicate Hit_f)**.** *Recall $M_0 = \mathsf{M}\setminus(\mathsf{C}\cup\mathsf{K})$ and $C_0 = \mathsf{C}\setminus(\mathsf{M}\cup\mathsf{K})$ (Sec. 2.2). For a functional operator $f \in \mathsf{M}\cap\mathsf{K}$ and a target c, define the* discovery predicate

$$\mathrm{Hit}_f(c) \;:=\; \exists\, m \in M_0,\ \exists\, k \in (\mathsf{M}\cap\mathsf{K})\cup(\mathsf{C}\cap\mathsf{K}) \text{ such that } f(m,k) = c.$$

When $c \in C_0$ and $\mathrm{Hit}_f(c)$ holds, c is reclassified $C_0 \to M_0$*: an element becomes understood once it is representable as the output of a known function applied to known arguments. All three region conditions are part of the predicate — one on the message, one on the key, and one on the operator itself — and it is the last that makes the function a* known *one rather than merely a function.*

Remark 5.2 (Why the operator carries a region condition)*.* The condition $f \in \mathsf{M}\cap\mathsf{K}$ is load-bearing, and dropping it would make the reading above unsafe. Without it, Hit_f could be witnessed by an operator drawn from $\mathsf{C}\cap\mathsf{K}$: functional, and opaque to the agent that applies it. Such an operator may be perfectly deterministic and may take its argument and return its value inside M_0, so the exhibited representation $f(m,k) = c$ would be in order in every respect but the one that matters — the agent would hold no account of the operator by which the representation was obtained, and the reclassification would then read as understanding on the strength of something the agent does not understand. Requiring the operator to be M-bearing as well as K-bearing excludes exactly that configuration. The condition is necessary and is not claimed sufficient: it constrains where the operator sits, not what the agent can do with it, and Sec. 5.6 takes up separately what would keep the exhibited certificate from being trivial.

One consequence deserves saying outright, because the canonical instance below invites the opposite reading. *Being able to execute an operator is not what M-bearing asks for.* Bitwise XOR is algebraically transparent, computable in either direction, and defined by a table one can write down — and none of that is the condition. What M asks is that the agent hold an account of the operator: what its arguments stand for in the domain at hand, when it is the right operator to apply, and what its application licenses one to conclude. An agent may compute $m \oplus k$ faultlessly and hold no such account, and the placement $\oplus \in \mathsf{M}\cap\mathsf{K}$ is accordingly a *hypothesis of the construction below*, supplied by assumption and not established by the operator's simplicity. Three things that are easily run together are therefore kept apart throughout: executability, algebraic transparency, and the semantic integration that M names.

Conjecture 5.3 (Functional discovery)**.** *Along a discovery trace on which the system searches for representations, an element $c \in C_0$ eventually satisfies $\mathrm{Hit}_f(c)$ and is reclassified $C_0 \to M_0$. On the discovery trace this is the cosafety claim $\mathsf{F}\,\mathrm{Hit}_f$ over runs whose positions are successive exhibitions of a candidate pair (m,k) for the fixed c and f: a good finite prefix — a single exhibited representation $f(m,k) = c$ — suffices to establish it.* This fixes the scope of every temporal formula in the paper. *There are five, over two run families: this one and $\mathsf{G}\,(\mathit{mint} \to \mathit{recompute})$ of Sec. 4.2 over the alphabets just named, and the three $\mathsf{G}\,\mathsf{F}$/$\mathsf{F}\,\mathsf{G}$ statements of Secs. 3, 8 and 11 over iterations of the minting-and-recompute step. None carries a proof; the four-family vocabulary is used only to name the finite-prefix behaviour just described, and [2] is where it is defined. The alphabets are local*

to this paper and the runs are its own; what is not *settled here is the class of admissible extensions each formula quantifies over, and with it any monitor-side statement — soundness, completeness, or a verdict structure — which would need a monitor this paper does not construct. The formulas are accordingly locally typed and conjectural, not semantically closed, and the finite witnesses named above are the whole of what is claimed for them.*

5.3 Universal assimilation

Discovery (Def. 5.2) asks whether a *given* target is the output of a known function. Universal assimilation asks whether *every* target can be so represented, and answers constructively when the framework provides an involutive functional operator.

Definition 5.4 (Universal assimilation via an involutive operator)**.** *Suppose the framework provides a functional operator $\oplus \in \mathsf{M} \cap \mathsf{K}$ that is involutive on known messages: $m \oplus (m \oplus X) = X$ for all $m \in M_0$ and all X (the canonical instance is bitwise XOR). The quantifier on the first argument is restricted to M_0 because step 1 below chooses m there and the construction never applies $\oplus$ at a first argument outside M_0; requiring involutivity at every first argument would be a strictly stronger hypothesis than anything that follows uses. A* universal-assimilation verifier *V certifies an arbitrary target X as follows:*

1. *choose a known message $m \in M_0$;*
2. *form the key $k := m \oplus X$;*
3. *take $f := \oplus$, so that $f(m, k) = m \oplus k = m \oplus (m \oplus X) = X$.*

This yields a certificate $\pi_V(X) = (m, k, f)$ with $f(m, k) = X$. Hence

$$\forall X \;\; \exists\, m, k, f \text{ such that } f(m, k) = X,$$

i.e. every element can be represented as the output of a known function on known arguments. The construction is not special to XOR: any operator with the same involutive (group-inverse) property serves in its place (the substitution principle*).*

Two readings of that principle must be kept apart, and only one is adopted. The algebraic reading just stated is what the construction consumes — steps 2 and 3 use involutivity and nothing else — and it is the reading used below. An empirical *reading is also available and is not adopted: that the admissible substitutes are the operations the embedding itself performs consistently, such as vector difference or a declared similarity measure. The two do not coincide in either direction. A similarity measure is not involutive and does not have the arity the construction needs, so the empirical reading does not license the construction; and involutivity constrains the operator alone, so the algebraic reading says nothing about whether any given embedding realizes the operator. That second question is not dropped, only relocated: it is what Sec. 6 takes up, fixing offset regularity (Def. 6.1) as the formal reading of* the space performs this operation consistently*. The substitution principle as stated here is accordingly algebraic, and is not to be read as a claim about what any particular embedding does. The construction is definitional — it asserts the existence of a representing certificate, not that the certificate confers any epistemic status beyond U_0; promotion up the region ladder still requires the discovery of Def. 5.2.*

5.4 Assimilation fixes the symbol domain

The construction of Def. 5.4 is deliberately sweeping: read at face value it says that *every* target can be exhibited as the output of a known function on known arguments. It is natural to read a limit into that sweep — if the agent can so exhibit everything, then nothing stays opaque, so eventually nothing is left to ask about. This subsection shows that the reading is false, and false for a structural reason rather than a quantitative one: the reclassification that assimilation performs is defined on a set that is disjoint from the symbols, so it cannot reach a single one of them, however far it is iterated.

We first record the reclassification clause of Def. 5.2 as a map, so that it can be quantified over. Nothing is added: the definition below is that clause read functionally, and it is agent-relativized because the results of [3] that Cor. 5.7 invokes are stated per agent.

Definition 5.5 (Assimilation step)**.** *Fix an agent $a = (\mathsf{M}_a, \mathsf{C}_a, \mathsf{K}_a)$ and read Defs. 5.2–5.4 relative to a, with the agent-relativized predicates and regions of [3], the relativized domain $\Theta_a = \mathsf{M}_a \cap \mathsf{C}_a = R_4^a \cup R_7^a$ recalled from there (Sec. 2.3), and our notation $M_0^a = R_1^a$, $C_0^a = R_2^a$ (Sec. 2.2). An* assimilation step *is determined by a set*

$$S \subseteq \{\, c \in C_0^a \,:\, \mathrm{Hit}_f(c) \,\}$$

of targets it reclassifies, and produces the predicates

$$\mathsf{M}'_a = \mathsf{M}_a \cup S, \qquad \mathsf{C}'_a = \mathsf{C}_a \setminus S, \qquad \mathsf{K}'_a = \mathsf{K}_a,$$

i.e. each $x \in S$ acquires the triple $(1,0,0)$ of M_0^a and every other artifact keeps its triple. Write $\Theta'_a = \mathsf{M}'_a \cap \mathsf{C}'_a$ and $R_i^{a\prime}$ for the post-step regions. The step is left indexed by S rather than fixed to the set of all hits, because Def. 5.2 reclassifies target by target and says nothing about simultaneity; taking S to be the whole hit set recovers the maximal reading.

Theorem 5.6 (Θ-invariance of assimilation)**.** *Let a be an agent and let S determine an assimilation step (Def. 5.5). Then*

$$\Theta'_a = \Theta_a, \qquad R_4^{a\prime} = R_4^a, \qquad R_7^{a\prime} = R_7^a.$$

Consequently Θ is constant along every finite or infinite sequence of assimilation steps, and in particular $\Theta'_a = \emptyset$ if and only if $\Theta_a = \emptyset$. The conclusion is unchanged in the extremal case $S = C_0^a$.

Proof. The reclassification has domain C_0^a, and

$$\Theta_a \cap C_0^a = (\mathsf{M}_a \cap \mathsf{C}_a) \cap \big(\mathsf{C}_a \setminus (\mathsf{M}_a \cup \mathsf{K}_a)\big) = \emptyset,$$

since membership in Θ_a requires $M_a(x) = 1$ while membership in C_0^a requires $M_a(x) = 0$. The two statements now follow by cases on $x \in \mathcal{A}$.

If $x \notin S$ then, by Def. 5.5, the triple of x is unchanged, so $x \in \Theta'_a$ iff $x \in \Theta_a$, and x lies in the same one of R_4, R_7 — or in neither — before and after.

If $x \in S$ then $x \in C_0^a$, so $M_a(x) = 0$ and hence $x \notin \mathsf{M}_a \cap \mathsf{C}_a = \Theta_a$; after the step x has triple $(1,0,0)$, so $C'_a(x) = 0$ and $x \notin \Theta'_a$. Thus x belongs to neither Θ_a nor Θ'_a, and, both triples having a zero in a coordinate that R_4 and R_7 require to be one, to none of $R_4^a, R_7^a, R_4^{a\prime}, R_7^{a\prime}$.

Combining the cases gives the three displayed equalities. For a sequence of steps, each step is again of the form of Def. 5.5 with respect to the predicates it starts from, so the equalities propagate by induction; equality of sets gives equality of emptiness. The extremal case is covered because no hypothesis on S beyond $S \subseteq C_0^a$ was used. □

Remark 5.3 (What the proof does and does not use). The theorem is unconditional, and deliberately shallow. It does not depend on Conj. 5.3: that conjecture is what would be needed to reach the extremal case $S = C_0^a$, and the theorem holds whether or not that case is ever reached. It uses no property of f, of $\oplus$, or of the certificate — only the domain of the reclassification. Its force is therefore entirely structural: Θ is fixed *by construction*, not because assimilation is hard or slow. Correspondingly, the theorem is silent about any variant of Def. 5.2 whose reclassification acted on $\mathsf{C}_a \cap \mathsf{K}_a$, or on Θ_a itself, rather than on C_0^a; such a variant is a different operation and would need its own analysis.

Corollary 5.7 (Assimilation preserves symbol-possession)**.** *Let a be an agent with $\Theta_a \neq \emptyset$. After any sequence of assimilation steps:*

(i) *$\Theta'_a \neq \emptyset$, so a is not symbol-free and its denotation map retains a non-empty domain. By the symbol-free classification of [3], which characterizes symbol-freedom as exactly $R_4^a = R_7^a = \emptyset$, the agent retains index 4 or index 7 in its type throughout;*

(ii) *the hypothesis $\Theta_a = \emptyset$ of the query-horizon theorem of [3], under which an agent poses no queries at all, is never established by assimilation.*

In particular, no amount of assimilation converts a symbol-bearing agent into a symbol-free one.

Proof. (i) is Thm. 5.6 together with the cited characterization: $\Theta'_a = \Theta_a \neq \emptyset$, and $\Theta_a = R_4^a \cup R_7^a$ is non-empty iff at least one of the two cells is. (ii) is immediate from (i): the antecedent of the cited theorem's third clause is $\Theta_a = \emptyset$, which (i) denies at every stage. □

Corollary 5.8 (Realization-level transport of Θ-invariance)**.** *Let $\mathfrak{R} = (\mathfrak{M}_a, E, e)$ be an ideal MCK realization (Def. 3.1), let τ be an assimilation step (Def. 5.5) carrying stage t to stage $t+1$, and write $\Theta_{a,t}$ for the symbol domain at stage t. Then:*

(i) (Image transport; no realization hypothesis.) $e_{t+1}(\Theta_{a,t+1}) = e_{t+1}(\Theta_{a,t})$.

(ii) (Commutative transport; requires dynamic compatibility, Def. 3.2.) *If $\hat{\tau}$ realizes τ, then $e_{t+1}(\Theta_{a,t}) = \hat{\tau}(e_t(\Theta_{a,t}))$: the realized symbol region at $t+1$ is the image of the realized symbol region at t under the realized step.*

Proof. (i) By Thm. 5.6, $\Theta_{a,t+1} = \Theta_{a,t}$ as sets; applying the same map to equal sets gives equal images. (ii) Assimilation reclassifies only C_0^a, and $\Theta_a \cap C_0^a = \emptyset$ by the same theorem, so τ restricted to $\Theta_{a,t}$ is the identity and $\tau(\Theta_{a,t}) = \Theta_{a,t}$. Dynamic compatibility gives $e_{t+1} \circ \tau = \hat{\tau} \circ e_t$ where both are defined; evaluating on $\Theta_{a,t}$ and substituting $\tau(\Theta_{a,t}) = \Theta_{a,t}$ yields the claim. □

Remark 5.4 (What the transport does *not* say). Three statements must be kept apart, and only the first two are established. *Membership invariance* — an artifact is a symbol at $t+1$ exactly when it was one at t — is Thm. 5.6 and is unconditional. *Image transport* is clause (i), and it is weaker than it looks: it says that two *equal* sets have equal images, so it carries no geometric content by itself and must not be quoted as though it did. What clause (ii) adds is that the realized step carries the realized symbol region onto itself, and that is a statement about $\hat{\tau}$, purchased with the compatibility hypothesis.

Coordinate invariance — $e_{t+1}(x) = e_t(x)$ for $x \in \Theta_a$ — is **not** claimed, and does not follow. Nothing above prevents the realization from repositioning every point between stages; indeed Def. 4.2 contemplates a recomputed and possibly enlarged representation, in which coordinates need not be preserved at all. Pointwise geometric stability is a separate assumption, is not made

here, and would have to be argued rather than inherited. The distinction is exactly the one the foundations paper's staticity clause invites and does not settle: what is fixed is a membership, not a position.

Remark 5.5 (Two claims that are *not* being made). Clause (ii) is weaker than it may look, and the gap is worth stating. The query-horizon result of [3] is a one-way implication: symbol-freedom extinguishes queries. It supplies no converse, so $\Theta_a \neq \emptyset$ does not entail that a poses any query, and Cor. 5.7 must not be read as showing that assimilation leaves some question standing. What it shows is narrower and exact: assimilation cannot extinguish queries *by way of symbol-freedom*, because it cannot produce symbol-freedom. Whether assimilation might reduce or exhaust an agent's queries by another route is untouched here — and the question is live rather than idle, since a query in [3] requires a step that the agent cannot realize, and assimilation plausibly makes steps realizable. That is a separate mechanism and is not analysed in this paper.

Remark 5.6 (Where the opacity goes). The geometry behind Thm. 5.6 is worth stating plainly. Assimilation acts on $C_0^a = R_2^a$, pure opacity, and moves what it touches into R_1^a, pure understanding. It does not act on $R_4^a \cup R_7^a$, the symbols. So the opacity that assimilation removes is exactly the opacity that carries no meaning, and the opacity it cannot remove is exactly the opacity that is co-present with meaning — which is what a symbol *is*. The intuition this refutes fails at that junction: exhibiting an artifact as a known function of known arguments is an operation on unmeant opacity, and a symbol is not unmeant.

A mirror-image observation is already recorded upstream. The symbol-free classification of [3] notes that an agent with no symbols still retains its R_2 and R_6 cells, so that opacity does not vanish when the symbols do. Thm. 5.6 is the pairing in the other direction: the symbols do not vanish when the unmeant opacity does. Neither removal drags the other with it, and an agent that assimilated all of R_2^a would not be an agent without opacity — it would be an agent whose entire remaining opacity sits inside its symbols.

5.5 The key-membership gap

Def. 5.4 closes by deferring promotion to the discovery predicate of Def. 5.2. That deferral is correct, and this subsection states as a result what it currently states as a caution: the displayed conclusion of the universal-assimilation construction is strictly weaker than Hit_f, and the difference is exactly one membership condition. This sharpens the paper's own guard rather than repairing an error in it.

Proposition 5.9 (The gap is the key-membership condition)**.** *Write* (†) *for the display of Def. 5.4,* $\forall X\, \exists m, k, f.\ f(m,k) = X$*, and compare it with* Hit_f *of Def. 5.2. Then:*

(i) *the two differ by three requirements: that the first argument satisfy* $m \in M_0$*; that the second satisfy* $k \in (\mathsf{M} \cap \mathsf{K}) \cup (\mathsf{C} \cap \mathsf{K}) = \mathsf{K} \setminus R_3$*, i.e. be* K*-bearing and not purely so; and that the operator be given* and *lie in* $\mathsf{M} \cap \mathsf{K}$*,* Hit_f *being stated for a fixed* $f \in \mathsf{M} \cap \mathsf{K}$ *where* (†) *leaves* f *existentially quantified and unconstrained;*

(ii) *the construction of Def. 5.4 discharges two of the three —* $m \in M_0$ *by its step 1, and the operator by taking* $f = \oplus$*, which is moreover in* $\mathsf{M} \cap \mathsf{K}$ *by hypothesis — leaving exactly the key-membership condition; for that one it supplies no ground whatever for* $K(k) = 1$*: the key* $k := m \oplus X$ *is produced as a* datum*, the value of* $\oplus$ *at two arguments, and nothing in the construction makes it an operator;*

(iii) *the gap is not merely unproved but unclosable in general: there is an MCK predicate assignment on which* (†) *holds for every target while* $\mathrm{Hit}_\oplus$ *holds for no element of* C_0*.*

Hence (†) *does not establish* Hit_f*, and the universal-assimilation certificate does not by itself trigger the reclassification* $C_0 \to M_0$*.*

Proof. (i) and (ii) are read off the two definitions: $\mathrm{Hit}_f(c)$, at a fixed $f \in \mathsf{M} \cap \mathsf{K}$, is the conjunction of $m \in M_0$, $k \in (\mathsf{M} \cap \mathsf{K}) \cup (\mathsf{C} \cap \mathsf{K})$ and $f(m,k) = c$, while (†) asserts only the existence of some m, k, f with $f(m,k) = X$ — which is the three differences of (i), and no others, since the equation $f(m,k) = X$ is common to both. Of the three, step 1 of Def. 5.4 supplies $m \in M_0$ and its step 3 supplies the operator, which the hypothesis of that definition places in $\mathsf{M} \cap \mathsf{K}$; the key-membership condition is what remains. The identity $(\mathsf{M} \cap \mathsf{K}) \cup (\mathsf{C} \cap \mathsf{K}) = (\mathsf{M} \cup \mathsf{C}) \cap \mathsf{K} = \mathsf{K} \setminus R_3$ is distributivity together with the definition of R_3.

For (iii) we exhibit the assignment. Fix $n \geq 1$, let $D = \{0,1\}^n$, let $D = D_M \sqcup D_C$ be any partition into two non-empty parts, and let $\mathcal{A} = D \sqcup \{e\}$ with $e \notin D$ a single further artifact. Assign triples

$$\begin{aligned} (M,C,K)(d) &= (1,0,0) \quad (d \in D_M), \\ (M,C,K)(d) &= (0,1,0) \quad (d \in D_C), \\ (M,C,K)(e) &= (1,0,1). \end{aligned}$$

All eight triples are admissible a priori in [3], so this is a legitimate assignment; no artifact is placed in R_3, so it is legitimate whether or not the R_3 suspension is in force (Sec. 2.4). By construction $M_0 = D_M$, $C_0 = D_C$, and $e \in R_5 = (\mathsf{M} \cap \mathsf{K}) \setminus \mathsf{C}$, so

$$(\mathsf{M} \cap \mathsf{K}) \cup (\mathsf{C} \cap \mathsf{K}) = \{e\}.$$

Let $\oplus$ be bitwise XOR on D, extended to $\mathcal{A}$ by $x \oplus e = e \oplus x = e$, and let e be the artifact that is this operator; then $\oplus \in \mathsf{M} \cap \mathsf{K}$ as Def. 5.4 requires — so that $\mathrm{Hit}_\oplus$ is also well formed under the operator condition of Def. 5.2 — and $\oplus$ is involutive on known messages exactly as that definition asks: $m \oplus (m \oplus X) = X$ for every $m \in M_0 = D_M$ and every $X \in \mathcal{A}$ — for $X \in D$ this is involutivity of XOR, and for $X = e$ both sides are e. The restriction of the first argument to M_0 is used and not merely convenient: at $m = e$ one has $e \oplus (e \oplus X) = e \neq X$ for every $X \in D$, so this operator is *not* involutive at an unrestricted first argument. The countermodel is therefore an exact instance of the hypothesis, not an approximate one.

(†) holds on this assignment: given any target $X \in D$, choose any $m \in D_M$ and put $k := m \oplus X \in D$; then $\oplus(m,k) = X$. (For $X = e$, take $k := e$.)

$\mathrm{Hit}_\oplus(X)$ fails for every $X \in C_0 = D_C$: a witness would require $k \in (\mathsf{M} \cap \mathsf{K}) \cup (\mathsf{C} \cap \mathsf{K}) = \{e\}$ and $m \in M_0 = D_M$ with $\oplus(m,e) = X$; but $\oplus(m,e) = e$ by construction, and $e \notin D_C$. So no element of C_0 is a hit, while every one of them is certified by (†). □

Remark 5.7 (Reading of the separation). The countermodel is not pathological, and its shape is the point. It places the only K-bearing artifact outside the argument domain of the very operator it is, so that the certificates (†) produces — which are all data — are never of the type Hit_f asks for. That is the type mismatch stated abstractly in clause (ii), made concrete. The moral is not that (†) is wrong but that it is a statement about representability, whereas Hit_f is a statement about representability *by an operator the agent holds*; only the second is a promotion condition.

5.6 An admissibility criterion for functional certificates

The foundations paper records, as an explicit limitation, that it does not supply the constraint that keeps its semantic-content predicate from being vacuous: read without restriction, a certificate exhibiting an artifact as the value of a known function on known inputs can be produced for any

artifact whatever, so the predicate would hold everywhere and separate nothing [3]. Def. 5.4 is that worry in explicit form — it is precisely a recipe for producing such a certificate for an arbitrary target. It is therefore this companion's business to say what would rule the recipe out.

We propose one such constraint. It is a proposal, not a discovery, and it is tagged accordingly; it is also *a* candidate and not the unique one, and we claim neither that the five conditions are independent nor that they are jointly sufficient.

Definition 5.10 (Admissible functional certificate)**.** *Fix a description-length function ℓ on artifacts (bit-length under some fixed encoding), a class $\mathcal{T} \subseteq \mathcal{A}^+$ of intended targets, and a class of admissible perturbations of a target. Let $\pi = (m, k, f)$ with $f(m, k) = X$ be a* functional certificate *for $X \in \mathcal{T}$, and write*

$$\mathrm{Reach}(k, f) \;=\; \{\, f(m', k) \;:\; m' \in M_0 \,\}$$

for the set of targets the certificate's operator-and-key part attains as the message varies. A certificate scheme *for $\mathcal{T}$ is a map σ assigning to each $X \in \mathcal{T}$ a functional certificate $\sigma(X) = (m_X, k_X, f_X)$. The non-adaptivity the scheme must exhibit is carried by (A5) below, on the repertoire of operator-and-key pairs, and not by the choice of message: the message is a witness and may vary with the target (Rem. 5.8). Call π* admissible *when it satisfies (A1)–(A4), and call a scheme σ admissible when every $\sigma(X)$ is admissible and σ satisfies (A5):*

(A1) Compressibility. *$\ell(\pi) < \ell(X)$, where $\ell(\pi) = \ell(m) + \ell(k) + \ell(f)$. The certificate must be shorter than the thing it explains.*

(A2) Transportability. *$|\mathrm{Reach}(k, f) \cap \mathcal{T}| \geq 2$. The certificate must apply to more than the single instance it was built from.*

(A3) Predictiveness. *$\mathrm{Reach}(k, f) \subseteq \mathcal{T}$. The certificate must license* correct *expectations about instances not used in its construction — not merely reach some of them, but stay inside the class while doing so.*

(A4) Counterfactual stability. *Every admissible perturbation X' of X lies in $\mathrm{Reach}(k, f)$: the certificate survives perturbation of the instance without a fresh key.*

(A5) Non-extensionality. *The scheme's repertoire of operator-and-key pairs, $\mathcal{K}_\sigma = \{\, (k_X, f_X) : X \in \mathcal{T} \,\}$, is strictly smaller than the class it serves: $|\mathcal{K}_\sigma| < |\mathcal{T}|$. Agreeing on values is not enough: a scheme that needs a fresh pair for every target merely re-lists the targets under another name, and $|\mathcal{K}_\sigma| = |\mathcal{T}|$ is exactly that.*

Remark 5.8 (What is certified, and by what). The conditions above are easier to read once the proposition they bear on is named, and naming it fixes which component plays which role.

Write $f_k := f(\cdot, k)$ for the reconstruction map at a fixed key. Then $\mathrm{Reach}(k, f)$ as defined above is exactly its image,

$$\mathrm{Reach}(k, f) \;=\; \{\, f(m', k) : m' \in M_0 \,\} \;=\; \mathrm{im}(f_k),$$

and the proposition a certificate bears on is *image membership*:

$$\varphi_{k,f}(X) \;\equiv\; X \in \mathrm{Reach}(k, f).$$

The *reusable structure* is the pair (k, f); the message m_X is a *witness* for $\varphi_{k,f}(X)$, satisfying $f(m_X, k) = X$. A witness may depend on the proposition it witnesses — that is what witnesses do — so a target-dependent m_X is not by itself a defect, and (A5) is the condition that carries non-adaptivity, at the level where it belongs: the repertoire of *pairs*, not the choice of message.

Read this way, (A2)–(A4) are conditions on the image of f_k and (A5) is a condition on how many maps the scheme needs, while (A1) is the only condition that is not about the image at all. Two things the apparatus does *not* supply should be said in the same breath: there is no acceptance relation here and hence no soundness claim, and no completeness claim; $f(m, k) = X$ is an equation, not a verdict.

Remark 5.9 (Why (A5) is a condition on schemes). An earlier and more obvious formulation of non-extensionality — that $X \mapsto k$ must not be an invertible re-encoding of the target — is too strong to be usable, since it would reject any compressive certificate as well: a good compression scheme *is* an invertible re-encoding, and is valuable precisely because the re-encoding is shorter. What separates compression from re-listing is not invertibility but whether the machinery grows with the class, which is what (A5) measures. The length of the re-encoding is already governed by (A1), and the two conditions should not be made to do each other's work.

The form of the inequality also deserves a word, because a weaker-looking formulation would have no content. Since $\mathcal{K}_\sigma = \{(k_X, f_X) : X \in \mathcal{T}\}$ is indexed by $\mathcal{T}$, the bound $|\mathcal{K}_\sigma| \leq |\mathcal{T}|$ holds for every scheme whatever, so demanding only that the repertoire be "bounded in terms of $|\mathcal{T}|$" asks for nothing: Def. 5.10 fixes a *single* target class, and at a single class the only available demand is the strict one, that at least one pair be reused. The intended asymptotic reading — that the machinery does not grow with the class — is a statement about a parametrised family $\{\mathcal{T}_n\}$ of classes, namely $|\mathcal{K}_{\sigma_n}| = O(1)$, or at least $o(|\mathcal{T}_n|)$, as n grows. That family form is the stronger condition and is available whenever a family is in view; we state (A5) at one class because the definition is stated at one class. The certificate of Prop. 5.11 fails both forms, so nothing there turns on the choice.

Remark 5.10 (Relation to the upstream conditions). The foundations paper does not leave the field empty: its certificate of understanding is already required to demonstrate a stable relational position among other artifacts, enumerable and verifiable properties, and predictable behaviour under known operations [3]. What it declines to supply is a formulation precise enough to reject a candidate. The five conditions above are offered as one such formulation, and they line up with the three upstream desiderata rather than replacing them: (A2) and (A4) sharpen *stable relational position*, (A1) and (A5) sharpen *enumerable and verifiable properties*, and (A3) sharpens *predictable behaviour under known operations*.

Proposition 5.11 (The XOR certificate is inadmissible)**.** *Let $\pi_V(X) = (m, k, f)$ with $k = m \oplus X$ and $f = \oplus$ be the certificate of Def. 5.4, over targets $D = \{0,1\}^n$ with ℓ the bit-length, $M_0 = D$, and a target class $\mathcal{T} \subsetneq D$ with $|\mathcal{T}| \geq 2$. The operator itself is a further artifact carrying the triple $(1, 0, 1)$, which places $\oplus \in \mathsf{M} \cap \mathsf{K}$ as Def. 5.4 requires and, $\oplus$ being K-bearing, places it outside $M_0 = \mathsf{M} \setminus (\mathsf{C} \cup \mathsf{K}) = D$; the messages and the targets, which are the elements of D, carry the triple $(1, 0, 0)$. Then $\pi_V(X)$ fails* (A1)*,* (A3) *and* (A5)*, and is therefore inadmissible.*

Proof. (A1). Bitwise XOR is length-preserving on D, so $\ell(k) = \ell(m \oplus X) = n = \ell(X)$. Hence $\ell(\pi) = \ell(m) + \ell(k) + \ell(f) \geq \ell(k) = \ell(X)$, and $\ell(\pi) < \ell(X)$ fails. The failure survives the more generous accounting that charges nothing for m and f, since it already follows from $\ell(k)$ alone.

(A3). $\mathrm{Reach}(k, \oplus) = \{\, m' \oplus k : m' \in M_0 \,\} = \{\, m' \oplus m \oplus X : m' \in D \,\} = D$, since $m' \mapsto m' \oplus m \oplus X$ is a bijection of D. So $\mathrm{Reach}(k, \oplus) \subseteq \mathcal{T}$ holds only if $\mathcal{T} = D$, and fails for every proper target class.

(A5). The scheme is $\sigma(X) = (m, m \oplus X, \oplus)$ with m fixed by step 1 of Def. 5.4, hence independent of X. The map $\varphi_m(X) = m \oplus X$ is a bijection of D, so $X \neq X'$ gives $k_X \neq k_{X'}$ and the repertoire has $|\mathcal{K}_\sigma| = |\mathcal{T}|$: it carries exactly one pair per target, so $|\mathcal{K}_\sigma| < |\mathcal{T}|$ fails. The failure is not an artifact of stating (A5) at one class. For any parametrised family of target classes $\mathcal{T}_n \subsetneq \{0,1\}^n$ the same computation gives $|\mathcal{K}_{\sigma_n}| = |\mathcal{T}_n|$, which is neither $O(1)$ nor $o(|\mathcal{T}_n|)$, so the family form of the condition (Rem. 5.9) fails as well. □

Remark 5.11 (The target information cannot be displaced). The proof of (A5) uses that m is fixed before X. One might try to evade it by letting the message vary with the target instead — putting $m_X := k_0 \oplus X$ for a fixed k_0, which holds the key constant and reduces the repertoire to a single pair. This does not help, and the reason is worth stating: the same computation then shows $X \mapsto m_X$ to be a bijection, so the growth has moved from the key repertoire to the message. In either arrangement one of the two arguments of $\oplus$ is an invertible, length-preserving re-encoding of the target, which is what (A1) independently rejects. The construction has one degree of freedom and cannot spend it twice.

Remark 5.12 (Why the other two conditions are passed, and why that is the same defect). Prop. 5.11 claims failure for three conditions only, and the omission is deliberate. On the instantiation above the XOR certificate *satisfies* (A2) and (A4). Both are immediate from $\mathrm{Reach}(k, \oplus) = D$, computed in the (A3) step of the proof above: $|D \cap \mathcal{T}| = |\mathcal{T}| \geq 2$ gives (A2), and (A4) holds because the admissible perturbations of a target in D are again elements of D. But the fact that discharges (A2) and (A4) is precisely the fact that kills (A3) — the reach is maximal. A certificate that attains everything transports and survives perturbation for free, and explains nothing by doing so. Two consequences are worth recording. First, reach-based conditions cannot carry the anti-vacuity burden alone; the containment condition (A3) and the information condition (A5) are what do the work. Second, (A2) and (A4) are retained in Def. 5.10 anyway, because a certificate can fail them in the opposite direction — by being wholly instance-bound — and a criterion that rejected only the maximal-reach failure would let that one through.

6 Embedding Regularity and Comprehension Strength

Section 5 stratifies understanding into a structural level U_0 and two operational levels U_1, U_2, and Rem. 5.1 warns that reading a structural regularity as an operational witness inflates the status of an artifact. This section makes the two levels into two *measurable* axes and asks what each one bounds. The organizing claim is that they bound different things: the regularity of the embedding bounds how far M can reach, while the strength of the agent's comprehension function bounds how much of C assimilation can remove. Neither bound constrains the other, and removal stops at a floor that no strength can pass — a floor which Thm. 5.6 has already identified.

6.1 Offset regularity

The structural level U_0 of Def. 5.1 is stated as "the embedding space exhibits stable relational regularity". We fix a formal reading of that phrase. The reading is modelled on the regularity reported for continuous word representations, in which a relation between words corresponds to a consistent offset between their vectors [31, 30]. We take from that literature only the *form* of the regularity — relations realized as difference vectors — and nothing about which similarity measure recovers them; the definition below is therefore parametric in a declared similarity measure and tolerance, and we attribute no particular choice of measure to anyone.

Parametric is not unconstrained, however, and the constraint carries a cost that is better stated here than discovered later. The declared pair is required to be *separating*, which is a metric-like demand: the tolerance must bound a distance. A scale-invariant measure cannot meet it, and cosine similarity — the measure that literature standardly uses — is scale-invariant, so it admits no separating radius on any embedding of more than one point (Rem. 6.1). The reading fixed below is thus unavailable under the canonical measure of the work that motivates it. What follows is accordingly about embeddings equipped with a metric-like similarity-and-tolerance pair, and not

about embeddings under any similarity measure whatever.[1]

Definition 6.1 (Offset regularity and regularity basis)**.** *Let* emb *be the embedding of* $\mathcal{A}^+$ *(Sec. 2.6), let* sim *be a declared similarity measure on the embedding space and* ε *a declared tolerance. The declared pair is required to be* separating *on the embedding: in a norm fixed with the embedding, there is a radius* $\rho > 0$ *such that*

$$\mathrm{sim}\big(\mathrm{emb}(y),\ \mathrm{emb}(x) + \delta\big)\ \geq\ 1 - \varepsilon \qquad \Longrightarrow \qquad \big\|\mathrm{emb}(y) - \mathrm{emb}(x) - \delta\big\|\ \leq\ \rho$$

holds for all $x, y \in \mathcal{A}^+$ *and every vector* δ *of the embedding space, and such that* 2ρ *is strictly less than the least distance between two distinct difference vectors* $\mathrm{emb}(y) - \mathrm{emb}(x)$, $x, y \in \mathcal{A}^+$ *— a positive quantity, the domain being finite. The implication is quantified over exactly the pairs at which the offset condition below is evaluated, with* δ *left unrestricted because the definition restricts* δ_r *nowhere; both requirements are used as stated, and both in the proof of clause (ii) of Prop. 6.6. Tolerance is thus permitted and bounded by the geometry of the points it is applied to. A binary relation* $r \subseteq \mathcal{A}^+ \times \mathcal{A}^+$ *is* offset-regular *when there is a single vector* δ_r *with*

$$\mathrm{sim}\big(\mathrm{emb}(y),\ \mathrm{emb}(x) + \delta_r\big)\ \geq\ 1 - \varepsilon \qquad \textit{for all } (x, y) \in r,$$

and non-degenerate *when in addition* r *contains at least two distinct pairs and* $\delta_r \neq 0$*. A* regularity basis *is a finite set* Φ *of non-degenerate offset-regular relations; Rem. 6.1 says what the separation and non-degeneracy requirements are for — without the second, the word* generating *defined just below would hold of every finite configuration whatever, and without the first it would hold on the strength of relations the geometry does not realize. For* $P \subseteq \mathcal{A}^+$*, the* Φ-closure $\mathrm{cl}_\Phi(P)$ *is the least superset of* P *closed under the* partial, set-valued *steps*

$$x\ \longmapsto\ \mathrm{emb}^{-1}\big[\,\mathrm{emb}(x) \pm \delta_r\,\big]\ =\ \{\, y \in \mathcal{A}^+ : \mathrm{emb}(y) = \mathrm{emb}(x) \pm \delta_r \,\} \qquad (r \in \Phi),$$

each of which is empty at x *unless the displaced point lies in* $\mathrm{emb}(\mathcal{A}^+)$. The notation is an inverse image and not an inverse function: *injectivity of* emb *is nowhere assumed in this paper, so a displaced point may be the image of several artifacts or of none, and a step contributes all of them or nothing. Every statement below is read under this set-valued reading; where a step is spoken of as producing "the" point, one of the artifacts realizing it is meant, and no result turns on which. The restriction to* $\mathcal{A}^+$ *is part of each operation and not a filter applied afterwards to a closure taken first in the ambient space: a step whose result embeds no artifact is discarded, and the point that step would have produced is not available as the input of a later one. The two readings genuinely differ, which is why the restricted one is stated rather than left to the grammar — a* multiple $n\,\delta_r$ *of an offset may land on an artifact when* δ_r *itself does not, and the ambient reading would admit that point while the reading fixed here does not. We call* Φ trivial *when* $\mathrm{cl}_\Phi(P) = P$ *for every* P*, and* generating *when* $\mathrm{cl}_\Phi(\{p\}) = \mathcal{A}^+$ *for some single point* p*.*

Both requirements restore content that the passage from the phrase in Def. 5.1 to Def. 6.1 would otherwise drop. For the non-degeneracy clause the case begins before any mathematics, and is given here rather than in the remark below because it is motivation and not a result. U_0 asks for

[1]The empirical literature on representation geometry independently characterises embedding-space anisotropy, with contextual representations occupying a narrow cone [11], and representation degeneration, with word embeddings of generation models trained by likelihood maximisation with tied weights collapsing into a narrow cone [13], rather than filling a uniformly distributed region. This affects the reliability of cosine-similarity-based similarity judgments and constitutes independent motivation for the separating pair adopted below; the exact-ideal offset-regularity condition (Def. 6.1, Rem. 6.2) is not claimed to hold on any specific learned embedding, and its empirical adequacy is delegated to the companion treatment.

stable relational regularity, and what makes the regularity reported in the cited literature stable is that *many* pairs of one relation share *one* offset [31]. That is received understanding of those results, not a finding of this paper; it is reported here as such, and Rem. 3.3 is where the manuscript records it under a tag. A relation with a single pair has its offset fitted to it and carries no such content. The formal reading, having kept the single vector δ_r, has to keep the multiplicity as well; the two-pair clause is that multiplicity, not a new demand. What the remark below establishes is a separate matter — that without the clause the word *generating* would divide nothing — and that is a claim about the definition, provable and proved.

Remark 6.1 (Why separation and non-degeneracy are required). What the non-degeneracy clause protects is the discriminating power of the word *generating*. Without it, a generating basis exists over every finite domain and every embedding whatever. Enumerate $\mathcal{A}^+ = \{x_1, \dots, x_L\}$, put $p := x_1$, and for each $j \geq 2$ take the singleton relation $r_j := \{(x_1, x_j)\}$ with $\delta_{r_j} := \mathrm{emb}(x_j) - \mathrm{emb}(x_1)$. Each r_j meets the offset condition exactly, so it holds under any tolerance; $\Phi = \{r_2, \dots, r_L\}$ is finite, the definition bounding the cardinality of a basis nowhere; every point $\mathrm{emb}(x_1) + \delta_{r_j}$ realizes an artifact, so the restriction to $\mathcal{A}^+$ discards no step; and $\mathrm{cl}_\Phi(\{p\}) = \mathcal{A}^+$, one step per element. The domain of this paper being finite by declaration (Sec. 8), a term true of every configuration in scope would separate nothing.

The separation requirement blocks a degeneracy at the other end, and one the non-degeneracy clause leaves standing. Take cosine similarity on the collinear embedding $\mathrm{emb}(x_j) = j\,v$ used in clause (i) of Prop. 6.6 below, and let r be the *total* relation $\mathcal{A}^+ \times \mathcal{A}^+$ with $\delta_r := v$. Every embedded point is a positive multiple of v, hence so is every $\mathrm{emb}(x) + v$, and cosine similarity of two positive multiples of v is 1; so $\mathrm{sim}(\mathrm{emb}(y), \mathrm{emb}(x) + v) = 1$ for every pair whatever and r is offset-regular. It is non-degenerate as well, having many pairs and $\delta_r \neq 0$, and $\Phi = \{r\}$ is generating, closing $\{x_1\}$ to all of $\mathcal{A}^+$ one step at a time. The clause of the preceding paragraph therefore does not exclude it. What is wrong with it is that its offset is determined by nothing: $2v$, $3v$ and every other positive multiple of v serve equally well, and r is offset-regular under each, although the pairs it relates have actual difference vectors $-v$, 0 and $2v$ among others. Nothing in Def. 6.1 would separate r from the chain relation of clause (i), which the geometry does realize.

The source of this is that cosine similarity admits no separating radius at all, on this or any embedding of more than one point: for any $y \in \mathcal{A}^+$ with $\mathrm{emb}(y) \neq 0$ and any $\lambda > 0$, the instance $x := y$, $\delta := (\lambda - 1)\,\mathrm{emb}(y)$ of the separation requirement has $\mathrm{sim}(\mathrm{emb}(y), \lambda\,\mathrm{emb}(y)) = 1$ while $\|\delta\| = |\lambda - 1| \cdot \|\mathrm{emb}(y)\|$ grows without bound, so no radius bounds it and the tolerance bounds no distance. The offset equation therefore carries no content by itself; what content Def. 6.1 has is content of the declared pair $(\mathrm{sim}, \varepsilon)$, and leaving that pair unconstrained would rest the definition's whole weight on an undeclared parameter.

The same defect has already been diagnosed in this manuscript, on the other axis. Rem. 5.12 observes that a certificate whose reach is maximal discharges (A2) and (A4) for free and explains nothing by doing so, and concludes that reach-based conditions cannot carry an anti-vacuity burden alone. The two notions of reach are unrelated — there, the reach of a certificate scheme; here, the closure of a regularity basis — and no result about one is being imported into the other. It is the argument that transfers, and the clauses above are what keep it from applying to Reach_Φ.

Remark 6.2 (Def. 6.1 is an exact ideal notion). The separating requirement has a consequence sharper than tolerance-talk suggests, and stating it fixes what kind of notion the definition is.

Let r be offset-regular with offset δ_r and let (x_1, y_1), $(x_2, y_2) \in r$, writing $d_i = \mathrm{emb}(y_i) - \mathrm{emb}(x_i)$. The offset condition gives $\mathrm{sim}(\mathrm{emb}(y_i), \mathrm{emb}(x_i) + \delta_r) \geq 1 - \varepsilon$ for $i = 1, 2$, so the separation requirement — quantified over *all* $x, y \in \mathcal{A}^+$ and every δ, not over the pairs of r — yields $\|d_i - \delta_r\| \leq \rho$ and hence $\|d_1 - d_2\| \leq 2\rho$. Since 2ρ is strictly less than the least distance between two *distinct*

difference vectors, d_1 and d_2 cannot be distinct:

$$d_1 \;=\; d_2.$$

So any two pairs of an offset-regular relation carry *exactly* equal difference vectors, and the tolerance ε does no work in relating one pair to another — it only relates each pair to the fitted δ_r. *What the two clauses do and do not exclude.* The non-degeneracy clause constrains the fitted parameter, $\delta_r \neq 0$, and not the difference vectors of the pairs; those are two different quantities and the clause should not be read as bounding the second. Under the separation regime it does not need to: the display above forces $d_1 = d_2$, so a relation cannot mix a zero difference with a nonzero one, and the mixed case is unavailable rather than merely unintended. What survives is the single case in which *every* pair has zero difference — two distinct pairs (x_1, x_1), (x_2, x_2) with $\|\delta_r\| \leq \rho$ — which the clause admits. No result below excludes it and none needs to: such a relation relates each element to itself, so its closure adds nothing and it is generating for no configuration. The clause is therefore *parameter* non-degeneracy, and pairwise nontriviality is a strictly stronger condition this paper neither proves nor uses. Outside the separation regime not even $d_1 = d_2$ holds; the cosine example of Rem. 6.1 is exactly that, and is the reason separation is required rather than optional.

Non-degeneracy, which demands two distinct pairs, therefore demands an exact coincidence of difference vectors; on four distinct points that is an exact parallelogram.

Three consequences are recorded, and the third is the one that matters for how this paper is read.

First, the definition is an *exact ideal* notion: $\mathrm{emb}(y) - \mathrm{emb}(x) = \delta_r$ up to the fitted offset, holding of every pair of r on the nose. It is nontrivial only on representations containing exact repeated offsets under the declared separation regime.

Second, a representation in general position contains no such repetition, so the notion is not one that an arbitrary learned representation will satisfy. That is a property of general position, not a defect of the definition, and it is the reason the results below are stated for representations equipped with a declared metric-like pair rather than for representations at large — a restriction already forced above, since the canonical similarity measure of the motivating literature admits no separating radius at all.

Third, and consequently, *no claim is made here that this definition models the approximate behaviour reported for distributional embeddings.* The approximate notion — $\mathrm{emb}(y) - \mathrm{emb}(x) = \delta_r + \eta_{x,y}$ under a declared error model and a held-out validation regime — is a different definition with a different burden, and it is not introduced in this paper. It is delegated to the companion treatment of empirical realization, together with the question of whether any learned representation supports it. Rem. 3.3 records the corresponding limit on what the citation there is doing.

Definition 6.2 (Structural reach)**.** *Let a be an agent and $\mathrm{Den}_a \subseteq \mathcal{A}^+$ the set of points at which some token of a denotes (Sec. 2.5). The* structural reach *of a under a regularity basis Φ is $\mathrm{Reach}_\Phi(a) = \mathrm{cl}_\Phi(\mathrm{Den}_a)$. This is the formal reading of U_0 adopted here: an artifact is within a's structural understanding exactly when it lies in $\mathrm{Reach}_\Phi(a)$. It is a reading, not a theorem — it stipulates what the phrase in Def. 5.1 is to mean, and the substantive claims below are made relative to it.*

This is the first axis. It is not a property of what the agent can execute: it depends on the embedding and on the basis Φ, and on nothing the agent does. Both dependencies are real and neither is eliminable. In particular Φ is declared by whoever fixes the reading, so a statement about structural reach is a statement about an embedding *together with* a declared basis, unless it quantifies over all bases — which clause (ii) of Prop. 6.6 does and clause (i) does not. Everything the difference-vector discovery of Rem. 3.3 can produce lies in $\mathrm{Reach}_\Phi(a)$ by construction, and a

token-less point reached that way (Cases B and C of Def. 4.1) is a point of $\mathrm{Reach}_\Phi(a)$ that is not in Den_a. Both cases serve here: reaching the point is what places it in $\mathrm{Reach}_\Phi(a)$, and whether the verifier then accepts (Case B) or declines (Case C) governs only whether it acquires a token, not whether it was reached.

6.2 Comprehension functions and their strength

The second axis is a property of the *agent*.

Definition 6.3 (Comprehension function)**.** *A* comprehension function *for an agent a is a partial map γ_a that assigns to an artifact $x \in \mathcal{A}^+$ a functional certificate $\gamma_a(x) = (m, k, f)$ which is*

1. *a witness for* $\mathrm{Hit}_f(x)$ *in the sense of Def. 5.2 — so $m \in M_0^a$, the key-membership condition $k \in (\mathsf{M}_a \cap \mathsf{K}_a) \cup (\mathsf{C}_a \cap \mathsf{K}_a)$ of Prop. 5.9, and the operator condition $f \in \mathsf{M}_a \cap \mathsf{K}_a$; and*
2. *admissible in the sense of Def. 5.10, as part of an admissible scheme.*

Its domain $\mathrm{dom}(\gamma_a)$ *is the set of artifacts it certifies.*

Definition 6.4 (Comprehension strength)**.** *For comprehension functions γ_a, γ'_a of the same agent, say γ_a is* at least as strong as *γ'_a, written $\gamma_a \succeq \gamma'_a$, when* $\mathrm{dom}(\gamma'_a) \subseteq \mathrm{dom}(\gamma_a)$*. Strength is thus a preorder, not a number; we introduce no numerical measure and none is needed below. The* residual opacity *of a under γ_a is*

$$\mathrm{Res}(a, \gamma_a) \;=\; \mathsf{C}_a \setminus \mathrm{dom}(\gamma_a),$$

the opacity the comprehension function does not certify.

Two cautions are owed. First, possession of γ_a is an operational fact and membership in $\mathrm{Reach}_\Phi(a)$ is a structural one; Rem. 5.1 forbids inferring either from the other, and Prop. 6.6 below is that prohibition made precise in one instance. Second, γ_a is required to produce *admissible* certificates, which is what keeps Def. 6.3 from being satisfied by every agent at once: by Prop. 5.11 the universal-assimilation scheme of Def. 5.4 does not supply a comprehension function, even though it supplies a certificate for every target.

6.3 What each axis bounds

Proposition 6.5 (Strength lowers residual opacity; assimilation stops at a floor)**.** *Let a be an agent.*

(i) *Residual opacity is antitone in strength: $\gamma_a \succeq \gamma'_a$ implies* $\mathrm{Res}(a, \gamma_a) \subseteq \mathrm{Res}(a, \gamma'_a)$.

(ii) *Let C'_a be the opacity surviving any assimilation (Def. 5.5) driven by any comprehension function whatever. Then*

$$\mathsf{C}'_a \;\supseteq\; \mathsf{C}_a \setminus C_0^a \;=\; \mathsf{C}_a \cap (\mathsf{M}_a \cup \mathsf{K}_a) \;=\; \Theta_a \cup R_6^a,$$

and this lower bound is independent of γ_a: no increase in strength lowers it.

The two clauses concern different objects and are not to be conjoined. Clause (i) is about certification *— what the comprehension function carries in its domain — and clause (ii) about* removal *— what an assimilation step takes out of C_a. The floor is claimed for the second only. It is not claimed for* $\mathrm{Res}(a, \gamma_a)$*, and the reason is that Def. 5.2 defines* $\mathrm{Hit}_f(c)$ *at an arbitrary target c, only its reclassification clause being restricted to C_0^a: a comprehension function may therefore certify artifacts of Θ_a, so* $\mathrm{dom}(\gamma_a)$ *may meet Θ_a and nothing above bounds* $\mathrm{Res}(a, \gamma_a)$ *below by $\Theta_a \cup R_6^a$.*

Proof. (i) is immediate from Def. 6.4: if $\mathrm{dom}(\gamma'_a) \subseteq \mathrm{dom}(\gamma_a)$ then $\mathsf{C}_a \setminus \mathrm{dom}(\gamma_a) \subseteq \mathsf{C}_a \setminus \mathrm{dom}(\gamma'_a)$.

(ii) By Def. 5.5 an assimilation step removes from C_a only elements of a set $S \subseteq C_0^a$, whatever selected them, so $\mathsf{C}'_a = \mathsf{C}_a \setminus S \supseteq \mathsf{C}_a \setminus C_0^a$. The same holds after any sequence of steps, because the pure-opacity cell never grows: from the post-step predicates of Def. 5.5,

$$C_0^{a\prime} = \mathsf{C}'_a \setminus (\mathsf{M}'_a \cup \mathsf{K}'_a) = (\mathsf{C}_a \setminus S) \setminus ((\mathsf{M}_a \cup S) \cup \mathsf{K}_a) = C_0^a \setminus S \ \subseteq \ C_0^a,$$

so every step of a sequence removes a subset of the *original* C_0^a, and the accumulated removal is contained in C_0^a. For the two identities, $C_0^a = \mathsf{C}_a \setminus (\mathsf{M}_a \cup \mathsf{K}_a)$ gives $\mathsf{C}_a \setminus C_0^a = \mathsf{C}_a \cap (\mathsf{M}_a \cup \mathsf{K}_a) = (\mathsf{C}_a \cap \mathsf{M}_a) \cup (\mathsf{C}_a \cap \mathsf{K}_a) = \Theta_a \cup (R_6^a \cup R_7^a) = \Theta_a \cup R_6^a$, using $\Theta_a = R_4^a \cup R_7^a$ (Sec. 2.3). Nothing in the argument mentions γ_a, so the bound is uniform over comprehension functions. □

Remark 6.3 (Reading the floor). The floor $\Theta_a \cup R_6^a$ is exactly the opacity that is co-present with meaning or with function. It bounds what assimilation *removes* from C_a, and it says nothing about which artifacts a comprehension function certifies; the two are separated in the statement of Prop. 6.5 and are separated again here, because certification is not removal. Its Θ_a part is Thm. 5.6 restated in the vocabulary of this section: strengthening the comprehension function is one more thing that does not touch the symbols. Its R_6^a part — opaque operators — is a second obstruction of the same shape, and it is worth naming separately because it is not a consequence of the symbol result. Fig. 2 separates the two cells the statement keeps apart, and adds the case distinction the prose leaves implicit: an artifact in R_6^a is opaque and functional without being a symbol, and assimilation leaves it alone for the same reason, namely that the reclassification is defined on C_0^a only. Consequently the honest summary of the second axis is not "strength removes opacity" but "strength removes *unmeant, non-functional* opacity, and only that" — a statement about what assimilation removes, not about what a comprehension function certifies.

6.4 The two axes are independent

Proposition 6.6 (Independence)**.** *Neither axis bounds the other. Specifically:*

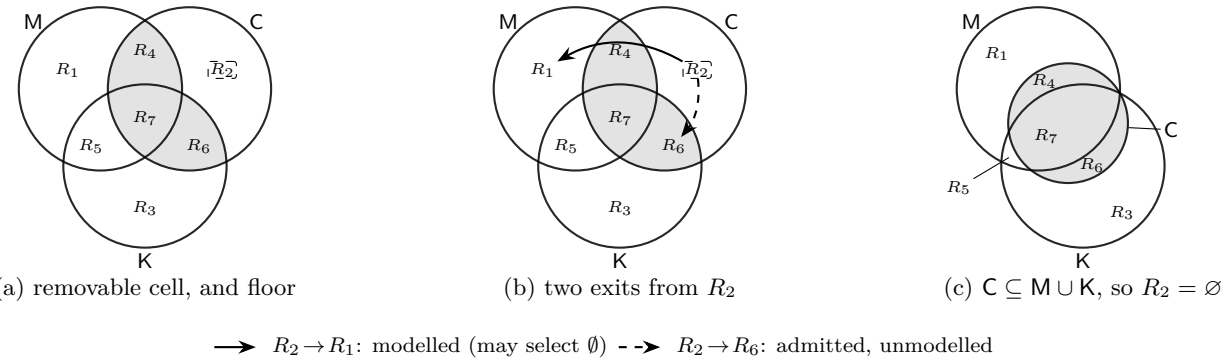
(i) *there is a configuration, and a generating regularity basis* Φ *for it, in which* $\mathrm{Reach}_\Phi(a) = \mathcal{A}^+$ *and no comprehension function is defined anywhere on* C_0*;*

(ii) *there is a configuration in which* every *regularity basis is trivial, so that* $\mathrm{Reach}_\Phi(a) = \mathrm{Den}_a$ *for every* Φ*, carrying a comprehension function total on its target class.*

The two clauses are not quantified alike, and deliberately so: (i) asserts that some basis is generating, (ii) that no basis is. Only the second is a property of the configuration alone, and only Def. 6.1's non-degeneracy clause makes it available: without that clause a generating basis exists over every finite embedding (Rem. 6.1) and no configuration could have the property (ii) asserts.

Proof. (i) Take the predicate assignment constructed in the proof of Prop. 5.9 — $\mathcal{A} = D \sqcup \{e\}$ with $D = \{0,1\}^n = D_M \sqcup D_C$, $M_0 = D_M$, $C_0 = D_C$, and e the sole K-bearing artifact, sitting in R_5 — and extend it by one further artifact s carrying the triple $(1,1,0)$, so that $s \in R_4$. The extension changes none of the sets the Hit-failure argument uses: s is neither in $M_0 = \mathsf{M} \setminus (\mathsf{C} \cup \mathsf{K})$ nor in $C_0 = \mathsf{C} \setminus (\mathsf{M} \cup \mathsf{K})$, since it is in $\mathsf{M} \cap \mathsf{C}$, and it is not K-bearing, so $M_0 = D_M$, $C_0 = D_C$ and $(\mathsf{M} \cap \mathsf{K}) \cup (\mathsf{C} \cap \mathsf{K}) = \{e\}$ all still hold. Its role is to give the agent a symbol, so that its denotation map has non-empty domain.

The configuration is admissible. Its type is $\tau(a) = \{1,2,4,5\}$: $R_1^a = D_M$, $R_2^a = D_C$, $R_4^a = \{s\}$, $R_5^a = \{e\}$, and the remaining cells are empty. This meets each of the three conditions of the



(a) removable cell, and floor (b) two exits from R_2 (c) $\mathsf{C} \subseteq \mathsf{M} \cup \mathsf{K}$, so $R_2 = \varnothing$

Figure 2: The ceiling of assimilation, in three separate structural cases — *not* three times of one process. Shaded cells are the floor $\Theta \cup R_6$ of Prop. 6.5; the dashed cell is $R_2 = \mathsf{C} \setminus (\mathsf{M} \cup \mathsf{K})$, the only opacity an assimilation step removes. Panel (c) is drawn, not asserted: C is redrawn small enough and far enough into $\mathsf{M} \cup \mathsf{K}$ that no C-only region survives, since $R_2 = \varnothing$ *is* the containment and not a label one may place on an unchanged picture. All three floor cells are still shaded there, and that is the point of the panel — what disappears is R_2 and nothing else. That the drawn C meets the other two circles at one of their crossing points is a convenience of the drawing, chosen so that R_5 stays one region; the containment is what the panel asserts and it does not depend on that tangency. Emptying R_2 requires $\mathsf{C} \subseteq \mathsf{M} \cup \mathsf{K}$, which two disjoint routes satisfy: acquiring M, giving $R_2 \to R_1$, which Def. 5.5 models; and acquiring K without M, giving $R_2 \to R_6$, admitted by the identity and left unmodelled here. *The modelled route need not move anything:* by Prop. 5.9(iii) an assignment exists on which the reconstruction certificate holds at every target and no element of R_2 is selected. Hence panel (c) is a case, not an outcome, and $R_2 = \varnothing$ entails neither $\mathsf{C} = \varnothing$ nor $\mathsf{C} \subseteq \mathsf{M}$.

classification of [3] — $\tau(a)\cap\{1,4,5,7\}=\{1,4,5\}$, $\tau(a)\cap\{2,4,6,7\}=\{2,4\}$ and $\tau(a)\cap\{3,5,6,7\}=\{5\}$ are all non-empty — so M_a, C_a and K_a are all inhabited and $\tau(a)$ is one of the realizable types. Since $\Theta_a = R_4^a \cup R_7^a = \{s\} \neq \emptyset$, the denotation map is defined somewhere and $\mathrm{Den}_a \neq \emptyset$.

Now the embedding. Every artifact here is classified, so $\mathcal{A}^+ = \mathcal{A}$; enumerate it as $\mathcal{A}^+ = \{x_1, \ldots, x_L\}$ with $L = 2^n + 2$, fix a non-zero vector v of the embedding space, and set $\mathrm{emb}(x_j) = j\,v$. This is injective, since the $j\,v$ are pairwise distinct for $v \neq 0$. Let Φ consist of the single relation $r = \{\,(x_j, x_{j+1}) : 1 \leq j < L\,\}$. Then r is offset-regular with $\delta_r = v$, and exactly so: $\mathrm{emb}(x_{j+1}) = \mathrm{emb}(x_j) + v$ holds on the nose, so the inequality of Def. 6.1 holds for any declared pair $(\mathrm{sim}, \varepsilon)$ normalized to $\mathrm{sim}(w, w) = 1$, separating or not. A separating pair is available here, and we exhibit one rather than assert its existence, since Def. 6.1 admits no configuration without one: the difference vectors are the multiples $(j - i)\,v$, of which two distinct ones differ by at least $\|v\|$, so $\rho := \|v\|/4$ meets the requirement $2\rho < \|v\|$, and

$$\mathrm{sim}(u, w) \;:=\; 1 - \min(1,\ \|u - w\|/\rho), \qquad \varepsilon := 1/2$$

is normalized and separating with that radius, since $\mathrm{sim}(u, w) \geq 1/2$ forces $\|u - w\| \leq \rho/2 \leq \rho$. It is non-degenerate as well: r has $L - 1 \geq 2$ distinct pairs, since $L = 2^n + 2 \geq 3$, and $\delta_r = v \neq 0$. And Φ is generating: closing $\{x_1\}$ under δ_r and its inverse — every step from x_j with $j < L$ landing on x_{j+1}, and the restriction to $\mathcal{A}^+$ discarding only the step from x_L, whose value $\mathrm{emb}(x_L) + v$ embeds no artifact — yields all of $\mathcal{A}^+$. Hence $\mathrm{Reach}_\Phi(a) = \mathrm{cl}_\Phi(\mathrm{Den}_a) = \mathcal{A}^+$, structural reach being maximal, for Den_a is non-empty.

Yet by the proof of Prop. 5.9, unaffected by the extension, $\mathrm{Hit}_\oplus(X)$ fails for every $X \in C_0$: a witness would need $k \in (\mathsf{M} \cap \mathsf{K}) \cup (\mathsf{C} \cap \mathsf{K}) = \{e\}$ and $m \in M_0 = D_M$ with $\oplus(m, e) = X$, while $\oplus(m, e) = e \notin D_C$. And $\oplus$, being the only operator the model carries, is the only f available. So no comprehension function is defined at any point of C_0, by clause 1 of Def. 6.3, before admissibility of certificates is even consulted.

(ii) Let f be a fixed expanding operator, $f(\cdot, k_0) : M_0 \to \{0,1\}^N$ injective and non-constant with $N > \ell(m) + \ell(k_0) + \ell(f)$ for all $m \in M_0$, and put $\mathcal{T} := \mathrm{Reach}(k_0, f)$ and the perturbation class $:= \mathcal{T}$.

We first place the artifacts, since Def. 6.3 constrains where they may sit. The messages are the elements of $M_0 = R_1^a$, of which there are at least two, $f(\cdot, k_0)$ being non-constant. The key k_0 and the operator f are K-bearing and placed in $R_5^a = (\mathsf{M}_a \cap \mathsf{K}_a) \setminus \mathsf{C}_a$. This discharges *both* region conditions that clause 1 of Def. 6.3 carries, and both are named here because Hit_f constrains the operator as well as the key: R_5^a lies inside $(\mathsf{M}_a \cap \mathsf{K}_a) \cup (\mathsf{C}_a \cap \mathsf{K}_a)$, which is what is required of k_0, and inside $\mathsf{M}_a \cap \mathsf{K}_a$, which is what is required of f. The targets are the elements of $\mathcal{T}$, placed in $C_0^a = R_2^a$; they are artifacts distinct from the messages, from k_0 and from f, since N exceeds each of $\ell(m)$, $\ell(k_0)$ and $\ell(f)$, and being classified they lie in $\mathcal{A}^+$, which is what Def. 6.3 requires of $\mathcal{T}$. Finally one artifact s is placed in R_4^a, giving $\Theta_a = \{s\} \neq \emptyset$ so that the denotation map is defined somewhere. The type is thus $\tau(a) = \{1, 2, 4, 5\}$, which meets the three conditions of the classification of [3] — $\tau(a) \cap \{1, 4, 5, 7\}$, $\tau(a) \cap \{2, 4, 6, 7\}$ and $\tau(a) \cap \{3, 5, 6, 7\}$ are $\{1, 4, 5\}$, $\{2, 4\}$ and $\{5\}$ — so the configuration is an admissible agent.

Define $\gamma_a(X) = (m_X, k_0, f)$ for $X \in \mathcal{T}$, where m_X is the preimage of X under $f(\cdot, k_0)$, unique by injectivity. Each $\gamma_a(X)$ witnesses $\mathrm{Hit}_f(X)$: $m_X \in M_0^a$, k_0 satisfies the key-membership condition as just placed, and $f(m_X, k_0) = X$. So γ_a is total on $\mathcal{T}$, and its scheme is admissible: (A1) holds by the length inequality; (A2) holds since $f(\cdot, k_0)$ is non-constant, so $|\mathcal{T}| \geq 2$; (A3) holds because $\mathrm{Reach}(k_0, f) = \mathcal{T}$ by construction; (A4) holds because the perturbation class is $\mathcal{T}$; and (A5) holds because the single pair (k_0, f) serves every target, giving $|\mathcal{K}_\sigma| = 1 < 2 \leq |\mathcal{T}|$.

For the regularity bases, fix the embedding first. Enumerate the artifacts as $\mathcal{A}^+ = \{x_1, \ldots, x_L\}$, fix a non-zero vector v of the embedding space, and set $\mathrm{emb}(x_j) = 2^j\,v$; this is injective, the $2^j v$

being pairwise distinct for $v \neq 0$. Declare with it a separating pair $(\mathrm{sim}, \varepsilon)$ in the sense of Def. 6.1. One is available explicitly: the difference vectors here are the $(2^j - 2^i)v$, whose coefficients are even integers, so two distinct ones are at distance at least $2\|v\|$; putting $\rho := \|v\|/2$ gives $2\rho < 2\|v\|$ as required, and

$$\mathrm{sim}(u, w) \;:=\; 1 - \min(1,\ \|u - w\|/\rho), \qquad \varepsilon := 1/2$$

is normalized and separating with that radius. The argument uses only that *some* separating pair has been declared, not this one. We claim that on this embedding *every* regularity basis is trivial, and hence that no declaration is doing the work.

The load is carried by an arithmetical fact: a non-zero difference of two powers of two determines both exponents. If $2^j - 2^i = 2^l - 2^k \neq 0$, take $j > i$ and $l > k$ — two negative differences give the same argument with the pairs exchanged, and a positive difference cannot equal a negative one — so that $2^i\,(2^{j-i} - 1) = 2^k\,(2^{l-k} - 1)$ with both bracketed factors odd; the two sides are then divisible by exactly 2^i and 2^k, whence $i = k$ and then $j = l$. So the ordered difference vectors $\mathrm{emb}(x_j) - \mathrm{emb}(x_i)$ with $i \neq j$ are pairwise distinct.

Now let r be any relation admitted to a regularity basis, so that r has at least two distinct pairs and $\delta_r \neq 0$. Take any two of its pairs, (x_i, x_j) and (x_k, x_l), sharing the offset δ_r. Separation gives $\|\mathrm{emb}(x_j) - \mathrm{emb}(x_i) - \delta_r\| \leq \rho$ and $\|\mathrm{emb}(x_l) - \mathrm{emb}(x_k) - \delta_r\| \leq \rho$, so the two difference vectors lie within 2ρ of one another and must therefore be equal, no two distinct ones being that close. All pairs of r thus carry one difference vector d. If $d \neq 0$ then $d = (2^j - 2^i)v$ for exactly one ordered pair by the fact just proved, so r has a single pair and is degenerate. Hence $d = 0$: every pair of r is of the form (x, x), and $\|\delta_r\| \leq \rho$, non-zero though δ_r is by non-degeneracy. Such an offset moves no point of the embedding onto another — a non-zero difference vector is more than 2ρ from the zero one, hence of norm above 2ρ — so for every $x \in \mathcal{A}^+$ the point $\mathrm{emb}(x) \pm \delta_r$ lies outside $\mathrm{emb}(\mathcal{A}^+)$, and the corresponding step of Def. 6.1 is undefined at x and is discarded.

This last step is where the restriction built into the Φ-closure of Def. 6.1 carries the argument, and it is worth saying so. What has been established about δ_r bounds a *single* application: $\|\delta_r\| \leq \rho$ does not prevent some multiple $n\,\delta_r$ from having the norm of a difference vector and landing on an artifact, and on this embedding such multiples exist. Were the closure taken in the ambient space and intersected with $\mathcal{A}^+$ afterwards, that artifact would be reached and the claim would fail. Under the closure as defined it is not reached: the first step already leaves $\mathrm{emb}(\mathcal{A}^+)$, so it is discarded, and the point it would have produced never becomes available to carry a second. The triviality asserted here is a statement about the restricted closure and is not claimed for the ambient one. No point is ever added: $\mathrm{cl}_\Phi(P) = P$ for every P and every basis Φ this configuration admits, the empty basis included. Hence $\mathrm{Reach}_\Phi(a) = \mathrm{Den}_a$ for every Φ, which is the single point at which s denotes: structural reach is minimal, and minimal not by declaration but because the embedding permits nothing else. The two constructions are independent of one another, so both configurations exist. □

Remark 6.4 (What the independence does and does not settle)*.* Prop. 6.6 is a separation, not a general theory of the two axes. It shows that no implication holds in either direction between maximal structural reach and possession of a comprehension function, which is the precise content of Rem. 5.1 in the one case where both sides have been given formal readings. It does not discharge that remark, which is a broader claim about the structural and operational levels generally and remains conjectural. It also says nothing about the typical case: that a rich embedding often accompanies a strong comprehension function is entirely consistent with the two being logically independent, and this paper offers no measurement either way.

Clause (ii) of the proof deserves one explicit caveat. The target class there is *defined* as the reach of the scheme, which is what makes (A3) hold. That is legitimate for an existence claim —

it exhibits a configuration in which the criterion is met — but it is not evidence that the criterion is easy to meet for an antecedently given class, and it should not be read as such.

7 The Compositional Comprehension Lens

The discovery predicate of Definition 5.2 is already a statement about *several* artifacts at once: $\mathrm{Hit}_f(c)$ holds when a message $m \in M_0$, a key $k \in (\mathsf{M} \cap \mathsf{K}) \cup (\mathsf{C} \cap \mathsf{K})$ and an operator $f \in \mathsf{M} \cap \mathsf{K}$ jointly exhibit c. Three artifacts, three region conditions, one composite verdict. This section takes that pattern as far as it goes: a computation is a collection of typed artifacts standing in justification relations, each artifact carries its region, and — this is the point — so does each relation.

Nothing below introduces a new region, a new predicate, or a numerical measure of comprehension. The apparatus is the one already in place: γ_a (Definition 6.3), its residual opacity (Definition 6.4), the admissibility conditions (Definition 5.10), and the floor of Proposition 6.5.

7.1 Typed computation structures

Definition 7.1 (Computation structure)**.** *A* computation structure *is a finite directed acyclic graph $\mathcal{C} = (V, E)$ with $V \subseteq \mathcal{A}^+$ a set of typed artifacts and E a set of* justification relations*, each $e = (W \to v)$ with $W \subseteq V$ finite and $v \in V \setminus W$: the claim that the artifacts of W jointly justify v. Edges are n-ary and are themselves artifacts of $\mathcal{A}^+$.*

Each $v \in V$ carries a type, functional *($K(v) = 1$) or* non-functional *($K(v) = 0$), and the type constrains which regions are available to it: a functional artifact inhabits one of R_3, R_5, R_6, R_7, a non-functional one of R_1, R_2, R_4 (Section 2.2). Nothing else is stipulated; in particular no edge is required to exist, and the acyclicity is a modelling convention, not a claim about justification in general.*

Acyclicity and n-ary edges are together the smallest structure that carries the phenomenon of Section 7.2: the binary case cannot express "U and Rsd *jointly* justify the progress law", and a hypergraph would add generality nothing below uses.

Definition 7.2 (Status assignment)**.** *For an agent a, the* status assignment *on $\mathcal{C}$ is the pair of maps $\epsilon_a \colon V \to \mathcal{R}$ and $\epsilon_a \colon E \to \mathcal{R}$ sending each artifact and each justification relation to its MCK region, $\mathcal{R} = \{R_1, \ldots, R_7\}$. Both are instances of the ordinary region classification of Section 2.2 applied to artifacts of $\mathcal{A}^+$; the edge case is not a new kind of classification but the same one applied to an artifact that happens to be a justification.*

7.2 Composite status is not determined by component status

Proposition 7.3 (Composite non-determination)**.** *There is no function ϕ with $\epsilon_a(e) = \phi(\{\epsilon_a(v) : v \in W\}, \epsilon_a(v))$ for every computation structure, every agent and every edge $e = (W \to v)$.*

Proof. An edge is an artifact of $\mathcal{A}^+$ (Definition 7.1), and the three predicates are logically independent with all eight triples admissible a priori (Section 2.1). So for a fixed W, v and fixed component statuses, the triple $(M, C, K)(e)$ may be assigned in any admissible way, and two structures agreeing on every component status may disagree on $\epsilon_a(e)$. Hence no such ϕ exists. □

Remark 7.1 (The proposition is cheap; the consequence is not). The argument above is a one-line consequence of predicate independence, and we state it as such rather than dressing it up: it says only that nothing in the framework *forces* compositional behaviour. Its interest is that the

framework is right not to force it, and Proposition 5.9 (iii) is the existing evidence: there the display (†) holds for every target — a representation exists in each case — while $\mathrm{Hit}_{\oplus}$ holds for no element of C_0. Component conditions met, composite verdict false. The lens generalises that situation rather than discovering it.

7.3 Role-relative resolution

It is tempting to define full comprehension as the elimination of opacity, $C(x) = 0$ everywhere. That target is not merely unreachable but wrong in kind, and the paper's own results say why: $C(x) = 0$ for all x forces $\Theta_a = \mathsf{M}_a \cap \mathsf{C}_a = \emptyset$, and Θ_a is exactly the denotation domain (Section 2.3) — an artifact denotes precisely when it lies there. An agent meeting that condition would hold no symbols at all. Theorem 5.6 shows assimilation cannot produce it, and Corollary 5.7 records the consequence. Opacity co-present with meaning is not a defect to be removed; it is what a symbol is.

The correct target is therefore relative to the role an artifact plays.

Definition 7.4 (Role-relative resolution and closure)**.** *Fix a computation structure $\mathcal{C}$ and a distinguished $r \in V$, the result. Call $x \in V \cup E$* role-required *for r if x lies on some directed path terminating at r. An artifact is* resolved *when*

$$\epsilon_a(x) = R_1 \quad \text{if } K(x) = 0, \qquad \epsilon_a(x) = R_5 \quad \text{if } K(x) = 1,$$

*and $\mathcal{C}$ is M-*closed for *a at r, written* $\mathrm{FullyM}_a(\mathcal{C}, r)$*, when every role-required artifact* and every role-required justification relation *is resolved.*

Two features are deliberate. The functional target is $R_5 = (\mathsf{M} \cap \mathsf{K}) \setminus \mathsf{C}$, the transparent-operator region: functionality is *retained*, and an operator does not stop being an operator by becoming understood. And artifacts of Θ_a are not asked to resolve — a symbol that satisfied the condition would no longer be a symbol. Closure is a statement about what a computation requires, not a programme of universal transparency.

The edge clause is what Proposition 7.3 makes necessary. Without it, "every component resolved" would count as closure, and Example 7.1 below is a computation where that reading is false.

7.4 Worked instance: Hensel–Newton refinement

Refinement by Hensel lifting is a convenient instance because its parts are mathematically explicit and separable. We fix the artifact system

$$\mathcal{H} \;=\; (D,\ \mathrm{Rep},\ U,\ \mathrm{Rsd},\ P,\ A,\ \mathrm{Cert}),$$

typed as follows: D the domain and Rep the representation are non-functional; the update U, the residual *map* Rsd and the certificate *predicate* Cert are functional; the progress law P and the assumptions A are non-functional, being statements.

The distinction between the residual map and the residual *identity* is load-bearing and is kept throughout: the map is a functional artifact and can sit in R_6 (computable, unread), whereas the identity relating it to U is a justification relation and is classified as an edge.

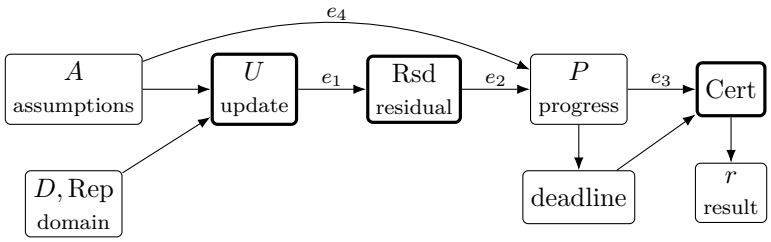


Figure 3: The Hensel–Newton computation structure. Thick boxes are functional artifacts ($K = 1$), thin boxes non-functional. The labelled relations are the ones that carry status independently of their endpoints: e_1 the residual identity (4), e_2 the passage from it to valuation doubling, e_3 the passage from the progress law to the certificate at the deadline, and e_4 the dependence of the progress law on the assumptions. Table 1 annotates both vertices and relations for five agents.

	D, Rep	U	Rsd	P	A	Cert	e_1	e_3
P0 black-box user	R_2	R_3	R_3	R_2	R_2	R_6	R_2	R_2
P1 symbolic executor	R_1	R_6	R_6	R_2	R_2	R_6	R_2	R_2
P2 certificate-aware	R_1	R_6	R_6	R_2	R_1	R_5	R_2	R_2
P3 progress-aware	R_1	R_5	R_5	R_1	R_2	R_5	R_1	R_1
P4 structural	R_1	R_5	R_5	R_1	R_1	R_5	R_1	R_1

Table 1: Status assignments for five agents on the structure of Figure 3. Only P4 is M-closed at r. P3 differs from P4 in one non-functional vertex, the assumptions, and that single cell is the whole of Example 7.3.

Modular inversion. Let p be prime, a a unit modulo p, and let the target be an inverse to precision n. Put

$$U(x) = x\,(2 - ax), \qquad \mathrm{Rsd}(x) = 1 - ax, \qquad \mathrm{Cert}_n(x) \;:\; ax \equiv 1 \pmod{p^n}.$$

The assumptions A are that a be a unit, that the initial x_0 satisfy $ax_0 \equiv 1 \pmod p$, and that a target precision n be fixed. The justification connecting U to Rsd is the identity

$$\mathrm{Rsd}\big(U(x)\big) = 1 - ax(2 - ax) = (1 - ax)^2 = \mathrm{Rsd}(x)^2, \tag{4}$$

from which the progress law P follows as $\nu_p(\mathrm{Rsd}(U(x))) = 2\,\nu_p(\mathrm{Rsd}(x))$, and from that the deadline: $\nu_p(\mathrm{Rsd}(x_k)) \geq 2^k$, so $k \geq \lceil \log_2 n \rceil$ steps suffice for Cert_n. Since $\mathrm{Cert}_n(x)$ holds exactly when $\nu_p(\mathrm{Rsd}(x)) \geq n$, the certificate is checkable by one multiplication and is independent of the path that produced x.

7.5 Agent profiles

A computation structure with 7 vertices over the admissible regions admits several thousand status assignments, most of them without interest. We quotient by the only distinctions the lens uses — which role-required artifacts are unresolved, whether the operators are executable, whether the certificate is checkable, and what closure obligations result — and five classes survive on this structure.

7.6 Four witnesses

Example 7.1 (Components resolved, interaction unresolved). Let $\epsilon_a(U) = \epsilon_a(\mathrm{Rsd}) = R_5$: the agent reads and executes both maps. Let $\epsilon_a(e_1) = R_2$: it cannot derive (4) and cannot say why squaring the residual is what the update accomplishes. Every vertex of the pair is resolved and the relation is not, so $\mathcal{C}$ is not M-closed at r by Definition 7.4. This is the configuration Proposition 7.3 licenses, and the reason the edge clause is in the definition.

Example 7.2 (Opaque component, sufficient composite). Let $\epsilon_a(U) = R_6$ — an implementation the agent runs and cannot read — while $\epsilon_a(\mathrm{Cert}) = R_5$ and $\epsilon_a(e_3) = R_1$. The agent verifies $ax \equiv 1 \pmod{p^n}$ directly and so holds the result on grounds independent of the process. The composite is adequate at the abstraction level the result requires, with an internal component opaque.

This must not be read as promotion. Hit_f requires $f \in \mathsf{M} \cap \mathsf{K}$, and Remark 5.2 excludes precisely an operator drawn from $\mathsf{C} \cap \mathsf{K}$ — functional, and opaque to the agent applying it. Here $U \in R_6 = (\mathsf{C} \cap \mathsf{K}) \setminus \mathsf{M}$ is that operator. The lens records that the agent is entitled to the result; it does not license the reclassification $C_0 \to M_0$, and the guard is unchanged.

Example 7.3 (Operator resolved, applicability not). Let $U, \mathrm{Rsd}, P, \mathrm{Cert}$ all be resolved but $\epsilon_a(A) = R_2$. The agent iterates faultlessly with a not a unit modulo p. Every functional artifact is in R_5 and the output is worthless. Knowing an operator is not knowing when it applies, and on this structure the distance between P3 and P4 in Table 1 is exactly this one cell.

Example 7.4 (Result resolved, process not). Let $\epsilon_a(\mathrm{Cert}) = R_5$ with $U, \mathrm{Rsd}, e_1, e_2$ all in R_6 or R_2. The agent certifies the inverse and cannot account for the refinement that produced it. Result-comprehension and process-comprehension come apart, and the certificate is what makes the first available without the second.

7.7 Closure obligations

Definition 7.5 (Closure obligation)**.** *For an unresolved role-required x, an* obligation $\mathrm{Obl}_a(x)$ *is a demand paired with the artifact that discharges it. We admit only those for which the discharging artifact is already an object of this framework:*

verify *discharged by a checked certificate — for $\mathcal{H}$, an evaluation of* Cert_n*;*

establish domain conditions *discharged by a checked hypothesis of A;*

connect to invariant *discharged by a derivation of the identity labelling the edge, here* (4)*;*

connect to certificate *discharged by a derivation of* Cert *from P at the deadline;*

disambiguate *discharged by a resolved target set $T(s)$ for a symbol $s \in \Theta_a$;*

derive *discharged by an admissible certificate in the sense of Definition 5.10.*

Three further demands suggest themselves — to *explain*, to *ground*, to *test counterfactually* — and none is admitted here, because this framework supplies no artifact that discharges them. Admitting them would give the plan below entries that no evidence could close.

Definition 7.6 (M-closure plan)**.** $\mathrm{Plan}_M(\mathcal{C}, a, r)$ *is the set of obligations* $\mathrm{Obl}_a(x)$ *over role-required $x \in V \cup E$ that are unresolved in the sense of Definition 7.4.*

The plan is what makes the classification diagnostic rather than descriptive: given a partially understood computation it names which artifacts, and which justifications between them, still have to be discharged, and with what. For P1 of Table 1 it returns CONNECT TO INVARIANT on e_1, CONNECT TO CERTIFICATE on e_3 and ESTABLISH DOMAIN CONDITIONS on A; for P3 it returns the last of these alone.

Remark 7.2 (The plan has a floor, and must say so). No plan driven by assimilation can promise arbitrary resolution. Proposition 6.5 (ii) bounds the opacity surviving any sequence of assimilation steps, under any comprehension function, below by $\Theta_a \cup R_6^a$. The Θ_a part is as it should be (Section 7.3). The R_6^a part is a genuine limit and bears directly on the material above: R_6 is where the opaque operator of Example 7.2 sits, so the natural remedial path $R_6 \to R_7 \to R_5$ is *not* available by assimilation, which acts on $C_0 = R_2$ alone. A plan that listed such a transition as an assimilation step would promise what Proposition 6.5 forbids. Whether some other mechanism effects it is not settled here, and Remark 5.5 is the corresponding caution for queries.

7.8 A second instance: certification that withholds

Hensel–Newton is a procedure whose closure plan one would like to *discharge*. The next instance is one where part of the plan is deliberately left undischarged, and the construction is engineered so that it stays that way. It is worth including because it separates two things the first example holds together: obtaining a warrant for a proposition, and obtaining the object the proposition is about.

A zero-knowledge argument lets a verifier become entitled to a statement without receiving the data that makes the statement true. We use the pairing-based argument of Groth [15], whose relation is a quadratic arithmetic program over a field $\mathbb{F}$: statements are $(a_1, \dots, a_\ell)$ and witnesses $(a_{\ell+1}, \dots, a_m)$ satisfying $\sum_i a_i u_i(X) \cdot \sum_i a_i v_i(X) = \sum_i a_i w_i(X) + h(X)t(X)$. Setup picks $\alpha, \beta, \gamma, \delta, x$, publishes a common reference string σ and retains the simulation trapdoor $\tau = (\alpha, \beta, \gamma, \delta, x)$; the prover picks r, s and returns three group elements $\pi = (A, B, C)$; the verifier accepts iff

$$A \cdot B \;=\; \alpha \cdot \beta \;+\; \Big(\textstyle\sum_{i=0}^{\ell} a_i(\beta u_i(x) + \alpha v_i(x) + w_i(x))/\gamma\Big) \cdot \gamma \;+\; C \cdot \delta. \tag{5}$$

Three objects, not one. The lens forces a distinction the informal reading elides. Write w_P for the field elements the prover actually holds; $\widehat{w}$ for the *witness-role*, "a value satisfying $R(x, \cdot)$"; and $E_x := \exists w.\, R(x, w) = 1$ for the existential proposition. These are three vertices, and for the verifier they carry three different regions. Before any proof, w_P satisfies $M_V = C_V = K_V = 0$ and so lies outside $\mathcal{A}^+$ altogether — not in R_2, which requires $C = 1$ and therefore semantic presence, and not in R_3, which requires $K = 1$. After a valid proof the verifier holds a partial certificate for $\widehat{w}$: a stable relational position (it satisfies R with x) and one enumerable property, with the reference unresolved. That is $\mathsf{M} \cap \mathsf{C}$, so $\widehat{w}$ enters Θ — it becomes a symbol (Section 2.3) — while w_P itself does not move. The proposition E_x is a third matter again, and is resolved for the verifier only if the warrant below is.

Why π carries nothing further. Groth's zero-knowledge argument is that real and simulated proofs both have A, B uniformly random, with C then determined by (5); the two distributions are identical, and the simulator is given the trapdoor and the *statement only* [15]. The randomisers r, s buy exactly this. So no admissible evidence about w_P is carried by π, and the failure of w_P to move is a property of the construction rather than an omission in the verifier's reasoning. The notion goes back to Goldwasser, Micali and Rackoff [14]; in the simulation-based form Groth states (Def. 2 of [15]), it asserts not that nothing is learned, but that nothing is learned that could not have been generated without the witness.

artifact	type	ϵ_P	ϵ_V	role-required for V?
R relation	non-functional	R_1	R_1	yes
x statement	non-functional	R_1	R_1	yes
w_P witness value	secret	R_1	*outside* $\mathcal{A}^+$	**no**
$\widehat{w}$ witness-role	non-functional	R_1	$R_4 \subset \Theta$	no
E_x existential	non-functional	R_1	R_1 / R_2	yes
σ reference string	functional	R_5	R_6	yes
π proof	certificate	R_1	R_2 / R_4	yes
Vfy	functional	R_5	R_6 / R_5	yes
e_{verify}	relation	R_1	R_1	yes
e_{sound}	relation	R_1	R_2 / R_1	yes
τ trapdoor	secret	—	*outside*	no
simulator, extractor	meta-theoretic	—	—	no

Table 2: The zero-knowledge instance under the lens. Paired entries are the black-box and the protocol-literate verifier. Only $\epsilon_V(w_P)$ is constant across every verifier profile, including the one that understands the construction completely.

Two edges, and only one of them is a computation. The verifier's structure has $e_{\text{verify}} : (\sigma, x, \pi) \to b$, which is the evaluation of (5), and $e_{\text{sound}} : b = 1 \to E_x$, which is not. The second holds only relative to a security model, and in this construction the knowledge soundness is statistical *against adversaries using a polynomial number of generic bilinear group operations* [15], on a setup whose trapdoor was discarded. An agent may resolve e_{verify} completely — it ran it — and leave e_{sound} unresolved.

Remark 7.3 (Closure that is complete without being total)*.* Read $r := E_x$ in Definition 7.4 and compute the role-required set for the verifier. Its paths run $x, \sigma, \pi \to \mathsf{Vfy} \to b \to e_{\text{sound}} \to E_x$, and w_P lies on none of them. So FullyM_V does not require $M_V(w_P) = 1$ — not by a policy exempting it, but because the quantifier in the definition never reaches it. $\text{Plan}_M(V)$ accordingly contains VERIFY (discharged by evaluating (5)), ESTABLISH DOMAIN CONDITIONS (the setup, and the security model under which the warrant is stated) and CONNECT TO CERTIFICATE (the passage from $b = 1$ to E_x), and it does not contain a demand to recover w_P. For the prover the same construction gives a different plan, in which w_P is role-required, since every path to π runs through it.

Two consequences are worth stating. First, a verifier whose Vfy sits in R_6 — executed, unread — is not entitled to a promotion on the strength of it: Hit_f requires its operator in $\mathsf{M} \cap \mathsf{K}$, and Remark 5.2 excludes exactly the $\mathsf{C} \cap \mathsf{K}$ case. The correct bit does not confer the account. Second, the configuration in which every vertex is resolved and e_{sound} is not is the situation Proposition 7.3 describes, here with an operational consequence: an unsound argument system yields accepting proofs, and $b = 1$ then warrants nothing.

The two instances of this section are therefore not variations on one theme. Hensel–Newton exhibits a plan whose obligations one discharges; the zero-knowledge argument exhibits a plan that is complete for its role while an artifact adjacent to it stays permanently outside the role-required set. Only the second shows why closure had to be made role-relative rather than global.

Remark 7.4 (What is not settled here)*.* Zero knowledge itself is not a property of a single execution: it compares the verifier's view against a simulated view, and in the perfect case asserts equality of two distributions. Properties relating several executions fall under what Clarkson and Schneider describe as hyperproperties, sets of trace properties [6], and the statistical and computational variants are quantitative and complexity-indexed respectively. Nothing in this paper classifies

them, and the region assignments in Table 2 are made per artifact, not per execution pair. The formal typing of the information boundary is left open.

7.9 A scoped remark on epistemic processing

Remark 7.5 (Epistemic processing). We propose the following conjecturally, and it is used nowhere below. Say that a computation is *epistemic* for a when some $e \in \Theta_a$ is role-required for r: a represented artifact whose reference is unresolved lies on a path to the result, so the computation must discriminate something it already carries.

Two limits are stated with it. First, this is a *subclass* and not an analysis of thinking: a computation all of whose artifacts are resolved may be reasoning by any ordinary standard, and nothing here says otherwise. Second, the seed is $\Theta_a = \mathsf{M}_a \cap \mathsf{C}_a$ and not pure C_a — an artifact of R_2 is opaque *and* unrepresented, so it is not something the agent could be discriminating.

One consequence follows from existing results and is worth recording because it is narrow. Assimilation acts on C_0 and leaves Θ_a fixed (Theorem 5.6), so it cannot remove the condition above: an agent does not assimilate its way out of epistemic processing. This is a statement about assimilation only. Whether another mechanism could reduce the condition is untouched, exactly as Remark 5.5 leaves the parallel question open for queries.

7.10 Relation to existing accounts

That cognition and computation may be directed at reducing uncertainty rather than at producing an output is not new here and is not claimed. Kirsh and Maglio describe a distinction between *epistemic* and *pragmatic* actions — actions performed to uncover information or to reduce the difficulty of a computation, as against actions that advance toward a goal state [22]; Lindley gives a formal counterpart, the expected information provided by an experiment [25]. The verbs of Definition 7.5 are meant in a sense close to their epistemic actions, and we do not claim the underlying framing of Remark 7.5 as ours.

Understanding as a relation between a system and a target is likewise a framing we adopt rather than propose; Chen et al. set out the landscape of options for specifying such a relation and explicitly decline to defend a particular account of it [4]. What the present section adds to that framing is narrower and structural: within *one* computation there are many typed targets, they stand in justification relations that carry their own status (Proposition 7.3), the functional ones retain K when resolved, and the unresolved remainder yields an obligation set with a proven floor. Whether that structure is the right one is a question for the applications; that it is not delivered by a target-relative framing alone is the claim made here.

Remark 7.6 (Beyond Hensel–Newton). Nothing in Definitions 7.1 and 7.4 is specific to refinement. The Euclidean algorithm instantiates the same seven slots — $D = \mathbb{Z}$, $U(a, b) = (b, a \bmod b)$, $\mathrm{Rsd}(a, b) = b$, P the strict decrease of b in $\mathbb{N}$, A the condition $b > 0$, Cert the divisibility check — and carries its own instance of Example 7.1: an agent may execute the update and read the residual while being unable to justify the edge on which the whole rests, that gcd is preserved by the step. Comprehension may also be domain-relative in a further sense: the same agent holding $U, \mathrm{Rsd}, P, \mathrm{Cert}$ resolved for modular inversion may fail to produce the corresponding artifacts for root lifting or for factor lifting, and the artifact at which the reconstruction fails — typically A, the hypothesis under which lifting is licensed, rather than U — is what the lens localises.

8 The Token-less Region and the Scope of the Minting Claim

This paper straddles two scales: it mixes a finite substrate (a finite artifact domain $\mathcal{A}$, a finite token vocabulary) with an infinite target (the concept space that difference-vector closure proposes, the unemptiable generative potential of K; Rem. 3.4). This section states the discipline that governs every crossing between the two. It introduces no new claim object; it states how the paper's own minting claim (Conj. 4.3) is to be read.

(i) The gap between placement and naming. [2] The correspondence between tokens and geometric positions is partial: the positions of $\mathcal{A}^+$ that carry *no* token are exactly the gap between what the geometry can *place* and what the vocabulary can *name*. Every token-less point reached by difference-vector closure (Rem. 3.3) sits in this gap until a mint (Def. 4.2) names it — and a point the verifier declines (Case C of Def. 4.1) is not named at all, and stays.

(ii) Minting plus recompute as partial closure. Minting a token at such a point and recomputing the envelopes (Conj. 4.3) closes part of that gap *at the representation layer*: it brings the *currently* token-less region into the denoted space and restores the token↔geometry correspondence there. Crucially, one such step closes *only* the region reached by that recompute; an uncovered remainder — the portion of $\mathcal{A}^+$ still token-less — persists after each step. This is the residual already flagged in Conj. 4.3, stated here as discipline rather than as a new mechanism. Two things feed the remainder and only one of them is a matter of coverage: points the recompute did not reach, and points that were reached and declined. Iterating the step is a strategy against the first and none at all against the second, since a declined point is not a mint target.

(iii) Refusal of finite→infinite inflation. We explicitly refuse to inflate finite evidence into an infinite-reachability result. Each mint admits one operator and covers a finite region; *no finite sequence of mints is claimed to reach all of the concept space.* The assertion "minting eventually reaches every concept" would be finite evidence for an infinite-trace property and is not made. The reachability-forever reading of K (Rem. 3.4) and the conditional unlocking claim (Conj. 4.5) are both held at CONJECTURED for exactly this reason.

(iv) Fixpoint vs. eternal frontier: left open. Whether iterating this step reaches a fixpoint (a stage after which no token-less reachable point remains, $\mathsf{F}\,\mathsf{G}\,\neg$gap-exists) or the frontier stays non-empty forever ($\mathsf{G}\,\mathsf{F}$ gap-exists) is left open; every trace satisfies exactly one of the two, and each is at once a liveness *and* a coliveness property, so no finite prefix determines either [2]. It is taken up in Section 11. We record only that resolving it requires a decomposition that states which part of the space remains uncovered, not more finite examples.

(v) How far "generation" is carried here. The word *generation* spans a ladder of seven distinct stages, and this paper reaches two of them. Naming all seven prevents the slide by which

[2] *Tokenless* here means: no token of the agent's current vocabulary denotes the point (Def. 4.2) — it is a scoping predicate on the pair (vocabulary, artefact position), not a claim about the tokenizer architecture. In particular, *tokenless* in this sense is distinct from *tokenizer-free* models such as CANINE [5], ByT5 [44], and the Byte Latent Transformer [38], which operate on bytes or characters without a subword tokenizer. The tokenizer-free family varies in how it operates on the byte stream (ByT5 uses a fixed byte vocabulary; the Byte Latent Transformer uses dynamic byte-patching based on next-byte entropy); in each case the model reads a fixed input alphabet (characters or bytes), and whether a point of $\mathcal{A}^+$ is tokenless in the sense used here depends on the admitted vocabulary of denoting tokens, not on how the input is segmented.

a constructed point is described as a concept, and a concept as an acquired one.

(1) *Geometric point construction.* A point is produced by Φ-closure. **Established**, under the ideal reading of Def. 6.1 fixed by Rem. 6.2: this is what cl_Φ does, by construction.

(2) *Candidate-concept designation.* The point is *called* a candidate concept. **Definitional only.** Nothing below licenses reading the designation as more than a name for a position; in particular a constructed point is not thereby a concept.

(3) *Semantic validation.* The candidate is shown to have semantic content. **Not established, and not attempted.** No criterion of semantic validity is supplied here, and Def. 4.1 is explicit that the verifier's acceptance is a decision for which the definition offers no criterion.

(4) *MCK classification.* The candidate receives an observer-relative label. **Available**, as the three cases of Def. 4.1; note that Case C leaves the point unclassified rather than assigning it to C.

(5) *Lexical realization.* A token realizes the candidate. **Not established.** Def. 4.2 states an *admissibility condition* for seating an operator, not an operation that produces one.

(6) *Acquisition.* The agent obtains and can use the representation. **Explicitly declined**, both here and in the foundations, whose step relation holds the operator basis fixed [3].

(7) *Capability expansion.* Reach grows. **Conjectural, and conditionally so** (Conj. 4.5); its negative half is settled, since under a fixed basis and a fixed realization a mint extends nothing.

Stages (1) and (4) are what this paper delivers. Stages (3), (5) and (6) are open, and (6) is open by declaration rather than by omission. The gap between (1) and (3) is where the empirical companion begins.

9 The Denotation Graph and the Kinds of Symbol

The foundations paper fixes the denotation map and the target set $T(s)$ (Sec. 2.3) and then defers two things by name: the axiom system governing the graph on which T is computed, and the classification of symbols by the shape of $T(s)$ [3]. This section supplies both. It develops its own primitives and does not rely on the geometric material of Sections 3 and 6; the dependence runs the other way, Section 3 invoking Cor. 9.11 where it states what a recoverable realization would and would not deliver.

The section is deliberately conservative. Every MCK primitive it uses — Θ, the denotation map, T_{direct}, T_{indirect}, T, the classified domain $\mathcal{A}^+$, the regions — is recalled from [3] and cited, never restated as new. What is added is of three kinds: the graph carrying the deferred axioms (Sec. 9.1), the enumeration of symbol kinds (Secs. 9.3–9.5), and two negative results that mark the boundary of the apparatus (Secs. 9.6–9.7).

9.1 The denotation graph

Definition 9.1 (Denotation graph)**.** *Fix an artifact domain $\mathcal{A}$ with its predicates and regions, and let $\Theta = \mathsf{M} \cap \mathsf{C}$ be as recalled in Sec. 2.3. The* denotation graph *is the two-sorted directed graph*

$$G_D \;=\; (\Theta \sqcup \mathcal{R},\, E_D), \qquad E_D \;\subseteq\; \Theta \times (\Theta \sqcup \mathcal{R}),$$

whose first sort is the set of symbols Θ *and whose second sort* $\mathcal{R} \subseteq \mathcal{A}^+$ *is the set of* terminal referents*. An element of* E_D *is a* denotation edge*. The graph is subject to five axioms:*

(D1) Domain restriction. *Every edge leaves a symbol: only* Θ*-artifacts denote.*

(D2) Codomain confinement. *Every referent lies in the classified domain:* $\mathcal{R} \subseteq \mathcal{A}^+$*. Which regions of* $\mathcal{A}^+$ *may host a referent is a separate question, settled by stipulation in Rem. 9.2 and not by this axiom.*

(D3) Sort disjointness. $\Theta \cap \mathcal{R} = \emptyset$*: no referent is itself a symbol.*

(D4) Mediation. *A referent counts as reached indirectly from* s *only along a directed path all of whose internal nodes lie in* Θ*.*

(D5) Staticity. E_D *does not vary under the region dynamics.*

Two of these five axioms are recalled unchanged: (D2) is the codomain clause of the denotation map and (D5) its staticity clause, each read graph-theoretically and each recalled from [3] (Sec. 2.3). Two more are recalled but constrain nothing beyond the signature. (D1) is the domain-restriction clause of the same map, and it is also the typing $E_D \subseteq \Theta \times (\Theta \sqcup \mathcal{R})$ already imposed above. (D4) is the mediation clause of the recalled target set; given (D1) and (D3) it is automatic, since a node with an outgoing edge lies in Θ by typing, so every internal node of a directed path is symbolic of its own accord. Both are stated because the recalled definition states them and the reader should see where they land, not because they add content. Terminality — no edge leaves $\mathcal{R}$ — is likewise a consequence rather than an axiom: by (D3) an element of $\mathcal{R}$ is not a symbol, and by the typing of E_D no edge has a non-symbolic source.

(D3) is the axiom added here, and it is a genuine addition rather than a rendering. The recalled apparatus places no constraint on whether a referent may itself be a symbol; we stipulate that it may not. It is the second of the two departures this section makes from the recalled map, the first being the edge relation itself.

E_D *extends* the recalled denotation map rather than transcribing it, and the extension is in three places, each of which this section needs. First, the recalled map is a function into the referents and therefore supplies no edge whose head is a symbol; the mediating edges on which T_{indirect} depends are not in its range, so a graph carrying them is not obtained by reading that map as a relation. Second, the recalled map is single-valued, whereas E_D permits out-degree ≥ 2, which the plural labels of Table 3 require; the recalled definition anticipates this in collecting the full set of a symbol's targets into $T(s)$ rather than into the map. Third, the recalled map is total on Θ, whereas E_D permits out-degree 0, which Prop. 9.5 and label 0 require. On G_D the recalled target set (3) reads

$$T_{\text{direct}}(s) = \{\, r \in \mathcal{R} : (s, r) \in E_D \,\}, \qquad T_{\text{indirect}}(s) = \left\{ r \in \mathcal{R} : \begin{array}{l} s \to s_1 \to \cdots \to s_k \to r \text{ in } G_D, \\ k \geq 1,\ s_1, \ldots, s_k \in \Theta \end{array} \right\},$$

with $T(s) = T_{\text{direct}}(s) \cup T_{\text{indirect}}(s)$ as in [3]. The two clauses of the recalled target set are carried over verbatim; what carries them is the extended edge relation just described, not the recalled map, which cannot supply a mediating edge.

One consequence of (D3) should be recorded rather than left implicit. Since a symbol is never a referent, $T_{\text{direct}}(s)$ contains no symbolic successor of s: a symbolic successor contributes to $T(s)$ only through the referents it reaches, and only through T_{indirect}. A treatment admitting symbols as referents would compute a different $T_{\text{direct}}(s)$, a different $|T(s)|$, and hence a different class label, for every symbol with a symbolic successor — that is, for exactly the indirect and mixed symbols the taxonomy below exists to separate.

9.2 Denotation as a family of regional blocks

The edge relation can be read region-wise, and doing so makes visible a structural bound on how much of the transformation space denotation is able to occupy at all. Nothing in this subsection changes G_D; it decomposes the relation already defined.

Definition 9.2 (Region-indexed denotation blocks)**.** *For $i, j \in \{1, \ldots, 7\}$ let*

$$d_{i \to j} \;=\; E_D \cap (R_i \times R_j) \;\subseteq\; R_i \times R_j$$

be the denotation block from R_i to R_j*: the restriction of the edge relation to edges whose source lies in R_i and whose head lies in R_j. Each $d_{i \to j}$ is* partial *— not every artifact of R_i need have an edge — and* relational *rather than functional, since E_D permits out-degree ≥ 2. The family $\{\, d_{i \to j} \,\}_{i,j}$ recovers E_D, the regions being a partition of $\mathcal{A}^+$: $E_D = \bigcup_{i,j} d_{i \to j}$.*

On the name. *These are called* blocks *and not morphisms, and the choice is not stylistic: neither of the two structures a morphism requires is available.* There are no identities. *An identity would require $(x, x) \in E_D$ for every $x \in R_i$. For the five regions outside Θ this is impossible outright, since $d_{i \to i} = \emptyset$ there by Prop. 9.3. For the two regions inside Θ the blocks $d_{4 \to 4}$ and $d_{7 \to 7}$ may well be non-empty — symbolic heads are admitted — but nothing requires them to contain the diagonal, and an identity block would say that* every *symbol of the region denotes itself, which no axiom supplies and which (D4) would render inert, a self-loop contributing no referent to $T(s)$.* And there is no composition, *since E_D is not transitive and a two-edge path $s \to s' \to r$ is not itself an edge; indeed (D4) exists precisely because such a path must be treated separately from an edge. What the family is, exactly, is the block decomposition of a single relation under a partition of its carrier. The decomposition is faithful; the categorical vocabulary would not be.*

Two node kinds are distinguished by out-degree rather than by sort. A node is a leaf *of G_D when its out-degree is* 0*. By (D3) every element of $\mathcal{R}$ is a leaf; by the out-degree-*0 *allowance recorded above a* symbol *may also be one, and a symbol that is a leaf is a Θ-artifact denoting nothing — exactly the label-*0 *case of Prop. 9.5. Leaves therefore occur in R_4 and R_7 as well as in the referent regions, and leafhood is a property of a node's edges and not of its region.*

Proposition 9.3 (Denotation occupies two source rows)**.** *$d_{i \to j} = \emptyset$ for every j whenever $i \notin \{4, 7\}$. Consequently the denotation layer can carry edges in at most 2 of the 7 source rows of the region-to-region table, and the five rows R_1, R_2, R_3, R_5, R_6 are closed to it outright.*

Proof. By (D1) every edge of E_D leaves a symbol, so $E_D \subseteq \Theta \times (\Theta \sqcup \mathcal{R})$ and any edge with source in R_i requires $R_i \cap \Theta \neq \emptyset$. By (2), $\Theta = R_4 \cup R_7$, and the regions are pairwise disjoint, so $R_i \cap \Theta = \emptyset$ for $i \in \{1, 2, 3, 5, 6\}$. Hence $d_{i \to j} = E_D \cap (R_i \times R_j)$ is empty for those i and every j. □

Remark 9.1 (What the bound says, and why it is worth stating)*.* The proposition is elementary, and its value is that it settles by counting a question that would otherwise be settled by emphasis. Fig. 4(b) shows the two occupied rows against the five empty ones, and panel (a) beside it shows why they are not the only arrows one might draw over the same cells.

Denotation is the semantic relation this paper develops at length, and length invites the reading that it is *the* semantic relation — that a transformation between regions is, in the end, a denotational one. It cannot be. Its source is confined to Θ by (D1), so five of the seven source regions admit no denotation edge whatever, and no enrichment of $\mathcal{R}$ or of the target side changes that: the restriction is on where an edge may *begin*.

Two things the bound does *not* say are worth recording alongside it, since both are easy to read into a source-side restriction.

The target side is nearly unrestricted. By Rem. 9.2 the admissible heads are $R_1, R_2, R_4, R_5, R_6, R_7$ — six of the seven regions — and nothing in the definition restricts R_4 and R_7 to different targets, so each source class may reach any admissible region. Two of those targets deserve saying aloud. R_2 is an *opaque* head: a symbol may denote something the observer does not understand, so being denoted implies nothing about M. R_6 is an *opaque and functional* head: being denoted implies nothing about K either. *Denotational competence is a property of the source; the epistemic and functional status of the target is independent of it.*

Admissibility is not instantiation. That a regional pair is admissible says nothing about whether any edge of that type exists, and out-degree 0 is permitted throughout Θ — it is label 0 of Table 3. Nor is regional admissibility a semantic classification: it fixes where the endpoints of an edge may lie and leaves open which relations of that type are meaningful, realizable, or instantiated.

The contrast with the discovery predicate is exact and needs no new machinery. Def. 5.2 evaluates $f(m, k)$ at $m \in M_0$, and (1) identifies M_0 with R_1. So R_1 carries outgoing arrows of that predicate while $d_{1\to j} = \emptyset$ for every j: *the same region is a leaf under denotation and is not a leaf under functional application.* Being a leaf is therefore relative to the relation chosen, and two relations already present in this paper disagree about R_1. Neither is more fundamental than the other; they are different relations, and the region table records the effects of both.

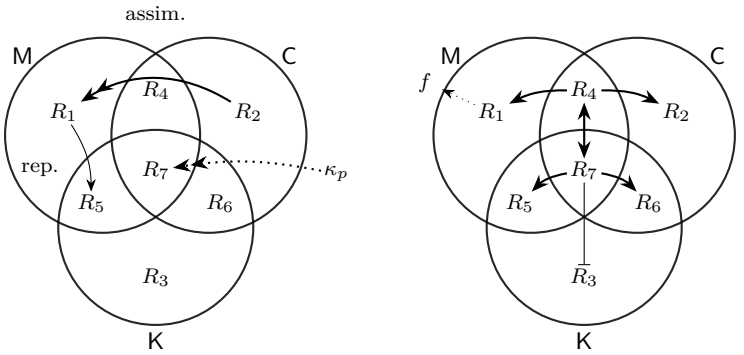
Scope. The bound is about G_D as axiomatized here. A treatment admitting non-symbolic sources would not be a treatment of this graph, and Prop. 9.10 already shows that the region data do not determine E_D in any case.

Remark 9.2 (Admissible heads, and one stipulation). On the target side the axioms leave a choice, and it is recorded here rather than left to the reader. By the typing of E_D a head lies in $\Theta \sqcup \mathcal{R}$; by (D3) the referent sort is disjoint from Θ, so $\mathcal{R} \subseteq \mathcal{A}^+ \setminus \Theta = R_1 \cup R_2 \cup R_3 \cup R_5 \cup R_6$, while symbolic heads — the mediating edges of (D4) — lie in $R_4 \cup R_7$. Heads are thus a priori available in all seven regions.

We stipulate that $\mathcal{R} \cap R_3 = \emptyset$, so that admissible heads lie in $R_1, R_2, R_4, R_5, R_6, R_7$. The reason is that $R_3 = \mathsf{K} \setminus (\mathsf{M} \cup \mathsf{C})$ carries neither meaning nor accessible structure, so a referent placed there would contribute nothing to what its symbol denotes: it would enlarge $T(s)$ by an element about which the target set can say nothing. This is a stipulation and not a consequence — (D2) confines referents to the classified domain, and R_3 is inside it, the more so because this paper declines the suspension convention (Sec. 2.4) and therefore cannot exclude R_3 by inheriting that convention. A treatment admitting R_3 referents is coherent and would change the class labels of Table 3 for exactly those symbols reaching one; nothing below depends on the choice, and it is reversible.

Remark 9.3 (Which indices the graph carries, and which it does not). A denotation graph invites indexing — by observer, context, time, modality, confidence, provenance, or interpretive regime — and the temptation is to carry all of them for generality. We add an index only where a statement this paper actually makes would otherwise be unstatable, and record the rest as declined.

Carried. (i) The *sort* distinction between Θ and $\mathcal{R}$ is required by (D4): the phrase "all of whose internal nodes are symbolic" has no content unless nodes are sorted. (ii) An *agent* index is the one exception to the criterion just stated, and we flag it as such: no statement in this section is made in terms of G_D^a, so the index is not needed to make anything here statable. It is carried for downstream compatibility, because the rest of this paper relativizes — Θ_a, R_4^a and R_7^a appear in Def. 5.5 and Thm. 5.6 — and a graph that could not be read as G_D^a would not compose with that material. Nothing is added to make this available: G_D^a is obtained by relativizing Θ and the predicates to a exactly as the regions already are. (iii) A *stage* index τ is required by Sec. 9.7 and by nothing else: staticity is a claim relating one stage to the next, so it cannot be stated without stages. Under (D5) the index is constant and therefore eliminable, and we suppress it everywhere



(a) three kinds of arrow (b) the denotes case

Figure 4: Regional relations and the denotation case, drawn on the same cells; the circle placement is the convention of [3]. *The two panels carry separate legends, and the reclassification arrow of panel (a) is double-headed precisely so that no transition is drawn in the single-headed style panel (b) reserves for relations.* Panel (a) shows three relations of three different kinds, one arrow each: a representative source–target pair, the assimilation reclassification $R_2 \to R_1$ (Def. 5.2), and the seating of an operator κ_p (Def. 4.2), drawn from outside the cells because it is an admissibility condition on an operator held elsewhere. Panel (b) shows the denotation edges of Def. 9.2 as a fan of *regional types*: sources lie only in $\Theta = R_4 \cup R_7$ (Prop. 9.3) and the fan is *complete over admissible target classes and representative over ordered source–target pairs*: it touches all six admissible heads. *Branches are relation types, not regional transitions:* a branch says a symbol in one region may denote an artifact in another, never that an artifact moves between them, and it asserts no instantiation. The barred arrow towards R_3 is the panel's one non-standard mark and denotes a head excluded by the *stipulation* of Rem. 9.2 — reversible and local to this manuscript. The dotted stub at R_1 carries no denotation: R_1 is a legitimate first argument position of f in Def. 5.2, so being a source is relative to the relation chosen.

except in that subsection.

Declined. A *context* index would be needed to state that an utterance situation selects among targets; no statement here selects, so it is not added (Rem. 9.5). A *confidence* index would be needed to state that one target is likelier than another; a *modality* index to state that a target is possible rather than actual; a *provenance* index to state who introduced an edge; an *interpretive-regime* index to state that two readings of the same graph disagree. None of these supports a statement made in this paper, and each would have to be discharged by further axioms, so each is left out. This is a scope decision, not a claim that the corresponding structures are unavailable.

9.3 Denotation modes

Definition 9.4 (Denotation mode)**.** *The* mode *of a symbol $s \in \Theta$ is determined by which of the two components of $T(s)$ is inhabited:*

direct	$T_{\text{direct}}(s) \neq \emptyset$,	$T_{\text{indirect}}(s) = \emptyset$;
indirect	$T_{\text{direct}}(s) = \emptyset$,	$T_{\text{indirect}}(s) \neq \emptyset$;
mixed	$T_{\text{direct}}(s) \neq \emptyset$,	$T_{\text{indirect}}(s) \neq \emptyset$;
non-denoting	$T_{\text{direct}}(s) = \emptyset$,	$T_{\text{indirect}}(s) = \emptyset$.

Proposition 9.5 (The four modes partition Θ)**.** *Every symbol has exactly one mode, so the four modes of Def. 9.4 partition Θ. Moreover a symbol is non-denoting if and only if $T(s) = \emptyset$, and this includes every symbol of out-degree 0 in G_D.*

Proof. The four cases are the four truth-assignments to the pair of independent conditions $T_{\text{direct}}(s) \neq \emptyset$, $T_{\text{indirect}}(s) \neq \emptyset$; being an enumeration of a two-element Boolean product, they are mutually exclusive and jointly exhaustive, and each symbol realizes exactly one. Since $T(s) = T_{\text{direct}}(s) \cup T_{\text{indirect}}(s)$, the union is empty precisely when both components are, which is the fourth case. If s has no outgoing edge then $T_{\text{direct}}(s) = \emptyset$ immediately, and no path of the form required by (D4) begins at s, so $T_{\text{indirect}}(s) = \emptyset$ as well. □

Remark 9.4 (Modes by provenance, not by successor sort)*.* Def. 9.4 reads the mode off the *targets*. An implementation may instead read it off the *successors*, calling s direct, indirect, mixed or non-denoting according to whether it has a referent successor, a symbol successor, both, or neither. Write d for $T_{\text{direct}}(s) \neq \emptyset$, i for $T_{\text{indirect}}(s) \neq \emptyset$, and g for "s has a symbol successor". By (D3) and the typing of E_D, d holds exactly when s has a referent successor, and i implies g, since an indirect path begins with a symbolic edge.

The two readings therefore agree whenever $\neg g$ — both give direct if d and non-denoting otherwise — and whenever $g \wedge i$ — both give mixed if d and indirect otherwise. They diverge on exactly the condition $g \wedge \neg i$: a symbol with a symbol successor whose symbolic successors reach no referent at all. That condition splits in two. With d, the successor reading gives mixed and Def. 9.4 gives direct; without d, the successor reading gives indirect and Def. 9.4 gives non-denoting. We take the target reading because it is the one under which Prop. 9.5 holds without a side condition, and because T — not the successor relation — is the object the foundations paper defers to us.

A hybrid of the two readings is easy to write by accident and worth warning against: taking mixedness from the successors while taking indirectness and non-denotation from the targets, and requiring positive out-degree for non-denotation, leaves a symbol of out-degree 0 matching no case at all. That combination contradicts the final clause of Prop. 9.5, which assigns such a symbol label 0. The proposition is stated for the target reading and nothing here uses the successor one.

9.4 Acceptations

Definition 9.6 (Acceptation, specificity, plurality)**.** *An* acceptation *of a symbol $s \in \Theta$ is an element of its target set $T(s)$: acceptations are targets, and two acceptations are distinct exactly when they are distinct referents. A symbol is* specific *when $|T(s)| = 1$,* plural *when $|T(s)| \geq 2$, and non-denoting when $|T(s)| = 0$ (Prop. 9.5). A plural symbol is one with several acceptations, $T(s) = \{t_1, \ldots, t_n\}$ with $n \geq 2$.*

Remark 9.5 (An acceptation is a target, not a path). Def. 9.6 settles a choice among four candidates, and the choice is forced by the recalled apparatus rather than made freely.

Not a path. $T(s)$ is a set of referents, so two distinct denotation paths from s terminating at the same referent contribute one acceptation, not two. This is not an incidental consequence: it is why recording multiple routes to a shared target is a question distinct from counting acceptations, and why a refinement that tracked routes would add information the acceptation count does not carry.

Not an interpretation function and not an equivalence class. An interpretation function would be an assignment selecting a target; the apparatus contains no such selector, as (D5) leaves nothing to select on. An equivalence class would require a relation identifying targets, and the graph supplies none: distinct referents are distinct nodes.

Plurality, not uncertainty, contradiction, or underspecification. These four readings of ambiguity are different, and the apparatus supports exactly one. $T(s)$ is a set whose members are all simultaneously reachable and none preferred; the several acceptations of a plural symbol therefore remain simultaneously active, and none is selected by context, because no context index is carried (Rem. 9.3). Reading plurality as *uncertainty* would require a measure on $T(s)$; as *contradiction*, an inconsistency relation on referents under which two targets may conflict; as *underspecification*, a refinement order on referents under which a target may be a coarsening of another. The apparatus supplies no measure, no inconsistency relation and no refinement order, and we add none. Plurality here means multiplicity of simultaneously admitted targets, and nothing stronger.

9.5 Redundancy and the canonical symbol classes

Definition 9.7 (Shared targets and redundancy)**.** *Fix a set $\Sigma \subseteq \Theta$ of symbols against which comparison is made. A referent r is* shared *when $r \in T(s')$ for at least two distinct $s' \in \Sigma$, and* unique *otherwise. A symbol s with $T(s) \neq \emptyset$ is* fully redundant *when every element of $T(s)$ is shared,* partially redundant *when $T(s)$ contains both a shared and a unique element, and* non-redundant *when every element is unique. The comparison set Σ is a parameter of the definition and must be declared: for a symbol of direct mode it is customary to compare direct targets only, and otherwise to compare full target sets.*

Definition 9.8 (Canonical symbol classes)**.** *The* canonical classes *are the fourteen labels of Table 3, obtained by refining the mode (Def. 9.4) by specificity (Def. 9.6) and then by redundancy (Def. 9.7), with label* 10 *taking precedence over the redundancy split in the mixed mode.*

Remark 9.6 (Two asymmetries, and one gap). The enumeration is not the full product of its three coordinates, and the places where it is not are worth stating rather than presenting the table as uniform.

First, only the direct mode splits *specific* symbols by redundancy (labels 1 and 2); the indirect mode collapses that split into the single label 6. Second, label 10 is per-component specificity — one direct and one indirect target — and not global specificity, despite reading like the latter. Third,

Label	Mode	Specificity	Redundancy
0	non-denoting	$T(s) = \emptyset$	—
1	direct	specific	non-redundant
2	direct	specific	fully redundant
3	direct	plural	non-redundant
4	direct	plural	partially redundant
5	direct	plural	fully redundant
6	indirect	specific	—
7	indirect	plural	non-redundant
8	indirect	plural	partially redundant
9	indirect	plural	fully redundant
10	mixed	plural, one target per component	—
11	mixed	plural	non-redundant
12	mixed	plural	partially redundant
13	mixed	plural	fully redundant

Table 3: The fourteen canonical symbol classes of Def. 9.8. Label 10 is the mixed symbol with $|T_{\text{direct}}(s)| = |T_{\text{indirect}}(s)| = 1$ and $|T(s)| = 2$; it is tested before the redundancy split, so a symbol meeting its condition receives it whatever its redundancy. Dashes mark a cell the enumeration does not split.

and consequently, the fourteen labels do *not* cover the mixed mode: a mixed symbol with $|T(s)| = 1$, that is one whose single referent is reached both directly and through mediation, satisfies no row of Table 3. Checking the rows settles this: rows 1–5 and 6–9 fix a mode other than mixed, row 0 requires $T(s) = \emptyset$, row 10 requires $|T(s)| = 2$, and rows 11–13 require plurality, which $|T(s)| = 1$ denies. Such symbols are covered only by the structural refinements below. Any statement that the canonical classes are a complete classification of symbols is therefore false as it stands, and the correct statement is that they classify every symbol except the globally specific mixed ones.

We do not repair the gap here. A refinement covering the globally specific mixed case would have to split on the shape of the mediation, and several such splits suggest themselves; none is defined in this paper, and naming them without definitions would put the completeness of the classification on objects a reader cannot check. The honest statement is therefore the negative one already made — Table 3 is exhaustive over the modes it covers and silent on the globally specific mixed symbol — and closing that silence is left open. The enumeration, both canonical and refined, is the one our classifier implementation uses; it is offered here as a definition, and no claim in this paper rests on any measurement made with that implementation.

9.6 Θ is not characterized by plurality

The vocabulary invites an inference. Θ is the structured ambiguity domain, plural symbols are the ambiguous ones, so plurality ought to be what Θ-membership consists in. The inference is unsound, and since it would if adopted silently license a redefinition of a term fixed elsewhere, we record why.

Proposition 9.9 (Plurality is neither a criterion for, nor a consequence of, Θ-membership)**.** *Let* $\Theta = \mathsf{M} \cap \mathsf{C}$ *and let* T *be as recalled in Sec. 2.3. Then:*

1. *$|T(s)| \geq 2$ is not a criterion of Θ-membership. The target set is defined only for $s \in \Theta$, so the condition presupposes Θ-membership instead of establishing it, and the implication "$|T(s)| \geq 2$ implies $s \in \Theta$" holds vacuously and carries no information. The same holds of "every plural symbol lies in Θ".*

2. *The converse fails, and strictly: $\{\, s \in \Theta : |T(s)| \geq 2 \,\} \subsetneq \Theta$ whenever a specific or a non-denoting symbol exists.*

3. *Θ-membership is a function of $(M(x), C(x))$ alone. Consequently no alteration of $T(s)$ alone can move an artifact into or out of Θ; in particular, were a disambiguating operation added that reduced $|T(s)|$ to 1, the symbol would remain a symbol.*

Proof. (1) The target set is introduced as the target set *of a symbol* $s \in \Theta$ [3], and by (D1) only Θ-artifacts have outgoing denotation edges. The expression $T(x)$ is therefore undefined for $x \notin \Theta$, so $|T(x)| \geq 2$ is not a predicate on $\mathcal{A}$ whose extension could be compared with Θ. Restricted to its domain of definition the implication is trivially true, which is the vacuity asserted.

(2) Specific symbols exist: the foundations paper exhibits the classical single-referent case as a symbol whose target set is the singleton of a fully resolved referent, for instance an R_1 artifact [3]. Non-denoting symbols are admitted by Prop. 9.5, a symbol of out-degree 0 being one. Either witnesses an element of Θ outside the left-hand set, and the inclusion is immediate from the definitions.

(3) By $\Theta = \mathsf{M} \cap \mathsf{C}$, membership is decided by the two predicate values $M(x)$ and $C(x)$ and by nothing else. By Prop. 9.10 below, T is not a function of those predicates, so the two are logically independent: fixing the predicates leaves T free, and altering T while holding the predicates fixed leaves every region, and Θ, unchanged. □

Proposition 9.10 (The regions do not determine the denotation graph)**.** *There are two denotation graphs over the same artifact domain, with identical predicates M, C, K — hence identical regions $R_1, \ldots, R_7$ and identical Θ — and a symbol whose target set differs between them.*

Proof. Take $\mathcal{A} = \{s, t_1, t_2\}$ with $M(s) = C(s) = 1$, $K(s) = 0$, so $s \in R_4 \subseteq \Theta$, and $M(t_i) = 1$, $C(t_i) = K(t_i) = 0$, so $t_1, t_2 \in R_1 \subseteq \mathcal{A}^+$ and both are admissible terminal referents by (D2). Put $E_D = \{(s, t_1)\}$ and $E'_D = \{(s, t_2)\}$. Both satisfy (D1)–(D5): each edge leaves the only symbol, each target is classified, $\Theta = \{s\}$ and $\mathcal{R} = \{t_1, t_2\}$ are disjoint, no path has an internal node, and neither edge set varies. The predicate assignment is the same in both, so the seven regions and Θ coincide. But $T(s) = \{t_1\}$ while $T'(s) = \{t_2\}$. □

Remark 9.7 (What the separation buys). Prop. 9.10 is elementary, and its value is that it fixes two things that would otherwise be easy to assume.

First, it explains why staticity is not read off the region data. Since those data leave T free, no argument from the regions can *determine* T, and so staticity cannot be derived from them; the claim is about that route and not about every route, since a rule constraining when T may change — rather than what its value is — would be untouched by the countermodel, in which both graphs are separately static. Clause (D5) is a stipulation of the foundations paper [3], and Sec. 9.7 treats it as one rather than as a theorem.

Second, it delimits Thm. 5.6. That theorem shows Θ_a, R_4^a and R_7^a to be fixed under assimilation: what it constrains is the *domain* of denotation, the set of symbols, and thereby the region structure. It is silent about targets, and by the separation it could not be otherwise — no statement about the predicates constrains T. Domain invariance and denotation invariance are two properties, and the first does not imply the second.

Corollary 9.11 (Label recovery does not yield denotation)**.** *Let $\mathcal{D}$ and $\mathcal{D}'$ be the two denotation structures of Prop. 9.10: the same artifact domain, identical predicates M, C, K — hence identical regions and identical Θ — and $T(s) \neq T'(s)$. Let $e\colon \mathcal{A}^+ \to E$ be any realization map (Def. 3.1); since the domain and the predicates coincide, the same e realizes both. Then no map*

$$F\colon E \times \{0,1\}^3 \longrightarrow \mathcal{P}(\mathcal{R}), \qquad x \longmapsto F\big(e(x), \lambda(x)\big),$$

satisfies $F(e(s), \lambda(s)) = T(s)$ in $\mathcal{D}$ and $F(e(s), \lambda(s)) = T'(s)$ in $\mathcal{D}'$.

Proof. The pair $(e(s), \lambda(s))$ is the same in both structures — e is the same map and $\lambda(s)$ is determined by $M(s), C(s), K(s)$, which coincide by hypothesis. So F returns one value on that argument. But $T(s) \neq T'(s)$, so it cannot equal both. □

Remark 9.8 (What this buys, and what it does not)*.* The corollary bears directly on the strongest realization property named in Def. 3.2. Suppose a *recoverable* realization were exhibited — a map Ψ_a reconstructing the label triple perfectly from a position and its context. Its output is λ_a, and λ_a is exactly what the two structures share. *Perfect geometric recovery of the MCK labels would therefore still not deliver denotation.* Label recovery and semantic recovery are two problems, and solving the first leaves the second untouched; a realization programme that reached recoverability would have reached a strictly weaker goal than it might appear to have.

The scope of the impossibility must be stated precisely, because it is easy to inflate. What is shown is that denotation is not a function *of the information* $(e(x), \lambda(x))$. It is *not* shown that no enriched representation could carry denotation: a carrier whose points recorded denotation edges, or an observer parameter supplying them, would evade the argument immediately and trivially — and that is the point. Denotation must be *carried* as additional structure, or recovered from additional contextual or observer information; it cannot be read off the region data, nor off any realization determined by them. This is the realization-level reading of Prop. 9.10, and it inherits that proposition's scope exactly: it is a statement about one route, not about every route.

9.7 Static and dynamic denotation

Definition 9.12 (Stage-indexed targets)**.** *Let τ range over stages (Rem. 9.3) and write $T_\tau(s)$ for the target set computed on the stage-τ denotation graph. Denotation is* static *when*

$$T_{\tau+1}(s) \;=\; T_\tau(s) \qquad \textit{for every symbol } s \textit{ and every stage } \tau,$$

and dynamic *when it is governed by an update rule*

$$T_{\tau+1}(s) \;=\; \mathrm{Update}\big(T_\tau(s),\, e_\tau,\, c_\tau,\, o_\tau\big),$$

where e_τ is the event occurring at stage τ, c_τ the context, and o_τ the observer. The null event $\perp$ *is the event that adds no edge and removes none; this is a property of $\perp$ itself, stated here so that its effect on the update rule is derived rather than assumed.*

The framework stipulates the static case: clause (D5) of Def. 9.1, which is the staticity clause of the denotation map recalled in Sec. 2.3 from [3]. *This paper does not adopt the dynamic case.* It is stated here because the deferral we are discharging concerns the axioms of the denotation graph, and the reader is entitled to know which of them is a choice; adopting the dynamic reading would contradict a clause of a definition fixed elsewhere, which a downstream paper may not do. What follows therefore states what a dynamic version would have to guarantee, and stops there.

The candidate events are easy to enumerate and that is exactly the danger: new evidence about a referent, a new observer, a new context of use, an explicit stipulative definition, a metaphorical extension, adoption by a community, a correction of a previous assignment, an adversarial reframing, and a refinement of an existing target into a finer one. Nine such events, an update rule taking them, and no further constraint would let T move anywhere at all: an update rule without invariants is not a theory of semantic change but a licence for it. The content of a dynamic reading lies entirely in what it forbids.

Definition 9.13 (Admissible update)**.** *An update rule* Update *is* admissible *when every stage of the resulting family of denotation graphs satisfies (D1)–(D4) of Def. 9.1 — so that domain restriction, codomain confinement, sort disjointness and mediation are preserved — and in addition:*

(U1) Domain preservation. *An update acts on E_D only: it adds and removes denotation edges and does not alter the predicates M, C, K. Three consequences follow, and they must be kept apart rather than treated as one clause. Membership in Θ is preserved, so a symbol remains a symbol and a non-symbol acquires no targets. Each individual predicate is preserved, none being touched. And the region of every artifact is preserved, the regions being determined by the predicate triple.* The implications run in that direction only. *Preserving $\Theta = \mathsf{M} \cap \mathsf{C}$ as a set does **not** preserve the individual predicates: an artifact carried from R_4 to R_7 stays in Θ throughout while K flips, and likewise $R_1 \to R_5$ and $R_2 \to R_6$ each fix Θ — vacuously, both being outside it — while changing K. The clause stated is therefore the strong one, and the weaker readings are consequences of it rather than reformulations.*

(U2) Derived-component coherence. *T_{indirect} is recomputed from the updated edge set rather than set independently, so that $T = T_{\text{direct}} \cup T_{\text{indirect}}$ continues to hold and a change to one symbol's edges propagates to the target sets of the symbols that reach it.*

(U3) Groundedness. *Every difference between $T_\tau(s)$ and $T_{\tau+1}(s)$ is licensed by the event e_τ: each acquired target is contributed by an edge that e_τ adds, and each lost target by an edge that e_τ removes.*

Proposition 9.14 (Staticity is the null-event case of any grounded dynamics)**.** *Let* Update *be admissible in the sense of Def. 9.13 and let the event stream be null, $e_\tau = \bot$ for every τ. Then $T_\tau(s) = T_0(s)$ for every symbol s and every stage τ: denotation is static in the sense of Def. 9.12.*

Proof. By induction on τ. The base case is an identity. For the step, suppose $T_\tau = T_0$. The null event adds no edge and removes none (Def. 9.12), so by (U3) there is no edge available to contribute an acquired target and none to account for a lost one; hence no target is acquired and none is lost, and $T_{\tau+1}(s) = T_\tau(s)$ for every s. By (U2) the indirect component is recomputed from an unchanged edge set and so is itself unchanged, and by (U1) the symbol domain over which the equality is asserted is the same at both stages. Hence $T_{\tau+1} = T_\tau = T_0$. □

Remark 9.9 (The bearing on the stipulation)*.* Prop. 9.14 is the reason the distinction can be stated here without disturbing anything. A grounded dynamics *recovers* the static stipulation as its null-event case: on a null event stream the apparatus of Def. 9.13 reproduces exactly what (D5) asserts. That is the whole of the compatibility, and it should not be read for more. A treatment adopting a non-null event stream is a different matter: any $e_\tau \neq \bot$ that adds or removes an edge makes E_D vary, which (D5) forbids outright. Such a treatment would have to *supersede* (D5) explicitly, stating that it does so and on what authority, and that is outside the present paper. What such a treatment would additionally owe, and what we do not supply, is a semantics for the events

themselves: (U3) refers a change to an event without saying what makes an event licence one edge rather than another. Until that is given, the enumeration of update events above is a list of names, and we have deliberately not built on it.

10 Related Work

This section covers the interfaces this paper actually uses. It is deliberately narrow: adjacent literatures that the manuscript neither imports from nor constrains are left out rather than surveyed, and where our relation to a body of work is a distinction rather than a reduction we say so in those words. Secs. 10.7–10.10 record boundaries with adjacent literatures only so that they are not mistaken for imports.

10.1 Having, understanding, and being able to use

The distinction the three predicates draw — between an artifact an agent merely holds, one whose content it grasps, and one it can put to work — is not new as a concern, and two lines in the epistemic-logic literature address it directly enough to fix what is and is not ours.

The first is *awareness*. Possible-worlds semantics makes an agent logically omniscient, and the standard repair introduces a syntactic awareness filter: an agent must be aware of a concept before it can hold beliefs about it [12]. The second is *algorithmic knowledge*, where an agent is equipped with a knowledge algorithm and counts as knowing a proposition when that algorithm answers affirmatively, in the formulation of Halpern, Moses and Vardi [17] as presented by Halpern and Pucella [18]; knowledge is then bounded by what the agent can actually compute.

Both separate possession from availability, and the present framework is continuous with that concern rather than a departure from it. Two differences are worth stating precisely, because they are what makes the classification here a different object and not a variant. First, awareness and algorithmic knowledge are filters on a *propositional* attitude: they say which formulas an agent holds. The predicates here apply to *artifacts*, and an artifact is classified without being assigned a truth value at all — an opaque operator in R_6 is not a proposition the agent fails to know but a thing it can run and cannot read. Second, and consequently, the classification is a partition of a domain rather than a restriction on a modality: the seven regions arise as the cells of three predicates and are not defined by any accessibility relation. *We claim no embedding of either framework into this one, and none in the other direction*; the relation is a shared concern with different carriers, and calling it a generalization would require a translation that is not given here.

10.2 The imported structure, and what it does not supply

The region partition, the three predicates, and the group structure on region transitions are all taken from the foundations paper [3] and are not re-derived here. The relation is a strict import, and the boundary is the one drawn in Sec. 1: that result establishes, for any ordered pair of regions, that a unique transition between them exists, and it supplies no procedure that carries a given artifact along one. The distinction is the familiar one between an existence statement about a structure and a construction that acts on its elements, and it is the reason this paper studies four restricted mechanisms rather than claiming the general case. Nothing below closes that gap, and we do not describe the present results as *generalizing* the imported theorem: no embedding or reduction between the two settings is exhibited, and without one the word would overclaim.

10.3 Offset regularity in continuous word representations

Def. 6.1 takes its *form* from the observation that relations between words appear in continuous representations as consistent offsets between vectors [31, 30]. We take the form and nothing else: no claim about which similarity measure recovers such offsets is imported, and none is made.

That reservation is not ours alone, and the subsequent literature is the reason it has to be stated. Evaluations of the vector-offset method have found its apparent success to depend in part on the proximity structure of the neighbourhood rather than only on the offset itself, and to be sensitive to whether the query words are excluded from the candidate set at all [26]; a broader assessment reaches the same conclusion, that the method's success depends on vector similarity, which the authors take to make it a poor model of the relations it is supposed to encode [39]; and a third argues that, even granting that such regularities exist to be found, standard implementations of the test return distorted results, with low-dimensional projections poor evidence for the regularities they are used to display [41].

These three are methodological: they concern how the test is run and what its outcomes license, and none of them states a claim about the similarity measure as such. That is the boundary we are careful about below. Our result is of a different kind — a formal limitation, under this paper's declared geometric model, on what a scale-invariant measure can separate — and it is neither proved nor anticipated by any of them. The three analyses and ours support a common caution from different levels of abstraction, and that is the whole of the relation we claim.

Our own treatment reaches a related conclusion from the opposite direction, and the agreement is worth recording because the two routes are independent. We require the declared similarity-and-tolerance pair to be *separating*, which is a metric-like demand; Rem. 6.1 shows that cosine similarity admits no separating radius on any embedding of more than one point, and exhibits a collinear configuration on which every pair is offset-regular for every positive multiple of one vector. Where the empirical literature finds that cosine-based offset recovery reports neighbourhood structure, the same measure is shown here to be unable to separate a relation the geometry realizes from one it does not. We claim no empirical result and run no model: the offset condition adopted here is an exact ideal one, and Rem. 6.2 states exactly how narrow that makes it.

10.4 Trace vocabulary, and what is not claimed with it

Five statements in this paper are written with temporal operators, and the four-family vocabulary naming their finite-prefix behaviour is taken from [2]. The import is terminological. All five are conjectural or explicitly open, none is used in a proof, and their run domains are declared locally — the alphabets of Secs. 4.2 and 5.3, and iterations of the minting-and-recompute step for the remaining three. What is *not* settled here is the class of admissible extensions each formula quantifies over, and with it every monitor-side notion: no monitor is constructed in this paper, no soundness or completeness statement is made about one, and no monitorability result is proved. The vocabulary names a behaviour; it does not carry a verification apparatus with it.

10.5 Four adjacent literatures this paper does not draw on

Several fields are adjacent to the vocabulary used here, and the honest report is that the manuscript defines its own objects rather than importing theirs. We name them so that a reader does not mistake a coincidence of words for a borrowing.

Denotational semantics. The relation E_D of Sec. 9 is not an interpretation function into a domain of meanings. It is a binary relation between artifacts, plural by construction, with no compositional clause and no fixed-point apparatus; the fourteen classes classify its *source-side*

shape and not the semantics of any language. The word *denotes* is used in a project-specific sense that the definition fixes, and no result here bears on denotational semantics in the standard sense.

Refinement calculi. The gap between an algebraic transition and a procedure that carries an artifact along it is the same shape as the gap between a specification and its implementation, and that resemblance is why the distinction is easy to state. It is a resemblance and not an instance: no refinement relation is defined here, no simulation is exhibited, and the four mechanisms are not presented as refinements of the imported transition.

Closure-operator theory. The operator cl_Φ is defined directly in Def. 6.1 and used only through that definition, as the least superset closed under its steps. Extensivity, monotonicity and idempotence follow immediately from that description, and the proof of Prop. 4.4 uses the last two; what we do not import is any theory built on the axioms — no lattice of closed sets, no Galois connection, no representation result — and no claim here depends on one.

Certificate and proof-carrying frameworks. What Rem. 5.8 isolates is the *information content* of a candidate certificate relation: which propositions the pair (k, f) settles about which artifacts. It is not a verification architecture. No verifier is constructed, no proof system is fixed, and the remark states in its own text that soundness and completeness are absent — an equation $f(m, k) = X$ is an equation, not a verdict, and the manuscript is careful not to let the second reading in.

10.6 Vocabulary expansion, embedding initialisation, and continued pretraining

Modern engineering practice for extending a language model's representational substrate combines three operations that are individually standard but whose joint architectural placement is what we locate here. OFA [27] factorises the source embedding matrix, adopts a target-side tokenizer, initialises target-token embeddings from external multilingual word vectors via a bipartite word–subword graph (copy for shared tokens, similarity-weighted convex combination for non-shared, Gaussian fallback), and continues pretraining. HYPEROFA [37] replaces the convex-combination initialisation with a learned hypernetwork over external multilingual vectors. Mundra's CW2V [35] learns a stochastic weight matrix producing convex-hull-compatible initialisations and proves (Thm. 1–2 there) that greedy generation over the source-vocabulary prefix is preserved at initialisation, before continued pretraining — a formal guarantee about the initialisation step, which we cite rather than adopt; it addresses a different question from Conj. 4.3. Tejaswi et al. [42] quantify the capability-drop and capability-recovery curves across (vocabulary-size, CPT-budget, base-LM) triples for language-specific LLM construction. VRCP [36] isolates a distinct vocabulary-*replacement* axis (fixed-total vocabulary size, non-additive) rather than the vocabulary-*extension* axis the other four share.

Placement. Each of the four vocabulary-extension mechanisms produces admitted lexical items that Def. 4.2 would register as candidate admissibility acts, and each supplies a stage that we read as the recompute of Conj. 4.3. *What this paper can register:* that each mechanism admits new tokens and runs an initialisation (and, in most cases, CPT) step we read as recompute. *What this paper cannot infer from these mechanisms alone:* whether Def. 6.1's separating-similarity condition holds on the resulting embedding (an empirical question); whether the mint sequence reaches a fixpoint after a finite number of steps (Sec. 11); whether Conj. 4.5's antecedent (integrated acquisition) is satisfied in any specific realisation. Table 4 records the placement compactly.

Continued pretraining as such is engineered in Ke et al. [20] with soft-masking against catastrophic forgetting, sitting in the classical continual-learning line of LwF [24] and EWC [21]. Sec. 11 discusses our own open question about training-time status against this background.

Mechanism	Our reading (Def. or Conj.)	Not registrable from the mechanism alone
OFA [27]	Def. 4.2 admissibility for target vocabulary; Conj. 4.3 with initialisation (and, where performed, CPT), read as recompute	whether Def. 6.1 holds after CPT; whether Sec. 11 fixpoint reached
HYPEROFA [37]	same as OFA, with the initialisation produced by a hypernetwork	same as OFA
Mundra CW2V [35]	same as OFA; Thm. 1–2 (in source): convex-hull initialisation preserves greedy generation on source-vocabulary prefixes at initialisation; a different question from Conj. 4.3	whether initialisation counts as the recompute of Conj. 4.3
Tejaswi et al. [42]	empirical study of vocabulary size, CPT budget and base LM on downstream performance; relevant to, not an instance of, stages (6)–(7) of Sec. 8(v)	whether any specific triple satisfies Conj. 4.5
VRCP [36]	outside Def. 4.2 (no removal clause); adjacent case	whether replacement preserves any pre-replacement Reach_Φ

Table 4: Vocabulary-axis mechanisms placed in this paper's definitions and conjectures (our reading; not a result).

Remark 10.1 (The vocabulary-extension family read against Sec. 8(v)). Read against the seven-stage ladder of Sec. 8(v), the vocabulary-axis-extension family (OFA, HYPEROFA, Mundra CW2V and Tejaswi; VRCP by replacement at fixed vocabulary size) is engineering practice for the lexical step that stage (5) names, and it reports downstream-task performance after continued pretraining. That evidence bears on stages (6) and (7) without instantiating them as defined there: stage (6) is declined in this paper, and stage (7) is the consequent of Conj. 4.5, stated in terms of Reach_Φ. These works construct no point by cl_Φ (stage (1)) and supply no criterion of semantic validity (stage (3)). This is our reading and not a result.

10.7 Model editing and localised parametric change

A parallel line modifies model behaviour or knowledge through targeted parameter updates that leave the tokenizer and the embedding matrix dimensionally fixed. ROME [28] performs a rank-1 update to a mid-layer MLP to install a single factual association. MEMIT [29] distributes the same class of update across several layers and scales to thousands of associations. EMMET [16] unifies ROME and MEMIT under one preservation-memorisation objective. MEND [33] instead uses small learned editor networks that transform the fine-tuning gradient. Task Arithmetic [19] composes weight-space deltas rather than editing specific facts.[3] KnowledgeEditor [8] is a hypernetwork-

[3]We include Task Arithmetic here under the parametric-change umbrella because it shares the fixed-vocabulary weight-modification substrate.

editor baseline, evaluated alongside MEND in, e.g., [28]. Cohen et al. [7] formalise the induced ripple set $R(\delta)$ of implied triplet changes following an edit δ.

Placement. These operate on a distinct axis from the mint in its language-model reading (Rem. 4.1): none of them extends the vocabulary or the embedding matrix's row count; they modify weights under a fixed input-token axis. That reading of the mint changes the input-token axis; ROME/MEMIT/EMMET/MEND/Task Arithmetic change weight internals. Both are parametric transformations, and they are *adjacent* — one agent may perform both — but they are not instances of one operation. *This paper can locate:* parametric change under fixed vocabulary. *This paper cannot infer from an edit alone:* vocabulary-axis extension; denotation change (denotation is not recoverable from region predicates, Cor. 9.11); comprehension-function strength change; conservative capability expansion (the antecedent of Conj. 4.5 is not settled by any parametric edit under fixed vocabulary). Rem. 11.1 records that Cohen's ripple set $R(\delta)$, even read as a closure, and this paper's cl_Φ are different objects.

10.8 In-context and external-memory adaptation are outside the present substrate

Two families of adaptation mechanism sit outside the scope fixed here: the formal objects of this paper ($\mathcal{A}^+$, the realisation emb, the envelopes) carry no context-window or external-memory state, and we record the boundary rather than expand to cover them. In-context learning as studied by Min et al. [32] exploits label space, input distribution, and sequence format under a fixed model, without a persistent parametric transformation; this class of behaviour is therefore outside the present formalism. Retrieval-augmented generation [23] and Memorizing Transformers [43] similarly maintain a non-parametric external store that the base model consults at inference time; the persistence locus is external to the model's parametric substrate and again outside the present scope. Editing approaches that leave the base model's parameters unchanged, such as the in-context IKE [45] and the memory-based SERAC [34], are excluded on the same criterion: they operate via prompting or retrieval rather than a parametric update of the base model.

10.9 Evaluation-methodology adjacency

Adjacent to this paper's substrate is a methodology literature on evaluation under contamination and adaptive analysis. Sainz et al. [40] argue, in a position paper, that benchmark contamination inflates scores and must be measured per benchmark; Dwork et al. [9] initiate a formal study of statistical validity in adaptive data analysis; Blum and Hardt [1] propose the Ladder mechanism for reliable leaderboards. These works address different questions — how a model's exposure to benchmark test data during pretraining inflates its evaluation (contamination), and how adaptive reuse of shared holdout data corrupts inference on the underlying distribution — not what representational substrate the evaluated model carries. They are methodological adjacencies rather than importable machinery.

10.10 Mechanistic interpretability adjacency

Mechanistic interpretability studies how neural networks can represent more features than they have dimensions, in *superposition* — where one neuron or one direction carries several features — including, in toy models, when this occurs and whether decomposition such as sparse coding could recover the features [10]. The substrate (neural representations + learned features) differs from ours ($\mathcal{A}^+$ artefacts + envelopes), but the two share a concern: distinguishing what a model *operates on* (executability) from what a model *can be read as representing* (readability). This paper's R_5 /

R_6 / R_7 distinction and mechanistic interpretability's readable-vs-superposed distinction address adjacent phenomena at different abstraction levels; neither specialises the other, and no formal reduction is proposed here.

11 Discussion and Open Questions

The machinery developed above — envelope theory (Sec. 3), the verifier transition model (Sec. 4), and the functional-understanding hierarchy (Sec. 5) — overlays MCK region semantics on an embedding space and give a controlled account of how the space grows when a token-less concept is reached. Section 9 stands apart from them: it discharges a deferral the foundations paper makes by name, supplying the axioms of the denotation graph and the classification of symbols by the shape of $T(s)$, and it is independent of the geometric material in both directions. Its two negative results bear on the rest: denotation is not determined by the region data, so the staticity of the denotation map cannot be read off them and stands there as a stipulation; and Θ is not the set of symbols carrying several acceptations, so plurality of targets is no test of symbolhood. We close with the open questions this first draft deliberately leaves standing.

Training-time status of the embedding. Is the embedding space fixed *before* training or co-formed *during* it, and what is its relation to the trained MLP? This determines whether "recompute the envelopes" (Def. 4.2, Conj. 4.3) is a well-defined operation or an approximation. The question is unresolved in the source corpus and is not settled here.

Frontier non-emptiness. Is there always a next token-less reachable concept — $\mathsf{G}\,\mathsf{F}$ gap-exists, a property that is at once liveness *and* coliveness — or does iterating the minting-and-recompute step of Section 8 reach a fixpoint? We do not decide this, and the obstruction is structural rather than prudential: a property in that intersection is determined by no finite prefix, so no finite observation could settle the question either way [2]. Establishing frontier non-emptiness would be a crossing from finite evidence to an infinite claim, requiring a decomposition that states which portion of the concept space remains uncovered; it is tagged *conjectural at most* and carries the same caveat as Rem. 3.4. Finite evidence that "we always found a next concept" must not be inflated into the infinite claim.

The conditional unlocking claim. Whether minting new tokens extends what an agent can structurally reach (Conj. 4.5) is left open, and is stated there as a conditional whose antecedent this paper does not discharge. Two of its conjuncts — that the operator is acquired and that it is integrated — name the acquisition question that Def. 4.2 declines, so the claim cannot be settled here without the treatment of acquisition that neither this paper nor its foundations supplies. What *is* settled is the negative half, Prop. 4.4: under a fixed regularity basis a mint never extends structural reach, because the minted point already lies in the closure. Any extension under a fixed realization therefore requires the basis itself to change, and exhibiting one relation that becomes available only after a mint would be the first genuine evidence for the claim.

Continued pretraining placement. In the language-model reading of the mint (Rem. 4.1), continued pretraining is the standard engineering practice for adapting a model after vocabulary-axis extension. Ke et al. [20] study catastrophic forgetting in continual domain-adaptive pre-training and propose a soft-masking mechanism treating general-knowledge preservation as an importance-weighted gradient constraint; the classical continual-learning line — LwF [24] and EWC [21] — is

the surrounding family. This literature documents forgetting under continued training and proposes mitigations — soft-masking [20], distillation from the original model [24], importance-weighted regularisation [21]; none of the three addresses new vocabulary items specifically, and none guarantees conservative capability expansion. The two preceding paragraphs defer the fixpoint question and the acquisition question behind Conj. 4.5; this literature settles neither.

Remark 11.1 (Cohen's ripple set and cl_Φ are distinct objects). The ripple effect $R(\delta)$ of Cohen et al. [7] is the set of triplets a model needs to inject, modify or delete from its knowledge graph $\mathcal{G}$ (their K) to reflect the world state after a fact edit $\delta : (e, r, o) \to (e, r, o')$. It is defined on relational triples and fixed by semantic implication; Cohen et al. do not present it as a closure, though it can be read as one. The Φ-closure cl_Φ of Def. 6.1 is the least superset of a point set closed under difference-vector steps $x \mapsto \mathrm{emb}^{-1}[\mathrm{emb}(x) \pm \delta_r]$; it is defined on $\mathcal{A}^+$ via emb, and its closure operator is geometric. Read as closures, the two objects operate on different substrates and use different mechanisms; neither specialises the other, and no reduction between them is known to us or attempted here. The absence of a reduction is compatible with this paper's declined interest in closure-operator theory (Sec. 10.5).

12 Conclusion

The transition structure this paper imports settles, for any ordered pair of regions, that a unique transition between them exists [3]. It is an algebraic fact about the classification and not a procedure: it names what is structurally possible and moves no artifact. Everything below is an attempt to say what happens when one asks for the procedure instead.

Four mechanisms were studied, each restricted and each stated for a declared observer. Assimilation reclassifies opacity on the strength of a comprehension function. Operator admission seats a candidate in the fully intersecting region once a verifier accepts it. Denotation relates a symbol to a target without moving either. Reconstruction certification exhibits a target as the value of a known function at known arguments. They are not variants of one operation: their domains differ, and so do the reasons each one stops.

What they achieve is exact. Assimilation removes pure opacity, and the removal is genuine wherever a comprehension function is defined. Θ is exactly the set of admissible denotational sources — which is not to say every symbol denotes, degree zero being permitted throughout — so the two-row shape of the denotation graph is a consequence of the definitions and not a modelling choice (Prop. 9.3). Under a separating similarity-and-tolerance pair the offset condition is exact rather than approximate, which fixes precisely the class of representations on which the regularity notion has content (Rem. 6.2).

The limits are sharper than the achievements, and they are the paper's substance rather than its caveats. Assimilation stops at a floor: the opacity co-present with meaning or with function is a lower bound no comprehension strength lowers (Thm. 5.6, Prop. 6.5), so an agent that possesses symbols cannot assimilate its way out of possessing them. The step may also select nothing at all — there is an assignment on which the reconstruction certificate holds at every target and no element is reclassified (Prop. 5.9(iii)). Admission leaves structural reach unchanged under a fixed realization, so what extends reach is the recomputation and never the name (Prop. 4.4, Rem. 4.2). Denotation is not recoverable from the regional predicates, so a perfect reading of an artifact's label still leaves what it denotes undetermined (Cor. 9.11). And the similarity measure under which the motivating regularities are usually reported admits no separating radius at all (Rem. 6.1).

Read together these say one thing: *a resource introduced at one level need not change capability at another.* A name may leave reach unchanged, candidate certificate data may leave comprehension

unresolved (Prop. 5.11), and an admissible assimilation step may select no artifact. None of this precludes stronger mechanisms. It locates what they would have to supply, since the algebraic transition structure alone supplies none of it. The universal artifact-level realizer — a procedure that, given any admissible structural transition, carries an arbitrary artifact along it — is not established here and remains open.

References

[1] Avrim Blum and Moritz Hardt. The ladder: A reliable leaderboard for machine learning competitions. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, 2015. arXiv:1502.04585. https://arxiv.org/abs/1502.04585.

[2] Davide Bragetti. The intersection algebra of safety, cosafety, liveness, and coliveness over linear temporal logic, 2026. Zenodo, CC BY 4.0. Concept DOI: https://doi.org/10.5281/zenodo.20155195.

[3] Davide Bragetti, Carmelo Asaro, and Giuseppe Francesco Italiano. Meaning, opacity, and function: An observer-relative classification of artifacts, 2026. Zenodo, CC BY 4.0. Version 4 (61pp). Version DOI: https://doi.org/10.5281/zenodo.22016203.

[4] Huili Chen, Stephen R. Grimm, Olga Russakovsky, and Tania Lombrozo. Machine understanding. *Trends in Cognitive Sciences*, 2026.

[5] Jonathan H. Clark, Dan Garrette, Iulia Turc, and John Wieting. CANINE: Pre-training an efficient tokenization-free encoder for language representation. *Transactions of the Association for Computational Linguistics*, 10:73–91, 2022.

[6] Michael R. Clarkson and Fred B. Schneider. Hyperproperties. *Journal of Computer Security*, 18(6):1157–1210, 2010.

[7] Roi Cohen, Eden Biran, Ori Yoran, Amir Globerson, and Mor Geva. Evaluating the ripple effects of knowledge editing in language models. *Transactions of the Association for Computational Linguistics*, 2024. arXiv:2307.12976. https://arxiv.org/abs/2307.12976.

[8] Nicola De Cao, Wilker Aziz, and Ivan Titov. Editing factual knowledge in language models. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2021. arXiv:2104.08164. https://arxiv.org/abs/2104.08164.

[9] Cynthia Dwork, Vitaly Feldman, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Aaron Roth. Preserving statistical validity in adaptive data analysis. In *Proceedings of the 47th Annual ACM Symposium on Theory of Computing (STOC)*, 2015. arXiv:1411.2664. Companion in *Science* 349(6248), 636–638, 2015 ("The reusable holdout"). https://arxiv.org/abs/1411.2664.

[10] Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah. Toy models of superposition, 2022. Transformer Circuits Thread. https://transformer-circuits.pub/2022/toy_model/index.html.

[11] Kawin Ethayarajh. How contextual are contextualized word representations? comparing the geometry of BERT, ELMo, and GPT-2 embeddings. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 55–65, 2019.

[12] Ronald Fagin and Joseph Y. Halpern. Belief, awareness, and limited reasoning. *Artificial Intelligence*, 34(1):39–76, 1987. Author copy: `https://www.cs.cornell.edu/home/halpern/papers/awareness.pdf`.

[13] Jun Gao, Di He, Xu Tan, Tao Qin, Liwei Wang, and Tie-Yan Liu. Representation degeneration problem in training natural language generation models. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1907.12009. `https://arxiv.org/abs/1907.12009`.

[14] Shafi Goldwasser, Silvio Micali, and Charles Rackoff. The knowledge complexity of interactive proof systems. *SIAM Journal on Computing*, 18(1):186–208, 1989.

[15] Jens Groth. On the size of pairing-based non-interactive arguments. In *Advances in Cryptology – EUROCRYPT 2016*, volume 9666 of *LNCS*, pages 305–326. Springer, 2016.

[16] Akshat Gupta, Dev Sajnani, and Gopala Anumanchipalli. A unified framework for model editing. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024. arXiv:2403.14236. `https://arxiv.org/abs/2403.14236`.

[17] Joseph Y. Halpern, Yoram Moses, and Moshe Y. Vardi. Algorithmic knowledge. In *Proceedings of the Fifth Conference on Theoretical Aspects of Reasoning about Knowledge (TARK 1994)*, pages 255–266, 1994. Formulation cited as reported in [18].

[18] Joseph Y. Halpern and Riccardo Pucella. Probabilistic algorithmic knowledge. *Logical Methods in Computer Science*, 1(3:1), 2005. arXiv:cs/0503018. Also cited as the source for Halpern, Moses and Vardi's 1994 formulation of algorithmic knowledge (their Sec. 1).

[19] Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2212.04089. `https://arxiv.org/abs/2212.04089`.

[20] Zixuan Ke, Yijia Shao, Haowei Lin, Tatsuya Konishi, Gyuhak Kim, and Bing Liu. Continual pre-training of language models. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2302.03241. `https://arxiv.org/abs/2302.03241`.

[21] James Kirkpatrick, Razvan Pascanu, Neil Rabinowitz, Joel Veness, Guillaume Desjardins, Andrei A. Rusu, Kieran Milan, John Quan, Tiago Ramalho, Agnieszka Grabska-Barwinska, Demis Hassabis, Claudia Clopath, Dharshan Kumaran, and Raia Hadsell. Overcoming catastrophic forgetting in neural networks. *Proceedings of the National Academy of Sciences*, 114(13):3521–3526, 2017.

[22] David Kirsh and Paul Maglio. On distinguishing epistemic from pragmatic action. *Cognitive Science*, 18(4):513–549, 1994.

[23] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In

Advances in Neural Information Processing Systems (NeurIPS), 2020. arXiv:2005.11401. https://arxiv.org/abs/2005.11401.

[24] Zhizhong Li and Derek Hoiem. Learning without forgetting. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 40(12):2935–2947, 2018.

[25] D. V. Lindley. On a measure of the information provided by an experiment. *The Annals of Mathematical Statistics*, 27(4):986–1005, 1956.

[26] Tal Linzen. Issues in evaluating semantic spaces using word analogies. In *Proceedings of the 1st Workshop on Evaluating Vector-Space Representations for NLP*, pages 13–18, 2016. ACL Anthology W16-2503. https://aclanthology.org/W16-2503/.

[27] Yihong Liu, Peiqin Lin, Mingyang Wang, and Hinrich Schütze. OFA: A framework of initializing unseen subword embeddings for efficient large-scale multilingual continued pretraining. In *Findings of the Association for Computational Linguistics: NAACL 2024*, pages 1067–1097, 2024.

[28] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2202.05262. https://arxiv.org/abs/2202.05262.

[29] Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. Mass-editing memory in a transformer. In *International Conference on Learning Representations (ICLR)*, 2023. arXiv:2210.07229. https://arxiv.org/abs/2210.07229.

[30] Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. Efficient estimation of word representations in vector space, 2013. arXiv:1301.3781. https://arxiv.org/abs/1301.3781.

[31] Tomas Mikolov, Wen-tau Yih, and Geoffrey Zweig. Linguistic regularities in continuous space word representations. In *Proceedings of the 2013 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 746–751, 2013. ACL Anthology N13-1090. https://aclanthology.org/N13-1090/.

[32] Sewon Min, Xinxi Lyu, Ari Holtzman, Mikel Artetxe, Mike Lewis, Hannaneh Hajishirzi, and Luke Zettlemoyer. Rethinking the role of demonstrations: What makes in-context learning work? In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 11048–11064, 2022.

[33] Eric Mitchell, Charles Lin, Antoine Bosselut, Chelsea Finn, and Christopher D. Manning. Fast model editing at scale. In *International Conference on Learning Representations (ICLR)*, 2022. arXiv:2110.11309. https://arxiv.org/abs/2110.11309.

[34] Eric Mitchell, Charles Lin, Antoine Bosselut, Christopher D. Manning, and Chelsea Finn. Memory-based model editing at scale. In *International Conference on Machine Learning (ICML)*, 2022. arXiv:2206.06520. https://arxiv.org/abs/2206.06520.

[35] Nandini Mundra, Aditya Nanda Kishore Khandavally, Raj Dabre, Ratish Puduppully, Anoop Kunchukuttan, and Mitesh M. Khapra. An empirical comparison of vocabulary expansion and initialization approaches for language models. In *Proceedings of the 28th Conference on Computational Natural Language Learning (CoNLL)*, 2024. arXiv:2407.05841. https://arxiv.org/abs/2407.05841.

[36] Yuta Nozaki, Dai Nakashima, Ryo Sato, Naoki Asaba, and Shintaro Kawamura. VRCP: Vocabulary replacement continued pretraining for efficient multilingual language models. In *Proceedings of the 2nd SUMEval Workshop: Scaling Up Multilingual & Multi-Cultural Evaluation*, pages 48–59, 2025. https://aclanthology.org/2025.sumeval-2.5/.

[37] Enes Özeren, Yihong Liu, and Hinrich Schütze. HYPEROFA: Expanding LLM vocabulary to new languages via hypernetwork-based embedding initialization. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics: Student Research Workshop*, pages 79–96, 2025.

[38] Artidoro Pagnoni, Ram Pasunuru, Pedro Rodriguez, John Nguyen, Benjamin Muller, Margaret Li, Chunting Zhou, Lili Yu, Jason Weston, Luke Zettlemoyer, Gargi Ghosh, Mike Lewis, Ari Holtzman, and Srinivasan Iyer. Byte latent transformer: Patches scale better than tokens, 2024. arXiv:2412.09871. https://arxiv.org/abs/2412.09871.

[39] Anna Rogers, Aleksandr Drozd, and Bofang Li. The (too many) problems of analogical reasoning with word vectors. In *Proceedings of the 6th Joint Conference on Lexical and Computational Semantics (*SEM 2017)*, pages 135–148, Vancouver, Canada, 2017. Association for Computational Linguistics. ACL Anthology S17-1017. https://aclanthology.org/S17-1017/.

[40] Oscar Sainz, Jon Ander Campos, Iker García-Ferrero, Julen Etxaniz, Oier Lopez de Lacalle, and Eneko Agirre. NLP evaluation in trouble: On the need to measure LLM data contamination for each benchmark. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, 2023. arXiv:2310.18018. https://arxiv.org/abs/2310.18018.

[41] Natalie Schluter. The word analogy testing caveat. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 2 (Short Papers)*, pages 242–246, 2018. ACL Anthology N18-2039. https://aclanthology.org/N18-2039/.

[42] Atula Tejaswi, Nilesh Gupta, and Eunsol Choi. Exploring design choices for building language-specific LLMs. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, 2024. arXiv:2406.14670. https://arxiv.org/abs/2406.14670.

[43] Yuhuai Wu, Markus N. Rabe, DeLesley Hutchins, and Christian Szegedy. Memorizing transformers. In *International Conference on Learning Representations (ICLR)*, 2022. arXiv:2203.08913. https://arxiv.org/abs/2203.08913.

[44] Linting Xue, Aditya Barua, Noah Constant, Rami Al-Rfou, Sharan Narang, Mihir Kale, Adam Roberts, and Colin Raffel. ByT5: Towards a token-free future with pre-trained byte-to-byte models. *Transactions of the Association for Computational Linguistics*, 10:291–306, 2022.

[45] Ce Zheng, Lei Li, Qingxiu Dong, Yuxuan Fan, Zhiyong Wu, Jingjing Xu, and Baobao Chang. Can we edit factual knowledge by in-context learning? In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023. arXiv:2305.12740. https://arxiv.org/abs/2305.12740.